

PRIME PURITY - SOLAS TRAINING MANUAL

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SECTION 1 EMERGENCY INSTRUCTIONS

1.1 MUSTER LIST

Introduction

Muster List, 1s notice which display the names and rank of each member of the ship’s complement including supernumeraries, with their allocated muster points and associated duties in the event of an emergency on board.

Authors note: Muster List to be insert when available.

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1.2 EMERGENCY SIGNALS

Introduction

When an emergency arises, the crew will be alerted by the ship's whistle and alarm devices. Instructions on the public address system may also be provided. On hearing an emergency alarm, all personnel must go to their muster stations as listed on the Muster List.

The fire alarm consists of the continuous sounding of the alarm devices.

The actual signal generated by Fire Detection System, is different than the Fire Alarm. During ship's familiarisation and drills the crew is made aware of the different sounds.

Note: Fire Alarm may differ from the alarm generated by the fire detection system.

The general alarm is seven short blasts (each blast lasting about one second), followed by one prolonged blast (ten seconds or more) on the ship's whistle, with the same signal on the general alarm devices.

The man overboard signal consists of three long blasts (morse letter 'O' repeated at least four times on the ship's whistle, with the same signal on the general alarm bells.

Information regarding the emergency signals used on board can be found on the Muster List.

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1.3 EMERGENCY STATIONS

Introduction

The emergency station on the bridge is for overall command, communications and control of the situation, with continued navigational watch duties ongoing.

The emergency station is to meet all the needs of the emergency situation and the continued running of the machinery, including any changes in status of machinery.

The emergency stations have been selected as the places from where each emergency team can best prepare to deal with the emergency and carry out their duties. Proximity of equipment such as fire fighting materials, breathing apparatus, emergency control devices, medical supplies etc, are available for quick intervention by the designated teams when required.

A muster must be taken by each group to establish if there are any missing persons. Once completed, a report must be issued to the overall Command Group located on the bridge/safety centre.

The nature of the emergency will dictate the following actions and further instructions should follow from the overall command team on the bridge. However, each person should be prepared and ready to undertake their assigned tasks as detailed on the station bill.

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1.4 FIRE ALARM PROCEDURE

Introduction

On discovery of a fire, the alarm should be raised immediately, either by operating a manual call point, or by contacting the bridge so the duty officer can activate the fire alarm. The fire alarm will be indicated by a continuous ringing of the ship's alarm devices.

The automatic fire detection system also activates the fire alarm if a fire detection alarm is not acknowledged within a predetermined period. If this operates before a fire alarm is sounded, secure your workstation and go to your designated muster station.

The ship's crew is divided into teams as specified on the Station Bill, with each team having a designated muster position. Each team has a leader and the team members are allocated specific duties. Each team leader must direct their squad to carry out the instructions received from the bridge team.

Each leader must also keep the bridge team informed on what is happening, allowing a record of events to be kept and an overview maintained. Based on the information received, the bridge team can issue new instructions and make decisions regarding the safety of the vessel and personnel on board.

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1.5 SUPERNUMERARY EMERGENCY INSTRUCTIONS

Authors note: Please provide these instructions so they can be inputted in to the manual?

DRAFT MANUAL FOR REVIEW & COMMENT

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1.6 MAN OVERBOARD PROCEDURE

Introduction

If a person is seen falling or jumping over the side, an observer must:

- Throw a lifebuoy over the side, towards the person immediately.
- Shout ‘**MAN OVERBOARD! STARBOARD/PORT SIDE!**’.
- Instruct someone to notify the bridge immediately or do it personally if no one else is around. The bridge should be advised of the following:
 - Name of observer, location, contact number.
 - Describe what was observed and which side the person went overboard.
- Keep a lookout for the person in the water or the lifebuoy.

Upon sighting a man overboard, it is important to raise the alarm as soon as possible by informing the bridge. This will enable the officer of the watch to sound the man overboard alarm, comprising of three prolonged blasts of on the ship’s whistle and general alarm, and commence the initial actions as per their emergency procedures.

When under way, the officer of the watch must mark the position, release the bridge wing lifebuoy, post a lookout for the man overboard and take the necessary ship manoeuvre.

Manoeuvre

The Williamson Turn is a proven method of returning the ship, via a reciprocal course, to the position of the person overboard whilst at the same time considerably reducing the ship’s speed.

However, due to the heavy traffic or confined waters, this manoeuvre may not always be possible to carry out or indeed the be the best manoeuvre for the current environmental and traffic conditions. The decision on the most suitable manoeuvre to use will be made by the Captain.

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1.7 ABANDON SHIP PROCEDURE

Introduction

The order to abandon ship will be given by the Master or, if they are incapacitated, by their appointed deputy. The order to ABANDON SHIP will normally be given via the public address system and over the radios, but only after the team leaders have reported that all persons are present and accounted for, and the final distress message has been sent. When abandoning the ship, the designated portable VHF radios, radar transponder (SART) should be taken to the survival craft and activated.

After the order is given and when proceeding to the survival craft, every effort should be made to ensure all persons have suitable dry, warm clothing covering the whole body. In this respect, a head covering would also be beneficial.

The following points should be followed if at all possible:

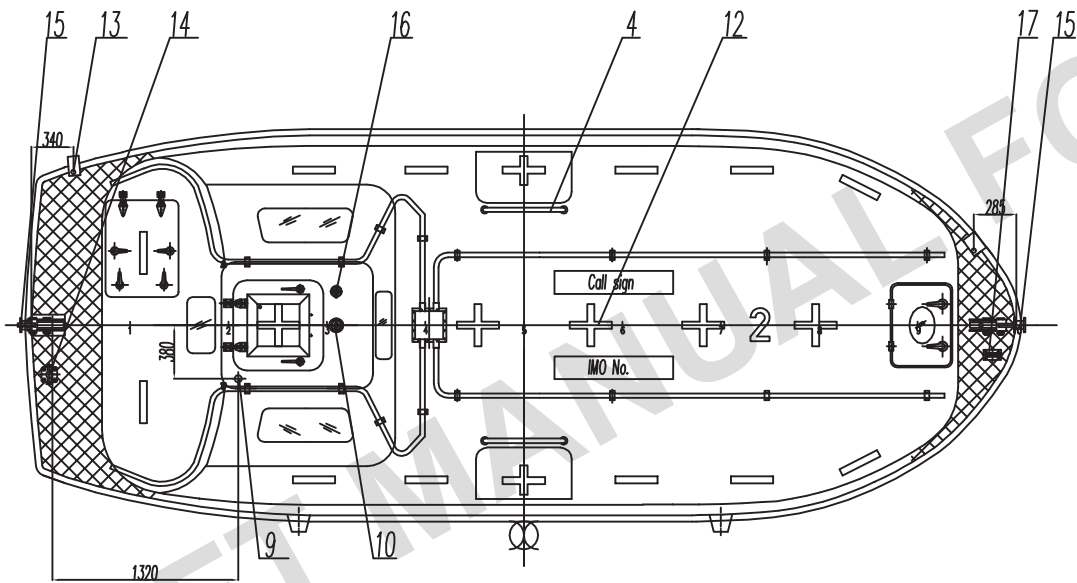
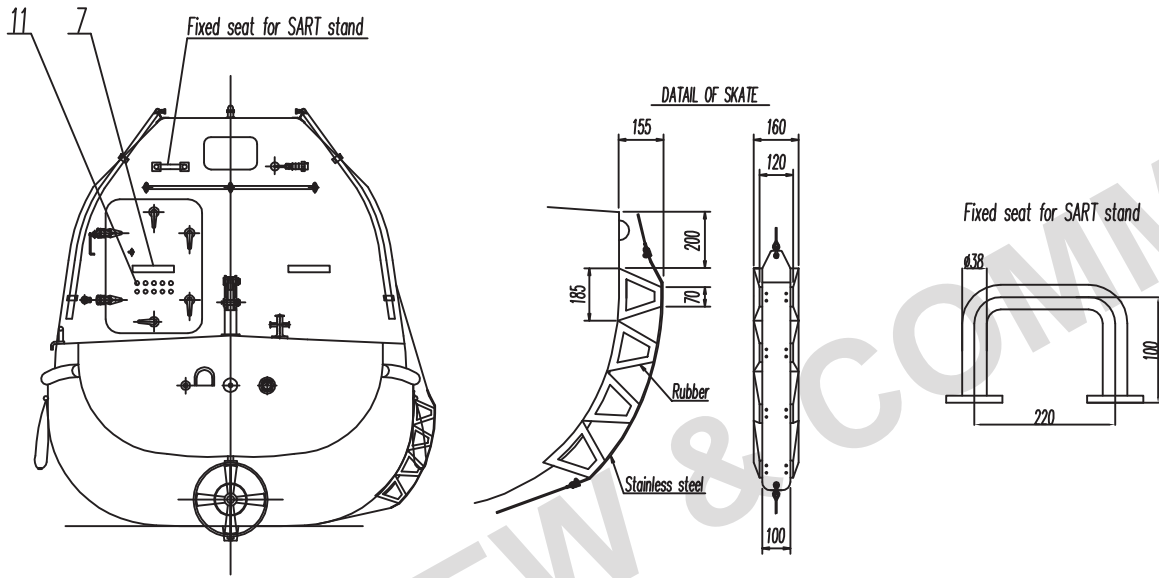
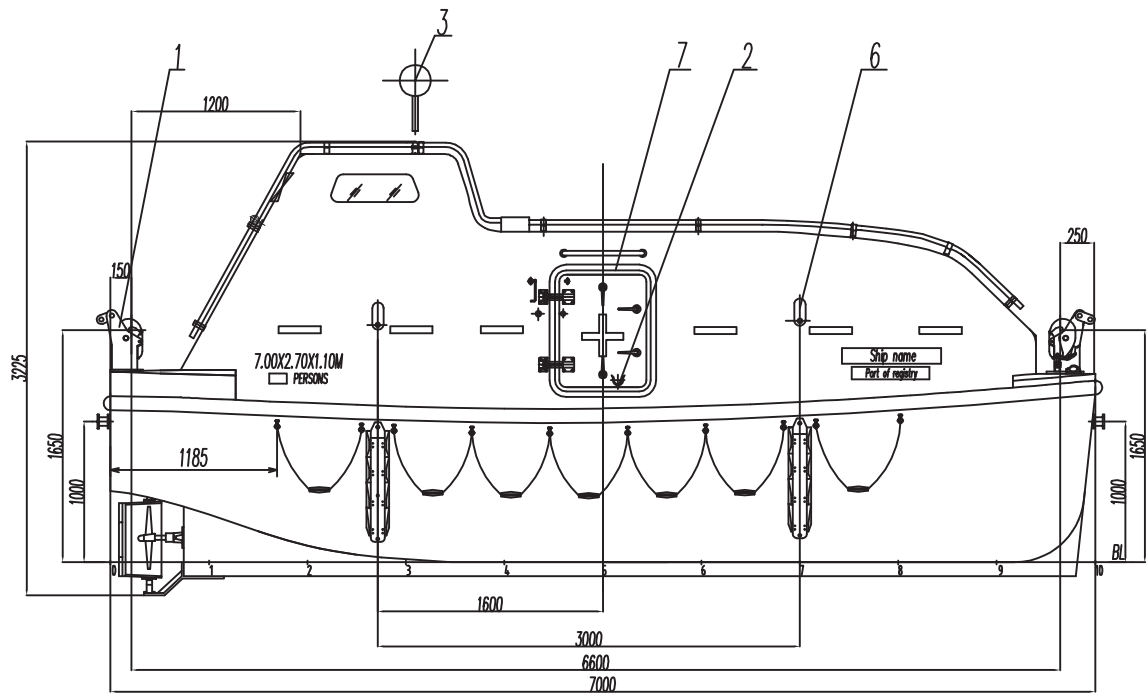
- Put on as many layers of warm clothing as possible, including foot protection, making sure to cover head, neck and hands. Clothing that fastens close to avoid ingress of water is preferable, but other types should not be discounted.
- Life jackets or immersion suits should be donned.
- All personnel should take anti-seasickness tablets after boarding the liferaft. Seasickness causes dehydration which will make hypothermia more likely.
- Avoid entering the water if at all possible; board the survival craft on the embarkation deck. If this is not possible, try to escape whilst keeping dry by using the overside embarkation ladders.
- Unless it is unavoidable, do not jump from higher than 5m (16.5ft) into the water. Try to minimise the shock of cold immersion. The sudden plunge into cold water can cause rapid death or an uncontrolled rise in breathing rate which may result in the intake of water into the lungs.
- Should it be necessary to jump into the water, the following method should be adopted:
 - Button-up clothing.
 - Ensure the life jacket is tightly secured.
 - Immersion suit correctly donned (if wearing one).
 - Keep your elbows tight in to your side.
 - Cover your nose and mouth with your left hand then cross over your left hand with your right hand and hold on to the life jacket collar.
 - Do not jump into the water astern of a liferaft in case the ship still has some headway on.
- Once in the water, try and orientate yourself and try and locate the ship, liferafts, other survivors or any floating objects.
- Confirm the life jacket light has been activated and use the whistle.
- Confirm the immersion suit light has activated and use the whistle (if wearing).
- While in the water, do not attempt to swim, unless to reach a nearby craft, fellow survivor or floating object on which to lean or climb.

2.1 Lifesaving and Safety Equipment

- 2.1.1 Lifeboat and Davit**
- 2.1.2 Liferafts**
- 2.1.3 Survival Craft Survival Guide**
- 2.1.4 Life Jackets**
- 2.1.5 Immersion Suits**
- 2.1.6 Water Survival Techniques**
- 2.1.7 Medical Emergencies**
- 2.1.8 Lifesaving Signals and Pyrotechnics**

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Illustration 2.1.1a Lifeboat



This drawing shows the port side boat.
The starboard side boat is to be symmetry.

(Quantity / One boat)						
17		Painter	Assemble	1		
16		Seat For Radar Reflector	Assemble	2		
15		Guide Device	Assemble	2		
14		Bitt	Stainless Steel	1		
13		Chock	Stainless Steel	2		
12		Reflect Light Strip	Plastic	36		
11		Ventilator	Assemble	1		
10		Position Light	Product	1		
09		Remote Control Line Hole	Assemble	1		
08		Pivor Duct&Steering Gear	Assemble	1		
07		Door Window&Cover	Assemble	1		
06		Skate Eye Plane	Assemble	4		
05		Lifeline Appliance	Assemble	1		
04		Rail	Stainless Steel	1st		
03		Radar Reflector	Product	1		
02		Rowlock	Assemble	2		
01		Hook Device Arrange	Stainless Steel	2		
No.	Code	Name	Material	Quantity	Mass	Remarks

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2.1 LIFESAVING AND SAFETY EQUIPMENT

2.1.1 LIFEBOAT AND DAVIT

Lifeboat

Manufacturer:	Jiangsu Jiaoyan Marine Equipment Co.,Ltd
Type	Fire protected, totally enclosed lifeboat and rescue boat
Model:	JY-QFN-7.00
No. of sets:	2
Dimensions (LxBxD):	7.000m x 2.700m x 3.225m
Release hooks:	JXN-1
No. of persons:	46
Weight of boat:	With equipment: 3,617kg With persons: 7,412kg
Water spray pump:	Marine centrifugal pump, model 90-80 CBPJ

Lifeboat Engine

Manufacturer:	Zhenjiang Siyang Diesel Engine Manufacturing Co., Ltd
Type:	Vertical, water-cooled, four stroke (3 cylinder)
Model:	380J-3 (with heater)
No. of sets:	1 per boat
Speed:	6 knots
Fuel capacity:	180 litres, stainless steel

Lifeboat Davit

Manufacturer:	Jiangsu Jiaoyan Marine Equipment Co.,Ltd
Type:	Gravity Rolling Type Davit
Model:	JYGR85
No. of sets:	1 per boat
Maximum Launching Load:	85 kN
Maximum Hoisting Load:	47 kN
Inclinations:	20° Heel + 10° Trim

Lifeboat Davit Winches

Manufacturer:	Jiangsu Jiaoyan Marine Equipment Co.,Ltd	
Type:	Electric	
Model:	JYW70L	JYW70R
No. of sets:	1 per davit	
Working Load:	70 kN	
Hoisting Load:	30 kN	
Minimum Lowering Load:	70 kN	
Rope Hoisting Speed:	≥ 10 m/min	≥ 36/8 m/min
Rope Lowering speed:	85 ~ 120 m/min	
Rope Capacity on Drum:	105 m	
Electric Motor (IP56)		
Model:	YZ160S-6H	YZ225L-4/16-H
Power Source:	440V/60Hz	440V/60Hz
Power:	8.5 kW	25/12 kW
Speed:	1140 r/min	1720/400 r/min
Duty Type:	30 min	10/5 min

Introduction

The lifeboats are stored on davits on the port and starboard sides of the accommodation block. In the event of an emergency, they can be boarded directly from their stowed positions. Both port and starboard lifeboats can be also used as Rescue boats.

The lifeboats are constructed from laminated fire-retardant polyester resin, and the space between the seats and the hull, and canopy liner and canopy are filled with polyurethane foam. This provides sufficient buoyancy to allow them to remain afloat and upright even if holed below the waterline. They are totally self-righting even when fully loaded with persons and flooded. Because of this, it is important everybody on board remains seated, with their seat belts fastened.

Each lifeboat is fitted with two lifting hooks, one forward and one aft. These are designed to be released simultaneously from inside the boat using an IMO Life-Saving Appliance (LSA) approved on-load/off-load release device. This is fitted with a hydrostatic interlock system, allowing the hooks to be released when the boat is fully water-borne.

WARNING

It is possible to release the hooks when the boat is out of the water, but this procedure is EXTREMELY DANGEROUS and must only ever be considered in exceptional circumstances. The crew must familiarise themselves with the operation of the lifting hooks and release mechanism, before attempting to operate.

The main entrance door to each lifeboat is on the side, but additional hatches have been provided at the helmsman’s position, and at the forward and aft ends of the canopy.

The steering position has a console containing the normal steering, engine instrument panel, and engine control lever. An emergency steering tiller has been provided which can be operated from inside the boat if normal steering fails.

The centre tank section contains water and provisions tanks, a fuel tank, and equipment tanks. It also gives access to the lifeboat drain plug. A fuel shut-off valve is located on the top of fuel tank.

The engine compartment has a removable cover providing access to the engine and propeller shaft. The engine is a fresh water cooled diesel engine with electric start. It drives the propeller via a reduction, reversing gearbox, allowing the boat to be driven ahead and astern. The engine is able to run for up to 5 minutes whilst secured in the davits.

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Batteries are located in a watertight box at the aft of the engine compartment. The No.1 12V battery is for engine starting via an electric starter. The No.2 battery is for starting via a totally independent 12V system. The battery selection switches are mounted on the aft side of the pillar by the steering position. A 42V AC/12V DC battery charger fitted in a cabinet inside the canopy on the same pillar, provides two channels capable of operating the two 12V circuits simultaneously or separately. The charger is connected to the ships 42V AC power supply via a plug on the side of the lifeboat near the hatches. The engines fuel tank has sufficient capacity to allow the engine to operate on full load for 24 hours at 6 knots. Authors note: To be checked on board vessel.

Natural ventilation is achieved via an automatic valve fitted on the aft end of the boat canopy. This prevents the cabin from becoming under-pressurised when the engine is running. If the engine fails or runs out of fuel, the lifeboat can be rowed using oars placed through the side hatches.

In the event of water leaking into the lifeboat, a manually operated bilge pump has been provided. This is mounted on the left side of the pillar by the steering position.

The helmsman position is equipped with a steering console which contains the engine instrument panel, engine throttle/gear control lever, steering, searchlight with plug, compass and switches for cabin lights and position light.

The launching arrangement for each lifeboat comprises a davit with davit arm, frame, platform, falls, suspension block, and gripes/lashing device. It also includes a winch mounted on main deck which has reduction gears, a hand brake, and a centrifugal brake. The swinging out and lowering operation can be controlled using a remote control wire inside the boat, or from a control lever on the ship's deck. During lowering, it is possible to stop and suspend the boat at any height. Both lifeboats are fitted with shock absorbing skates/fenders to provide protection to the inboard side of the boat during launching.

The lifeboat lashing arrangement comprises an automatically released hook, master link, PVC covered wire, turnbuckle and a bollard welded onto the davit arm. The lashing system automatically releases when the davit arms swing out.

Recovery of a lifeboat is done using the winch controlled from a pushbutton switch box. When the davit arm reaches a preset position, the winch is automatically stopped by a limit switch. After this, the winch has to be operated manually to wind the lifeboat up to its final position. The manual handle has a safety device to prevent accidental reverse operation.

Survival Equipment On Board Each Lifeboat/Rescue Boat

- 2 x Oars.
- 2 x Thole pins.
- 2 x boat hooks.
- 1 x buoyant bailer.
- 1 x bucket.
- 1 x survival guide manual.
- 1 x Compass.
- 1 x sea anchor.
- 2 x Painter.
- 2 x Hatchets.
- Water in plastic bags six persons (three litres per bag).
- 1 x Rustproof dipper with lanyard.
- 1 x Drinking vessel.
- Food (10 MJ per person).
- 1 x Pyrotechnic equipment.
- 1 x Waterproof torch and water proof box with batteries and spare bulb.

- 1 x Daylight signaling mirror.
- 1 x Copy of lifesaving signal on waterproof card.
- 1 x Whistle.
- 1 x Waterproof first aid kit.
- Anti-Seasickness medice (six does per person).
- Seasickness bag (one per person)
- 1 x Jackknife with lanyard.
- 3 x Can opener.
- 2 x Buoyant rescue quoit with 30 m rescue line.
- 1 x Manual pump.
- 1 x Set of fishing tackle.
- 1 x Engine tool kit.
- 1 x Fire extinguisher.
- 1 x Searchlight.
- 1 x Radar reflector.
- Thermal protective aids.
- 1 x Rope ladder.
- 1 x Rain water collector set.

Checking Closure of the Release Hooks

The safe use and operation of the lifeboats during drills, and inspection and maintenance is very important and dependent on knowing the release gear is properly reset.

WARNING

NEVER enter a lifeboat if the padlock symbol on either or both of the lifting hooks are out of the locked position. If they are out of the locked position, the lifeboat must be suspended using both maintenance lugs, and the reason investigated and repaired

Release Gear System

The release gear system comprises fore and aft hooks, a release handle located near the steering console, and a hydrostatic unit. The hydrostatic unit normally prevents the release of the hooks when the boat is not water-borne, however, the IMO LSA code requires that lifeboats be fitted with an on-load release device. This will allow release of a lifeboat while there is still load on the hooks. This is required to aid launching when there is a strong current, when the ship is making way, or if there are waves which cause the hydrostatic device to only release intermittently.

WARNING

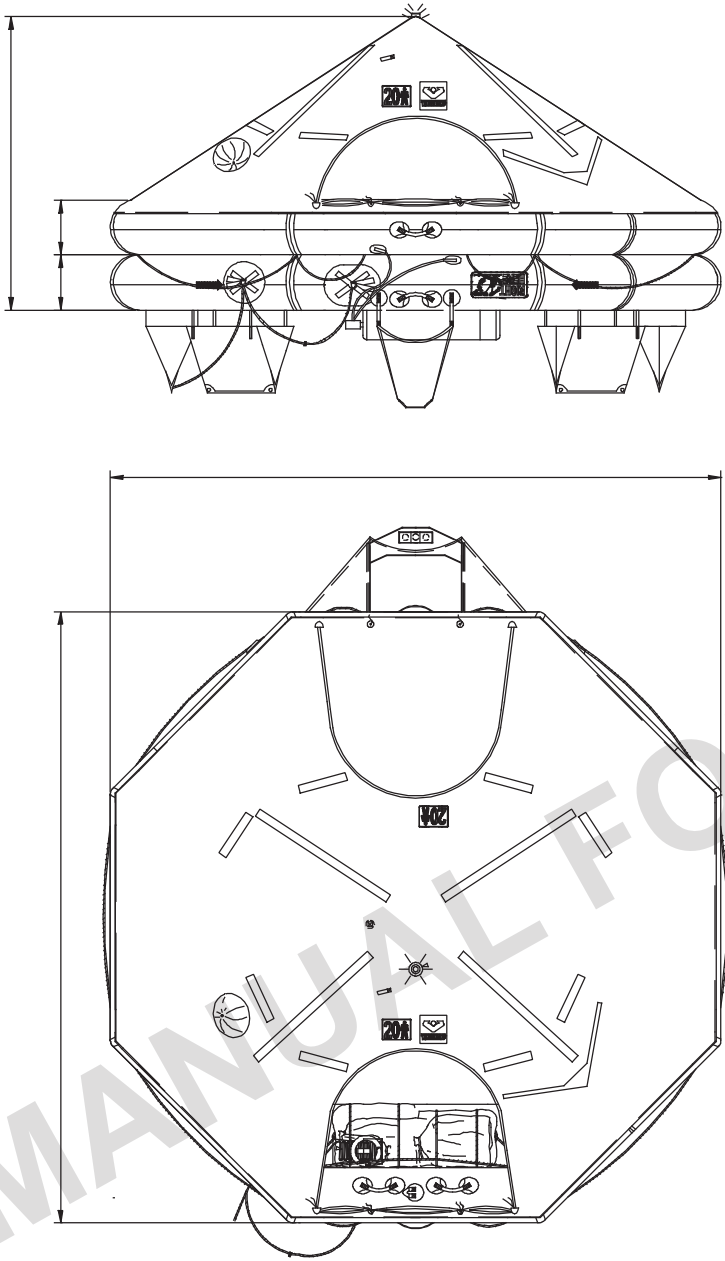
On-load release allows an empty or fully loaded boat to drop from any height, which can kill or seriously injure the occupants.

Operation of the release handle would normally only be done when the lifeboat was fully water-borne. To do this, remove the safety pin and pull the release handle fully and quickly to the open position. If off-load release is not possible, the lifeboat can be released on-load. To do this, break the interlock cover and force the lifeboat symbol down to the water surface. This overrides the interlock function of the hydrostatic unit, and allows the release handle to be operated.

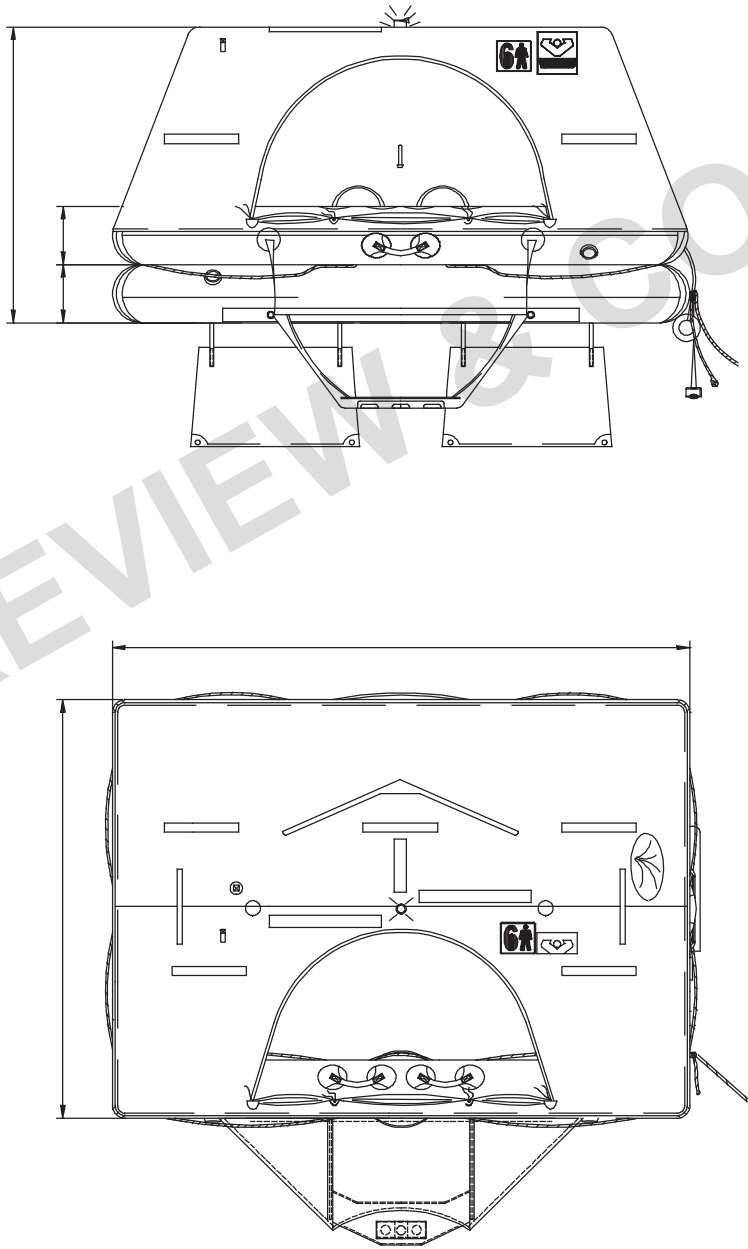
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Illustration 2.1.2a Liferats

Liferaft type 16-20-25 DK+



Liferaft type 4-6-8 DK+



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2.1.2 LIFERAFT

Liferaft (6 Person)

Manufacturer:	Viking
Type:	Throw overboard type
No. of sets:	1 x 6 persons
Weight:	70kg each

Liferaft (25 Person)

Manufacturer:	Viking
Type:	Throw overboard type
No. of sets:	4 x 25 persons
Weight:	172kg each

Introduction

All of the liferafts are constructed with twin buoyancy chambers, one above the other. The bottom and the canopy of the rafts are of double construction and may be inflated by bellows. The rafts are provided with boarding ladders, inside and outside gripping lines, capsize stabilisers, and a salt water activated battery for both internal and external lighting. Accessories supplied are:

- Rescue line with rubber quoit.
- Repair outfit.
- Hand bellows.
- Floating knife.
- Operational instructions.
- Sea anchor (drogue).
- Emergency pack to SOLAS standards.

The liferafts have two boarding openings in the canopy, both of which can be sealed using weatherproof coverings which are fitted to the raft. An observation opening is also provided in the canopy through which a lookout can keep watch whilst keeping their body inside the raft, warm and dry. Each liferaft is fitted with a waterproof identification tube, which contains the ship's identification name and reference numbers. Liferaft servicing involves an on board inspection and servicing ashore at intervals not exceeding 30 months. The hydrostatic release should be replaced upon the expiry date. Each liferaft is factory packed and stowed ready in a reinforced white glass reinforced fibre cylindrical storage container and is fitted with a waterproof identification tube, which contains the ship's name and reference numbers.

Release of Rafts

Hydrostatic Release Units (HRUs) are fitted to each of the 25 and 35-man liferafts and will be activated when submerged to a depth of between 1.5m and 4m; releasing the rafts to float towards the surface. After activation of the HRU, the raft will still be secured to the vessel by a weak link. However, after inflation of the raft, sufficient drag is applied to break the weak link wire and allow the raft to float free.

The throw-over type liferafts are launched by firstly ensuring the liferaft painter is securely fastened to the vessel, then removing the securing strap and throwing it into the sea. Further pulling on the painter line may be required to activate the carbon dioxide inflation cylinder. Boarding is ideally undertaken by ascending the boarding ladder before the painter is cut to free the liferaft from the vessel.

Hydrostatic release units require no maintenance, but must be replaced upon their expiry date. No hydrostatic release is fitted to the 6-person liferaft.

Forward Liferaft

Before moving the 6-man liferaft, it is necessary to unfasten the painter line and then the liferaft must carried to the ship's rail. Before launching, the painter must be secured to a strongpoint on the vessel and then launched from that point.

An emergency light is available at the stowage position and a portable light can be rigged to illuminate the launch area at night.

A wooden boarding ladder is secured under a waterproof temporary covering and this should be rigged at the rail side for boarding purposes. A knotted boarding rope is stowed in a steel box adjacent the liferaft for boarding purposes if the ladder is unavailable.

Initial Actions on Boarding a Liferaft

Step	Task
1	After launching and boarding the liferaft, the painter must be cut with the knife provided to avoid the liferaft being pulled under.
2	Paddle away from the danger zone using the paddles placed in a bag close to the entrance of the liferaft.
3	Alternatively, one of the lifeboats could be used to tow the liferaft clear.
4	When clear of the danger zone, if no other survival craft can be seen/reached, stream the sea anchor or drogue.

If other liferafts/survival craft are visible, act to move towards them or attract their attention if they are motorised. Once all survival craft are together, deploy all available sea anchors. The sea anchor stabilises the raft and helps to minimise drift.

Step	Task
5	Close the canopy openings. Inflate the canopy and the bottom of the liferaft as this gives excellent insulation against the cold. To do this, connect the bellows to each topping-up valve. Valves are placed in the liferaft floor and the inner canopy.

The bellows are located in a bag at the entrance. After a long stay in the liferaft, it may be necessary to top-up the two buoyancy tubes. Connect the bellows' plastic tube to the yellow topping-up valves.

Righting a Capsized Liferaft

Step	Task
1	Stand with the feet on the cylinder, hold onto the righting strap (placed across the bottom of the liferaft).
2	Manoeuvre the liferaft so that the opposite side is facing into the wind.
3	Throw the body backwards whilst holding onto the righting strap and keeping the feet on the cylinder.

Note: The side of the liferaft where the carbon dioxide cylinder is attached lies deepest in the water. The place is marked 'RIGHT HERE'

When the wind is very strong, the lifeline can be tied around the waist to prevent the liferaft being blown away. A non-swimmer should keep hold of the righting strap and allow the liferaft to fall back on him (the rubber raft will not injure him). He can then work his way back to the liferaft entrance under water, holding onto the strap of the lifeline.

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2.1.3 SURVIVAL CRAFT SURVIVAL GUIDE

Introduction

In the unfortunate event that the vessel has to be abandoned, it is necessary to make some very important decisions and carry out certain actions quickly. These are summarised as follows:

Procedure Prior to Abandonment

Step	Task
1	Put on extra clothing.
2	Take extra clothing or blankets, if possible.
3	Drink water if possible.
4	Take water in sealed containers.
5	Collect a life jacket and immersion suit.

In addition to the statutory lifeboat equipment such as emergency radio, water, rations, first aid kit etc, the following extra items would be useful:

- Extra life jackets.
- Extra survival bags.
- Small plastic bags.
- Extra medical supplies.
- Extra electric torches and batteries.
- Paper and pencil.
- Portable radio receivers, books, playing cards, etc.
- Navigational instruments, books, chart and chronometer.

Abandoning Vessel Procedure

Step	Task
1	All personnel should, if possible, board the lifeboat without getting wet.
2	If, for some reason, this is not possible and a jump into the water has to be made, remember: • Make sure it is clear to jump. • Hold your nose. • Hold down the life jacket. • Put your feet together. • Look ahead when jumping.

Additional duties, which should be allocated on the lifeboat muster list, include:

- SART to the lifeboat.
- GMDSS portable radios to lifeboat.
- Blankets and provisions.

Aboard the Survival Craft

First Actions:

- Elect a leader. This will normally be the most senior officer or the person appointed on the muster list.
- Take a muster of persons on board.
- Search the area for other survivors or survival craft.
- Liaise with any other survival craft to ensure that all persons are accounted for.
- Assess the situation. Is rescue likely and how long will it take?
- Assess the need to stay close to the position of the sinking or proceed towards the nearest land.
- Put the food and water under the control of one person who will be responsible for distributing the rations.
- Collect in all additional food, clothing and sharp objects or weapons that may have been brought into the survival craft.
- The leader must confirm to all that no food or water will be issued for the first 24 hours.
- The leader should nominate different people to the following positions; first aid, signalman, hull repairs, engine repairs, recorder of voyage log, navigator, helmsman and lookouts.
- Give an anti-seasickness tablet to all personnel.
- Minimise drift and rig the sea anchor or drogue.

With the improvements brought about by the GMDSS system in maritime search and rescue, staying close to the position of abandonment is the most likely decision that will be made. Prior to taking to the lifeboat, a distress alert would be sent out. In addition, there is the SART which should be taken to the lifeboat when abandoning ship.

The EPIRB, when activated, allows the MRCC to locate the position of survivors and guide vessels and aircraft to the scene of the casualty. Should the EPIRB not be in the survival craft when the vessel sinks, the HRU will automatically release the EPIRB which will then start its transmissions. Where possible, it is therefore beneficial for all the survival craft to stay together by tying the survival craft together. The SART should be positioned on the extension pole, switched on and mounted as high as possible.

To minimise drift, rig the sea anchor. Issue anti-seasickness tablets and ensure that any persons in the water are accommodated in the lifeboat as soon as possible.

Listen for whistles and look for survivors, signalling lights and lights of other rafts, ships or aircraft. The lookouts should be properly briefed in their duties regarding the collection of useful debris, how to keep a lookout, sector searches and the use of pyrotechnics, including when to use them.

Proceed Towards the Nearest Land

In some circumstances this will be the most obvious choice. Factors to take into consideration are:

- Was a distress alert sent?
- If there is no EPIRB in the boats, search the area of the sinking to see if it has surfaced.
- How far is it to the nearest land? Is the nearest land within the fuel range of your craft?

Indications of the proximity of land are changes in the wind direction around sunset and sunrise. The land and sea breeze effect can be quite distinct in some areas. A good indication of land is a single cumulus cloud or occasionally several appearing to be stationary close to the horizon whilst others are moving. There are many other indications such as a green and blue reflection on the underside of the clouds in low latitudes, the direction that birds fly either early in the morning or in the evening, and the change in colour of the sea from green or blue to a lighter colour.

Do not approach land at night unless you know exactly where you are and that the landing area or harbour entrance can be safely transited. During the hours of darkness, lookouts should keep a good watch for the sound of surf and report to the watch leader any visual or audible occurrences.

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Settling Down to a Period Before Rescue

Having made an assessment of how long it will be before rescue is likely, it is now necessary to decide how the available food and water will be divided and issued. The following are a few guidelines:

The minimum daily water ration should be around 450 to 500ml given in three separate issues at sunrise, noon and sunset. This quantity will be sufficient to avoid severe dehydration.

The daily food ration should consist of 800 to 850kJ of the emergency rations given in three equal amounts, this equates to around 500gm. To make the decision as to how much should be issued, take the total available, separate one third as emergency stock, and should rescue not be forthcoming when expected, then apportion the remainder on the above basis as a minimum.

In a lifeboat there should be a minimum 3 litres of water and 10,000kJ of food for each person that the boat is certified to carry. It should be noted that the emergency rations consist mainly of carbohydrates, some fat and minimal protein. These rations do not require the consumption of water or body fluid for them to be digested, which is of great importance.

Food and water should be issued in such a way that all can see that it is fair. Everyone will become thirsty, and as time passes, human nature will make the ration distribution a very difficult and harrowing experience and also the highlight of the day.

If a desalination plant is available, this should be put into operation immediately and its output used in preference to the internal water.

Passing the Time

The leader has to face and resolve the following problems:

Maintain morale. This is best approached by giving duties to each person which are meaningful and ensuring that they are carried out.

Duties such as lookout, helmsman and baler should be rotated at intervals of not more than one hour, as this will prevent boredom and lack of vigilance from setting in.

Continually show confidence that rescue will take place. Do not allow individuals to lapse into melancholy. Try to make everyone think of factors other than the situation that they are in by introducing games of various forms. If a portable radio is available, tune it in and listen to the various programmes. Playing card games is useful, as considerable concentration is required.

Maintaining the Mental and Physical Health of all On Board

Routines can be counter-productive and as such, duties should be rotated on a regular basis.

Where possible, restrict movement to a minimum, as all movement consumes body fluid. Body fluid is probably the most significant single factor to controlling whether or not you survive.

The initial withholding of food and water for 24 hours puts the body into a slightly dehydrated state, which is the ideal situation for a prolonged period in a survival craft. During this period, all persons should be encouraged to urinate, as this will assist in reducing urinary retention problems later. Do not consume food high in protein as this causes defecating which in turn causes body fluids to be used which will be irreplaceable. If possible, keep a good flow of fresh air through the boat as this will help to reduce seasickness. Ensure that all take the anti-seasickness tablets for the first two days, as after this most seamen will be acclimatised to the motion of the craft.

Hang out any damp clothing to ensure that it is dry for the evening chill in the tropics. This avoids the loss of body fluid as body heat dries the clothing and reduces the internal body temperature.

As thirst grows, the temptation to drink sea water increases. This must be prevented as ultimately death will certainly ensue.

All parts of the body should be shaded from the sun and the elements. This will reduce the loss of body fluid and/or the risk of sunburn or frostbite.

If the water ration is at least one litre per person daily, then fishing can be a worthwhile exercise. Remember that fish are high in protein which brings its own problems as previously mentioned.

The blood of sea birds is quite nutritious. To catch these, try putting some of the fish guts on a piece of wood with a hook in the middle and allow it to float a little way from the craft.

Do not encourage swimming as a form of exercise as this will use up energy and put the individuals at risk from sharks.

Injuries, Ailments and Treatment

Injuries

A first aid kit is supplied with every survival craft and a leaflet describing simple first aid is enclosed with each kit.

Dehydration

The healthy human body is over two-thirds water. Dehydration occurs when the normal water content of your body is reduced. This leads to a change in the vital balance of chemical substances in your body, especially sodium (salt) and potassium. In order to function properly, many of the body's cells depend on these substances being maintained at the correct levels. The amount of water in your body only has to decrease by a tiny proportion for dehydration to occur. The effects of dehydration can be serious, and in extreme circumstances can be fatal.

Dehydration occurs when there is a 1% - 2% or greater reduction in body weight due to fluid loss. Depending on the percentage of body weight lost, dehydration can be described as mild, moderate or severe.

The signs of dehydration in adults include:

- Dry mouth.
- Chapped or dry lips.
- Dry eyes.
- Dry, loose skin with a lack of elasticity.
- Sunken features, particularly the eyes.
- Clammy hands and feet.
- Headaches.
- Light-headedness.
- Dizziness.
- Tiredness.
- Confusion and irritability.
- Loss of appetite.
- Burning sensation in your stomach.
- Feeling of an 'empty stomach' or abdominal pain.
- Low urine output.
- Concentrated, dark urine with a strong odour.

Even if there is a relatively low level of fluid loss (causing a 1-2% loss of body weight), mild dehydration can cause the body to work less efficiently. However, mild dehydration carries few risks and can usually be easily treated by replacing lost fluids.

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Moderate dehydration is a 3-5% decrease in body weight due to fluid loss. This level of fluid loss can result in a decrease in strength and endurance and is the primary cause of heat exhaustion.

Severe dehydration is a decrease of more than 5% of body weight due to fluid loss. A 10% or greater reduction in body weight is extremely serious. If not treated immediately, this level of dehydration can be life threatening.

Dehydration is caused by not drinking enough fluid, or by losing fluid (through sweat, tears, vomiting, urine and stools) and not replacing it. If the body loses essential body salts, such as sodium and potassium, it will be unable to keep the body fluids in balance. The severity of dehydration can often depend on a number of factors such as climate, level of physical activity, fitness and your diet. Dehydration will often result from an illness, such as gastroenteritis, where there has been lost fluid through persistent diarrhoea and vomiting. Sweating excessively from a fever, heavy work in hot conditions or heatstroke will further increase the likelihood of dehydration.

It is therefore important to maintain an adequate fluid intake, including taking all precautions necessary to keep the body's demand for water down.

Frostbite

This usually occurs in extremities, ie, fingers, toes, ears. Wear protective clothing if possible. Reduce lookout periods in very cold weather and watch each other's conditions. Wiggle nose and cheeks and exercise hands and feet to aid circulation.

Do not massage affected area once signs of frostbite have appeared. Warm the area by holding a hand against it.

Urine Retention

This can be dangerous, so overcome 'mental blockage' early before urine production is reduced by rationing. Ensure that everyone urinates within the first 24 hours. If retention occurs, dangling hands in the water may help out, but whilst this is being done, be aware of the possibility of the presence of sharks. After a period in a survival craft, urine will appear dark and smoky. This is normal and no action is necessary.

Sunburn

Avoid excessive exposure to the sun by keeping under cover. Keep head, neck and other exposed areas covered.

Salt Water Boils

These are due to the skin becoming sodden with sea water. Do not squeeze or prick boils. Keep them clean and cover with a dry dressing. Keep the area as dry as possible to avoid chafing.

Dry Mouth and Cracked Lips

Swill water around the mouth prior to swallowing, suck a button, or smear the lips with cream or soft petroleum jelly.

Swollen Legs

This is common and due to long periods spent in a sitting position. It will subside without treatment after rescue.

Hypothermia

There is a risk of hypothermia in water below about 25°C. Extra clothing will delay the onset of hypothermia even if immersed, and of course will provide extra warmth for the survivor in the lifeboat even if immersion takes place. Totally enclosed or partially enclosed lifeboats provide far better protection from the elements than the older open type, but extra clothing is still essential for warmth in nearly all climates. If a survivor has been immersed in water and has hypothermia, strip off wet outer clothing and replace with any available dry garments. Warm the patient with extra layers of clothing and use life jackets as extra insulation. Use a thermal protective aid (plastic survival bag) if available.

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2.1.4 LIFE JACKETS

Manufacturer:	Jiangsu Huayan Marine Equipment Co.,Ltd.
Type:	Marine lifejackets with light and whistle
Model:	HYJ-A2&YDG1C
No. of sets:	58

Introduction

The life jackets are inherently buoyant, easy to don, and are specifically designed to be used by people with little or no experience of donning life jackets in an emergency situation. They are designed to be folded into a small package to take up minimal space when stored. There is no dead buoyancy across the chest which means better freedom of movement for the user, making it easier to board a liferaft. They have back panels which provide extra comfort, and a crotch strap that helps maintain a proper fit after impact with water. The waist belt is 175cm (69 inches) in accordance with the MSC.200 regulations, ensuring one-size fits all crew members. Instructions have been printed on one of the panels to aid donning.

Location

There are 58 life jackets on board and these are located as follows:

- 2 in engine control room 2nd deck.
- 8 on upper deck.
- 8 on A deck.
- 9 on B deck.
- 16 on C deck
- 13 on D deck.
- 2 on navigation deck.

Donning Instructions

Step	Task
1	Remove a life jacket from its stowage box or other location.
2	Inspect the life jacket to ensure it is free from damage and that the whistle and light are in place.
3	Untie the buckle.
4	Place the life jacket over the head.
5	Fasten the buckle and tighten the waist belt.

Donning instruction posters are placed on the bulkhead adjacent to where the life jackets are stowed in the following spaces: wheelhouse, engine room control room and bosun’s store forward.

Entering the Water

If it is necessary to enter the water to reach a survival craft, it is preferable to do so by means of one of the embarkation ladders. However, if it is necessary to jump into the water, it should be done from as low down on the vessel as possible and the following procedure should be adopted:

Step	Task
1	Place one arm across the chest and hold the neck of the life jacket to prevent it from being forced upwards when entering the water and possibly causing injury.
2	Place the other arm over the first and pinch the nose with fore finger and thumb.
3	After ensuring the water below is clear, look straight ahead and step off the vessel, do not jump.
4	Enter the water with both feet together.

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2.1.5 IMMERSION SUITS

Manufacturer:	Jiangsu Huayan Marine Equipment Co.,Ltd.
Type:	Cold water CR neoprene immersion suit
Model:	HYF-N2-R
No. of sets:	58

Introduction

The vessel carries a total of 58 immersion suits, located as follows:

- 2 in engine control room 2nd deck.
- 8 on upper deck.
- 8 on A deck.
- 9 on B deck.
- 16 on C deck
- 13 on D deck.
- 2 on navigation deck.

The immersion (survival) suit, model HYF-N2-R (red colour), is a highly visible one-piece, self-buoyant survival aid designed to provide thermal protection and buoyancy in cold water conditions, in accordance with SOLAS 1974, as amended.

The suit is manufactured from fully sealed 5 mm CR neoprene with insulating properties, providing protection against hypothermia. It is designed such that, after immersion in calm water at temperatures between 0°C and +2°C for a period of six hours, the wearer’s core body temperature will not decrease by more than 2°C.

The suit provides a minimum buoyancy of 142 N and is designed to be worn without a lifejacket. It ensures the wearer remains afloat while reducing heat loss in cold environments.

A waterproof full-length zipper is fitted to secure the suit, and high-visibility reflective tape is provided to enhance detection. The suit is equipped with an automatically activated light (triggered by seawater), a whistle, buddy line, and lifting beackets to assist in recovery operations.

For improved usability and protection:

- Rubber soles are fitted to the feet for durability and grip
- Adjustable straps are provided at the ankles for proper fit
- Integrated three-finger neoprene mitts are included for thermal protection of the hands

The immersion suit is available in the following sizes:

- M: suitable for height 150–170 cm, weight up to 150 kg
- L: suitable for height 170–190 cm, weight up to 150 kg
- XL: suitable for height 190–205 cm, weight up to 150 kg

Donning Instructions

Donning the immersion suit can be made either standing upright or sitting down on the deck.

Step	Task
1	Remove the survival suit from the bag and lay the suit flat down with the front face up and top entrance near your feet.
2	Remove shoes (if time permits) and insert each leg and tighten ankle straps
3	Pull the suit up to your waist and insert your non-dominant arm into the arm and adjust your fingers into the three-finger mitt.
4	Pull the hood up over your head, then inert your other arm and adjust your fingers into the three-finger mitt.
5	Pull the zipper upwards using the plastic zipper pull.
6	Secure the velcro neck flap.
7	Fit and secure life jacket when required to do so.

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2.1.6 WATER SURVIVAL TECHNIQUES

General

The publication ‘A Pocket Guide to Cold Water Survival’ gives an excellent guide on how to survive in cold water and should be studied if available. In calm water at 5°C, a normally dressed person has only about a 50% chance of surviving for one hour. Loss of body heat is a gradual process and simple self-help techniques can extend survival time. The body gives off heat to the surroundings in the following manner:

Conduction and Convection

The transfer of heat by direct contact with cold water or other materials. Heat is passed from the body, normally quite hot, to the cooler surrounding substances. Certain substances are better conductors of heat than others, for example, water conducts heat faster than air.

The transfer of heat by air or water currents. Moving air cools the body quicker than still air. The cooling effect of moving air is known as the ‘wind chill effect’. Similarly, moving water about the body cools faster than still water at the same temperature.

Radiation

The transfer of heat by rays of energy without direct contact with other substances.

Evaporation

The vaporising or drying up of liquid, such as sweat or moisture from wet clothing. When a body gets too hot, sweating occurs and the evaporation of the sweat has a cooling effect on the skin. Evaporation of moisture from wet clothing has the same effect and results in a cooling process and resultant loss of body heat. The body is normally heated by its own heat production, clothing acts as an insulation, trapping the warmed air between the skin and the clothing. If this layer of air is lost, then the insulation is reduced and the skin temperature will fall. Should the skin temperature continue to fall, then the body core temperature will also decrease.

Once the body temperature falls below 35°C, the person is suffering from hypothermia. At this stage, discomfort, impaired speech, disorientation and mental confusion are well established. When the temperature falls to 31°C, unconsciousness may occur, shivering is usually replaced by muscle stiffness and the pupils of the eye may be enlarged. The heartbeat becomes irregular and weak and the pulse barely detectable.

Death may occur at any stage of hypothermia, when the body temperature falls below 30°C, it is very difficult to be sure if someone is alive or dead. Death by hypothermia is defined as a failure to revive on re-warming.

Stages of Hypothermia

Initial immersion	Initial 2-3 minutes	Cold shock reactions
Short-term immersion	3-15 minutes	Exhaustion and drowning
Long-term immersion	30 minutes plus	Effects of progressive cooling
Post immersion (20% casualties rescued)		Post immersion collapse and secondary drowning

Initial Immersion Cold Shock Reaction

Sudden immersion into cold water, for those not accustomed to it, produces a dramatic increase in heart rate, blood pressure and breathing rate. These responses decline after a few minutes’ immersion, but it is not unknown for death or incapacitation to occur, particularly in persons already suffering from high blood pressure or heart defects.

Adequate clothing will reduce the effects of cold shock and it is considered better to slowly introduce a person into the water rather than jump, thus making a more gradual temperature change.

Short-Term Immersion

Even normally competent swimmers have been known to become quickly exhausted when immersed in cold water. The cold shock reaction causes a rapid gasping of breath, making it difficult to maintain adequate leg and arm movement, together with the possibility of the swimmer inhaling water.

Because swimming in these conditions is difficult, the wearing of a life jacket cannot be overstated. A life jacket gives additional buoyancy, and in the event of the wearer becoming unconscious, will automatically turn the wearer to face the wind and sea. This is because the lower half of the body acts as a sea anchor, the legs trailing to windward of the jacket.

Long-Term Immersion

Normal body temperature is about 36.9°C . If this falls to below 35°C, a person is said to be suffering from hypothermia. First signs of cold are shivering and increased numbness, particularly in the limb extremities. Hypothermia is also indicated by increasing slowness of physical and mental response accompanied with slurring of speech and problems with vision. Other symptoms are irritable and irrational behaviour, cramp, nausea and complete lethargy may also occur. Later, shivering is displaced with persistent muscular rigidity and finally the person will become unconscious, with depressed breathing and a slow pulse rate. The person will be very pale, cold to the touch, dilated pupils and tendon reflexes will be absent. Death will occur when the body temperature falls to about 24°C due to heart failure.

Water Temperature

Survival in water of between 14°C and 18°C, wearing ordinary clothing and in a life jacket, would be between 2.5 to 9 hours depending if lightly or heavily clad.

In water temperature of 6°C - 8°C, survival time would be reduced to between approximately 0.5 and 2 hours.

Body heat loss to the water from a naked person wearing a life jacket is about 26 times greater than if the person were in the same temperature conditions on land, discounting any wind chill factor.

The heat loss would increase if the person tried to swim or exercise in an attempt to keep warm. The blood supply to the muscles would be increased and subsequently heat is lost to the surrounding water. If dressed, any warm water trapped beneath the clothing would in turn be forced out and replaced with cooler water, again reducing the body temperature.

It is important to try and spend as little time in the water as possible, and should it be necessary to remain immersed, then the person should float quietly, allowing the life jacket to support the wearer.

By adopting the ‘HELP’ (Heat Escape Lessening Posture) it is possible to minimise the amount of heat loss to the water. The elbows should be kept close to the chest and the neck of the life jacket gripped. At the same time the legs should be crossed and brought up to the chest.

Should there be other persons in the water, get close together and form a circle facing inwards. This is the ‘huddle’ position, and as well as keeping everyone together, provides a bigger target for search and rescuers to spot.

The ‘HELP’ and ‘huddle’ positions are difficult to maintain in rough seas, and it is better to attempt to keep your back to the sea and wind in order to reduce the amount of wave splash into the face.

Amount of Clothing Worn

The more layers of clothing worn, the greater the survival times. Where oilskins or a survival suit are worn as the outer layer, the greater is the reduction in body temperature loss. Head covering, gloves and thick socks also help to reduce heat loss.

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Water Survival Techniques

Survival Times in Sea Water - Temperature 7°C

- Lightly clad - 40 minutes.
- Heavily clad - 80 minutes.
- Naval immersion suit - 6 to 8 hours.

Age and Physical Fitness

Old people, children and persons suffering from illness or injury are at higher risk.

Amount of Body Fat

Fat people are better protected from the cold than thinner people, having their own built-in insulation.

The Amount of Water Movement

In calm water, the amount of spray and drenching from waves is reduced, increasing the chance of survival.

Wind Effect

Once out of the water, the ‘wind chill’ factor becomes an additional hazard for those in wet clothing. Persons should seek to get out of the wind and shelter under the protective covers of the survival craft and huddle close to each other for warmth. If clinging to wreckage, the person should crouch or lie down to reduce the area of body exposed to the wind.

Exposure: Hypothermia

There is a risk of hypothermia in water below about 25°C. Extra clothing will delay the onset of hypothermia even if immersed, and of course will provide extra warmth for the survivor in the lifeboat even though immersion takes place. Totally enclosed or partially enclosed lifeboats provide far better protection from the elements than the older open type, but extra clothing is still essential for warmth in nearly all climates. If a survivor has been immersed in water and has hypothermia, strip off the wet clothing and replace with dry garments if available. Warm the patient with extra layers of clothing and use life jackets as extra insulation. Use a thermal protective aid (plastic survival bag) if available. Persons particularly at risk from hypothermia should be positioned nearer the engine, which will run for 24 hours at full power and much longer if kept at reduced speed. The engine can be a very valuable source of warmth in cold weather, but fuel should be conserved as much as possible, dependent upon sea conditions, by running at reduced speed or using the engine for certain periods only.

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2.1.7 MEDICAL EMERGENCIES

Introduction

The emergency squads on board the ship comprise the crew members listed in the Station Bill.

On raising the alarm, the bridge must be informed so that the duty officer can page all of the members of the emergency squads. The squads will muster and, depending on the type and location of the emergency, will proceed to the scene of the accident equipped with the correct equipment to deal with the specific emergency.

The medical emergency equipment consists of stretcher and first aid kit. It is important when raising an alarm to describe the type of incident and exact location. The way of raising the alarm is discussed in the following section.

Immediate Reaction on Finding a Casualty

On finding a casualty, the main steps of emergency action are to assess the situation, make safe the area for casualty and rescuer, give emergency first aid and raise the alarm to get help.

- **Assess the Situation:** Identify any risks to yourself, the casualty or any other persons nearby. If in immediate danger, assess how to remove the danger from the casualty or the casualty from the danger. Do not attempt to move a casualty unless there is a life threatening situation such as a fire or flooding.
- **Make the Area Safe:** You should make the area safe for both casualty and rescuer. The conditions that caused an accident may still be there when you arrive. If the casualty has suffered an electric shock, switch off the electrical source or break the circuit. Stand on some dry non-conducting material and use a dry piece of wood or other non-conducting material to remove the casualty from the source of electricity.

WARNING

Do not attempt any rescue if the casualty has collapsed in an enclosed space, such as a tank or cofferdam. Always assume that the atmosphere in an enclosed space is unsafe.

Entry into enclosed spaces with dangerous atmospheres, for rescue or any other purposes, can result in death, and should only be attempted with prior authorisation from the Captain or other responsible officer, after checking that the atmosphere is safe for entry and once all the procedures for entry into enclosed spaces have been complied with. Safe entry into enclosed spaces with oxygen deficient atmospheres can only be made by trained personnel using self-contained breathing apparatus.

- **Give First Aid:** Once the area is safe, make an assessment of each casualty. If there is more than one, prioritise in order of:
 - Not breathing/no circulation.
 - Unconsciousness.
 - Severe bleeding.

Check those who are not making any noise first, as these may be the more critically injured. Check if the casualty is conscious, ask a question and see if they answer by talking or moving. If there is no response, shake the shoulders gently to avoid causing further damage and/or pain. If the casualty is unconscious, follow the ABC of resuscitation explained later in this section.

- **Raise the Alarm:** Do not delay calling the medical emergency team. If other crew members are at the scene, ask them to raise the alarm, stating the location and type of emergency whilst you continue giving first aid. Make sure that anyone sent to raise the alarm reports back.

If you are on your own, you should give resuscitation (see ABC of resuscitation) for at least one minute to any casualty who may need it before seeking help. Do not leave any casualties on their own if there is a life threatening danger. Attract attention by shouting, banging the bulkheads with a hard object or activate a nearby fire alarm.

The diagram on the following page gives a summary of the emergency response to follow in the event of finding a casualty.

First Aid Procedures

Recovery Position

The recovery position helps a semi-conscious or unconscious person to breath and permits fluids to drain from the nose and throat which might otherwise be breathed in. It also helps to eliminate the possibility of the patient choking on their own vomit.

Once it has been established that the semi-conscious or unconscious patient is breathing, place them in the recovery position until medical help is available or they regain full consciousness.

In the event that the patient has a major injury such as to the back or neck, do not use this position.

This section gives a summary of first aid treatment to different situations that may occur on board ship to fellow crew members. Familiarise yourself with and practice the different procedures. Ask the safety officer or medical officer for further training or clarification on the procedures described below and any others that may not be covered in this manual.

ABC of Resuscitation

The ABC of resuscitation stands for **Airways**, **Breathing** and **Circulation**. These are the signs you should check on finding a casualty. Determine if the casualty is conscious and if there are signs of breathing and circulation. Follow the ABC procedure whenever you find a casualty and then treat for the specific injury. Refer to the following flow chart for a description of these techniques.

- **A-Airways:** Ensure the airway is open and clear from any obstructions such as food residues or loose dentures. Remove any obstruction from the mouth. To open the airways, lift the chin and press the forehead backwards to tilt the head back.
- **B-Breathing:** Check that the casualty is breathing for 10 seconds. If there are no signs of breathing or if in doubt, commence artificial respiration by using the mouth-to-mouth or mouth-to-nose technique. For mouth-to-mouth artificial respiration, open the airways and pinch the casualty's nostrils together. Seal your lips around the casualty's mouth and give two breaths. For mouth to nose artificial respiration, close the casualty's mouth and position your mouth on the casualty's nose, sealing the nostrils, and give two breaths.

If the artificial respiration is successful, the casualty's chest will rise. You should wait for the chest to fall before giving another breath. Continue to give artificial respiration until natural breathing is resumed or until the arrival of the medical emergency team. Give 10 breaths per minute.
- **C-Circulation:** Find the pulse in the neck or wrist and check for 10 seconds. If there are no signs of a pulse, commence chest compressions. To give chest compressions, kneel beside the casualty and place your middle and index finger on their breastbone. Position the heel of your other hand next to the fingers in the direction of the casualty's head. This is the place where the chest compressions will be applied. Interlock the fingers of your hands and apply pressure with the heel of the upper hand only. Remember to keep your arms straight and your shoulders parallel to the casualty's breastbone when doing chest compressions.

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In cases where circulation has failed, breathing will also stop. In this case you will have to alternate between artificial ventilation and chest compressions. This technique is called cardio-pulmonary resuscitation (CPR).

CPR should be given at a rate of 30 chest compressions (at a rate of 100 compressions per minute) to 2 ventilations. If there are more than two trained rescuers, you can take turns giving CPR or, more effectively, administer it using a two-person technique by which one rescuer gives artificial ventilation and the second one gives chest compressions. For the two-person technique, the rate of compressions to ventilations remains the same.

CPR should be continued until the arrival of the medical emergency team. If the casualty recovers breathing and circulation, they should be placed in the recovery position.

Blast Injuries

Explosions can cause fractures, wounds, severe bleeding, burns, asphyxia or any combination of these. Further injury may occur from falling wreckage or if the explosion starts a fire. When approaching the scene of an explosion, pay special attention to any of the hazards that may result from the explosion and be prepared to deal with badly injured casualties. If it is not safe to approach the scene of an explosion, wait for help to arrive.

If the casualty is unconscious, check for breathing and circulation. Start CPR if necessary. If the casualty is breathing, place them in the recovery position.

Following an explosion, conscious casualties may be incapable of moving, disorientated and unaware of what is going on around them. Guide those who are able to walk to safety and delegate someone to look after them.

An explosion may also cause injury to the lungs. Casualties who have suffered this type of injury may have difficulty in breathing, the skin of the face may be blue and they may cough bloodstained froth.

First aid for this type of injury should focus on loosening any tight clothing that impedes breathing. Take the casualty to fresh air if possible and support them in the half-sitting position. If their breathing fails, give them artificial ventilation.

Bleeding

Bleeding occurs when any of the blood vessels are ruptured. Bleeding can be external, caused from wounds, or internal if it is caused by trauma, fractures or ruptured organs. Severe bleeding causes low blood pressure and the casualty will enter shock. Bleeding is categorised according to the type of blood vessel affected:

- **Arterial Bleeding:** Bright red colour blood spurting from a wound following a heartbeat.
- **Venous Bleeding:** Dark red colour blood oozing from the wound.
- **Capillary Bleeding:** Blood will ooze from the blood vessel. This is the most common type of bleeding.

The following paragraph outlines some guidelines on the treatment of bleeding:

Step	Task
1	Examine the wound. If there is an object embedded, DO NOT attempt to remove it.
2	Apply direct pressure to the wound using a gauze to stop the bleeding. If there is an embedded object, apply pressure to either side of the wound.
3	If the bleeding is from a limb, raise it above the level of the head and support it in this position. For lower limb bleeding, lay down the victim and raise the affected limb.
4	Cover the wound with sterile dressing.
5	Check the casualty for shock and treat if necessary.

Whilst treating bleeding, the following precautions should be observed:

- **Do not** use a tourniquet unless bleeding is uncontrollable such as the case of a traumatic amputation of a limb. In this case, apply the tourniquet as near as possible to the point of amputation and make a note of the time when the tourniquet was applied.

- For cases of severe bleeding, apply pressure at the brachial artery (arm below biceps) or in the femoral artery (groin crease). In both cases, press the artery firmly against the bone. Pressure should not be applied for more than 10 minutes.
- Wear disposable gloves if available to reduce the risk of infection.

Burns

Burns may be caused by extreme hot or cold temperatures, extremely moist heat, chemicals, friction, radiation and electricity sources. Burns destroy the skin layers leading to infection and fluid loss.

Superficial burns involve only the outer layer of the skin. A partial thickness burn affects several layers of the skin. This type of burn requires medical treatment and, if affecting over 60% of the body's skin, can cause death. A full-thickness burn destroys all layers of the skin and can extend to damage nerve, muscle and fat.

The risk of shock and fluid loss is proportional to the area affected by the burn. As a rule of thumb, the size of the casualty's palm represents 1% of the body area. Extensive burns, 15% of skin surface in adults and 8% in children, can cause death. Another method to determine the extent of burns is the 'rule of nines' by which the body is divided in 9 equal parts. A guide to this method can be found in the 'Ship's Captains Medical Guide' kept on board.

The following paragraphs outline some guidelines on the treatment of burns:

Step	Task
1	Cool the affected area with copious amounts of running water. This has the effect of preventing further tissue damage and minimises swelling and shock. Do not over-cool a burn as this may result in shock.
2	Cover the injury with a sterile gauze or a lint-free sterile dressing. If no dressing is available, use cling film or a clean plastic bag to cover the area.
3	Treat the casualty for shock if necessary.
4	Non-alcoholic drinks may be given to casualties who are fully conscious to replace fluid loss.

Whilst treating burns, the following precautions should be observed:

- **DO NOT** remove any clothing sticking to the skin.
- **DO NOT** touch or interfere in any way with the injured area.
- **DO NOT** burst any blisters.
- **DO NOT** apply any ointments, adhesive tape or any other substance to the affected area.
- **DO** wear protective gloves in the event of a chemical burn.

Burns to the airway may occur when the casualty has inhaled very hot gases. In this case, maintain the airway open and be prepared to give artificial ventilation if necessary.

In the case of electrical burns, make sure that the contact with the electrical source is broken. Do not approach a casualty of a high voltage incident until you receive positive confirmation that the electrical source has been switched off and isolated. Failing to do so could result in electrocution when touching the casualty. For chemical burns, make sure that the chemical source is sealed. Ventilate the area of the accident and, if possible, remove the casualty from the area. Apply copious amounts of water to the affected area for at least 20 minutes. Remove any contaminated clothing.

Choking

Any foreign object that blocks off the windpipe can result in choking. It is vital to give prompt first aid to dislodge the object that is obstructing the airway.

The following paragraphs outline some guidelines on the treatment of choking:

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For Adults

Step	Task
1	Make the casualty bend forward. Give five sharp slaps with the palm of your hand to their back between the shoulder blades. Check the mouth for any foreign objects.
2	If the obstruction does not clear, stand behind the casualty and place a clenched fist below the ribs and give five thrusts by grasping your fist and pulling inward and upwards, compressing the casualty. Check the mouth for any foreign objects.
3	If choking persists, repeat slaps in the back at the ratio of five slaps to five abdominal thrusts until the emergency reserve team arrives.

For Children

Step	Task
1	Give five back slaps as per the method described above.
2	Give five chest thrusts by standing behind the child and pressing the breastbone inwards with your clenched fist. If the choking persists, give five more back slaps.
3	Give five abdominal thrusts as per the method described above.
4	If choking persists, repeat the whole cycle: back slaps – chest thrusts – abdominal thrusts until the medical emergency team arrives.

WARNING

Do not perform abdominal thrusts on a baby.

Crush Injuries

Crushing accidents may result on board ship from a variety of causes such as explosions, trapped limbs in machinery or other accidents in the workplace. Injuries from crushing include fracturing, severe bleeding and shock.

Prolonged crushing may result in severe damage to muscle tissue. Also, toxins will accumulate around the crushed tissue and if the pressure is released suddenly, the toxins may cause kidney failure.

The following paragraphs outline some guidelines on the treatment of crush injuries:

Step	Task
1	Release the casualty from the crush as soon as possible. If machinery is involved, press the emergency stop or cut the power supply.
2	Treat any external bleeding.
3	Immobilise any fractures.
4	Treat the casualty for shock if necessary.
5	Monitor their breathing and pulse. Be prepared to resuscitate.

Drowning

The ingress of water into the airways will stop breathing. Drowning can happen in water of any depth, and remember that the consumption of heavy meals before swimming may lead to drowning.

The following paragraphs outline some guidelines on the treatment of drowning:

Step	Task
1	Get the casualty out of the water.
2	Lay the casualty down on a horizontal surface.

Step	Task
3	Check the airways, breathing and circulation.
4	If there are no signs of breathing, initiate artificial respiration.
5	If there is no circulation, initiate CPR.
6	Treat the casualty for hypothermia. Remove any wet clothing and wrap them in a dry blanket.
7	A casualty should receive medical treatment, as the air passages may began to swell some hours later as an inflammatory response in the lungs, causing secondary drowning.

Epileptic Fit

Epilepsy is the result of abnormal electrical activity in the brain.

The following paragraphs outline some guidelines on the treatment of epilepsy:

Step	Task
1	Clear the area around the casualty and put a soft object under their head.
2	DO NOT restrain the casualty, just make sure that they are protected from injury.
3	Loosen any tight clothing on the casualty.
4	Once the fit is over, place the casualty in the recovery position.
5	Stay with the casualty to reassure them until the arrival of the emergency reserve team.

Eye Injuries

Most work-related eye injuries are preventable by using eye protection such as safety goggles.

The following paragraphs outline some guidelines on the treatment of eye injuries due to foreign bodies in the eye:

Step	Task
1	Wash the eye with an eye wash bottle or with copious amounts of running water.
2	Try removing the foreign body with a moist cotton bud or gauze. For metal objects, try using a strong magnet. If the particle is under the upper lid, grasp the lid by the lashes and pull it over the lower lid.
3	If the object is embedded, do not attempt to remove it. The eye should be covered and the casualty referred for specialist medical treatment.
4	The casualty must avoid rubbing the eyes, as this will make the injury worse.
5	Tell the casualty to keep their uninjured eye closed and still, as further movement may cause further damage to the injured eye.

The following paragraphs outline some guidelines on the treatment of eye injuries due to chemical substances:

Step	Task
1	In the case of eye injuries due to chemicals, wash the eye with water for at least twenty minutes. The casualty should hold their head with the affected eye lower to avoid contamination to the other eye.
2	Cover with a sterile pad and refer the casualty for specialist medical treatment.

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Applying Traction to Straighten a Fractured Lower Limb

Fractures

Whenever dealing with a fracture, the priority of the rescuer is to immobilise the fracture and avoid any unnecessary movement. Fractures are divided into two categories; closed, when the skin around the injury is not damaged, and open, when there is a wound caused by the fracture and the bone may be visible.

The following paragraph outlines some guidelines on the treatment of fractures:

Step	Task
1	Immobilise the fracture to prevent further injury. Steady the fracture by placing your hands above and below the site of the injury. Limb fractures can be immobilised by using a rigid object or, in the case of lower limbs, by securing the affected limb to the good one.
2	Treat any bleeding with a sterile gauze.
3	DO NOT attempt to push in any bone protruding from an open fracture.
4	If the casualty has a lower limb fracture, the leg can be straightened by applying gentle traction before immobilisation. Pull until the fracture is immobilised and then release the pressure gradually.

In case of a suspected back injury, do not move the casualty unless there is an imminent danger to life. Moving the casualty without extreme care may cause permanent damage to the spinal chord and nerves. Always suspect a spine injury if the casualty has fallen more than 2m (6'7") or if the accident involved a violent fall forward or backwards bending.

The following paragraphs outline some guidelines on the treatment of suspected spinal injuries:

Step	Task
1	If the casualty is conscious, tell them not to move.
2	Steady the casualty's head by placing your hands against the sides of the head.
3	If there is suspected injury to the neck, provide support with rolled-up clothing or other soft cushioning on either side of the neck and shoulders.

Frostbite

Frostbite is caused when exposed parts of the body, such as fingers or face, are exposed to extreme cold and freeze, resulting in injury. Frostbite may also be accompanied by hypothermia.

The following paragraphs outline some guidelines on the treatment of frostbite:

Step	Task
1	Warm the affected body part with your hands or armpits. Remove the casualty's shoes, jewellery and gloves. Do not massage the affected part.
2	Move the casualty into a warm area.
3	Place the affected area in warm water, but take care that the water is not too hot, as the casualty may have lost the sensation to temperature and burns may occur.
4	Apply a light dressing on the affected area.
5	Treat the casualty for hypothermia.
6	Refer to the medical emergency team.

Head Injuries

Any head injury is potentially serious, as damage to the brain may have been sustained and may not be visible.

The following paragraphs outline some guidelines on the treatment of head injuries:

Step	Task
1	Assess the level of consciousness of the casualty. Bear in mind that the head injury may have been sustained by the casualty falling as a result of another accident.
2	If the casualty is unconscious, check their ABC response as previously described. Be prepared to resuscitate if necessary.
3	If the casualty is unconscious but breathing and with a pulse, place them in the recovery position.
4	The casualty may have suffered a neck injury. Immobilise the neck with an improvised collar, made of paper or a stuffed sock loosely tied around the neck, prior to moving them. Only move them where there is a threat to their life.
5	Monitor the casualty for several hours and do not allow them to sleep soon after the trauma.

Heart Attack

A heart attack occurs when the blood supply to the heart muscle is interrupted. The symptoms of a heart attack are crushing pain in the centre of the chest that may radiate to the left arm or jaw, breathlessness, pale skin, cold sweat and an irregular pulse.

The following paragraphs outline some guidelines on the treatment of a heart attack:

Step	Task
1	Place the casualty in a comfortable position. This could be sitting with their back and shoulders supported by a cushion and knees bent and also supported by a cushion.
2	Monitor their breathing and circulation until the medical emergency team arrives. Initiate CPR if necessary.

Heatstroke

Heatstroke is caused by a failure of the heat-regulating system in the body due to prolonged exposure to heat, very high fever or exercising in very hot temperatures.

The following paragraphs outline some guidelines on the treatment of heatstroke:

Step	Task
1	Take the casualty to a cool area and remove as much clothing as possible. Make the casualty sit down with their legs extended.
2	Wrap the casualty in a wet cold sheet. Keep them wet to reduce their temperature below 38°C/100.4°F (measurement taken under the tongue).
3	Monitor the casualty's ABC response. If the casualty becomes unconscious, open the airways and initiate CPR if necessary.
4	After the casualty has cooled down, replace the wet sheet with a dry one. Monitor the casualty until the arrival of the medical emergency team.

If the casualty is suffering from heat exhaustion due to prolonged over-activity and sweating:

Step	Task
1	Lay the casualty down in a cool place and raise their legs.
2	Give them plenty of water.
3	Refer the casualty to the ship's medical officer.

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Hypothermia

Hypothermia occurs when the body temperature drops below 35°C/95°F. This can occur when the casualty has been immersed in cold water for a length of time or is exposed to cold wind.

The following paragraphs outline some guidelines on the treatment of hypothermia:

Step	Task
1	Take the casualty to a protected area.
2	Remove any wet clothing and replace with dry clothes.
3	Re-warm the casualty by wrapping them in a sheet, a thermal protection aid or by using the body heat of another person. If the casualty is conscious, they can be re-warmed in a warm bath if they are able to climb into the bath unaided.
4	If the casualty is fully conscious, give them a warm drink.
5	Look for signs of frostbite.
6	Monitor the casualty for breathing, pulse and temperature. Resuscitate if necessary.

Recovery Position

The recovery position prevents an unconscious casualty from choking on their own tongue or vomit. Unconscious casualties, who are breathing and have a pulse, should always be put in the recovery position.

The following procedure describes how to put a casualty into the recovery position.

Step	Task
1	Kneel beside the casualty and open their airways by lifting the chin and tilting the head backwards.
2	Straighten the casualty's legs. Remove any spectacles they may be wearing.
3	Straighten their arm nearest to you and place it palm up under their thigh.
4	Bring the other arm across the chest and place it palm down under their nearer cheek.
5	Pull the casualty's far leg towards you, keeping their foot on the ground.
6	Pull on the bent leg to roll them towards you. Make sure the hand under their cheek is supporting the head.
7	Make sure the other arm is fully stretched, palm facing upwards, and the casualty is not lying on it.
8	Adjust the bent leg so that the knee is at a right angle.
9	Ensure the airways remain open by tilting the head back.
10	Monitor the casualty's breathing and circulation.

Shock

Circulatory shock occurs when there is a loss in the volume of circulating blood leading to a fall in blood pressure. This type of shock is different from psychogenic shock, which causes fainting when a person suffers emotional stress.

The following paragraphs outline some guidelines on the treatment of shock:

Step	Task
1	Treat any possible causes of shock such as severe bleeding.
2	Lay the casualty down and raise the legs if they are not injured.
3	Keep the casualty warm by wrapping with a blanket or clothing item.
4	Check consciousness, breathing and pulse until the medical emergency team arrives. Be prepared to resuscitate if necessary.

Whilst treating shock, the following precautions should be observed:

- **DO NOT** give the casualty anything to eat or drink.
- **DO NOT** leave the casualty on their own.
- **DO NOT** try to warm the casualty with a direct source of heat such as a hot-water bottle.
- Constantly reassure the casualty and make them comfortable.

Stroke

Strokes occur when a clot or a blocked artery interrupts the blood flow to the brain. Strokes usually occur in elderly people.

The signs of a stroke are severe headache, dizziness, confusion, loss of consciousness, speaking difficulty and signs of paralysis.

The following paragraphs outline some guidelines on the treatment of a stroke:

Step	Task
1	Lay the casualty down, ensuring that the head is supported.
2	Tilt the head to one side and wipe any saliva dribbling from the mouth.
3	Do not give the casualty anything to eat or drink.
4	If the casualty is unconscious, open the airway and place them in the recovery position whilst waiting for the medical emergency team.

Transportation and Evacuation

First aiders should only move casualties if it is absolutely necessary, such as in a life threatening situation. The medical emergency team will have the appropriate equipment and training to move the casualty safely for further treatment. Transport of casualties can be by manual handling or using specialised equipment such as stretchers. In all cases, personnel involved need to plan the move and be trained in the techniques or in the use of equipment.

Manual Handling

The techniques used depend on the number of rescuers and if the casualty is conscious and able to walk. Always explain to the casualty your intentions so that they know what is going on and can assist if possible.

If the casualty is conscious and able to walk:

Human Crutch Method for One Rescuer

Step	Task
1	Stand on the casualty's weaker side and pass their arm around your shoulder and grasp their hand.
2	Grasp the waist with your other arm.
3	Take small steps at a time with your inside foot.

Human Crutch Method for Two Rescuers

Step	Task
1	As explained above, but each rescuer supports the casualty from either side.

If the casualty is conscious but is unable to walk:

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Piggyback Method for One Rescuer (Only Attempt with Children or Light Casualties)

Step	Task
1	Crouch in front of the casualty and ask them to put their hands around your neck.
2	Grasp the casualty's thighs and rise slowly. At all times, your back should be straight to avoid injury to yourself.

Drag Method for One Rescuer (May Aggravate Neck and Head Injuries)

Step	Task
1	Sit the casualty, then crouch behind them and cross their arms over their chest.
2	Pass your arms under the casualty's armpits and hold both their wrists.
3	Drag the casualty backwards by squat walking.

Cradle Method for One Rescuer (Only Attempt with Children or Light Casualties)

Step	Task
1	Squat beside the casualty. Pass one arm around the casualty's back towards their waist.
2	Pass the other arm under the casualty's thighs.
3	Rise, keeping your back straight.

Hand Seat Method for Two Rescuers

Step	Task
1	The two rescuers grasp each other's wrists to form an arm seat. For lighter casualties, form a three-hand seat and the rescuer who has the free arm can pass it to the shoulder of the other rescuer to form a back support for the casualty.
2	Four-hand seats are used for heavier casualties. In this case, the casualty will have to hold on to the shoulders of the rescuers.

Fore and Aft Carry Method for Two Rescuers

Step	Task
1	Crouch behind the casualty, sit them up and put their arms around their chest. Put your arms under their armpits and grasp their wrists firmly.
2	The second rescuer squats beside the casualty and passes their arms under the casualty's thighs, taking hold of their legs.
3	Both rescuers rise at the same time, keeping their backs straight.

Four Arm Method for Two Rescuers

Step	Task
1	Sit the casualty up.
2	Both rescuers squat on either side of the casualty. They pass their hands under the nearer thigh and hold the leg.
3	The other arm of the rescuers is used to support the casualty's back.
4	Both rescuers rise at the same time, keeping their backs straight.

If the casualty is unconscious and unable to walk:

- Use the drag method for one rescuer.
- Use the cradle method for one rescuer if the casualty is lightweight.

- Use the fore and aft method for two rescuers.
- Use the four arm method for two rescuers.

Stretchers

Stretchers are specialist equipment and should only be used by trained personnel. Serious injury can be caused if a casualty is transferred wrongly to a stretcher or if the stretcher collapses for any reason.

Neil Robertson Stretcher

This type of stretcher is not rigid and allows rescue from confined areas and narrow spaces where an ordinary stretcher with poles would not fit. It can be hoisted vertically through hatches, manholes and steep ladders. This stretcher does not require any assembly.

The following paragraphs outline some guidelines on the use of this stretcher:

Step	Task
1	Both rescuers should position themselves at the chest and shins of the casualty respectively with the casualty between their legs. The forward rescuer grabs the casualty from behind the armpits and puts the casualty's arms behind their neck (tie the wrists loosely for an unconscious casualty). The second rescuer gets hold of the legs just below and above the knees (tie the legs together if necessary). Lift the casualty at the same time and only to the minimum height necessary to lay them on the stretcher. If the casualty is suffering from spinal injuries, they should only be moved under the guidance of the medical emergency team.
2	Lower the casualty slowly and lay them on the canvas side of the stretcher with his forehead level to the head strap. Secure the head with the head strap.
3	Arms may be folded inside and secured by using the upper abdomen straps.
4	The legs are secured with the leg and ankle straps.
5	To stiffen the stretcher, in case of spinal injuries for example, pass rigid poles through the straps under the stretcher.
6	The casualty should be carried by four rescuers.
7	To lift the stretcher vertically, a rope of 1¾" diameter should be secured to the head grommet. Test the lifting rope and knots before use.

WARNING

The rope permanently attached to the head grommet of the stretcher is only a steadying rope and should not be used to lift the casualty. When lifting vertically, a second steadying rope should be attached to the galvanised ring at the foot end.

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2.1.8 LIFESAVING SIGNALS AND PYROTECHNICS

Rocket Parachute

Rocket Parachute Flares

Manufacturer:	Huahai Marine Signals Mfg. Ltd.
Type:	Red parachute signal rocket
Model:	JHR-4
No. of sets:	1 (12 each)
Height of launch:	300m (minimum)
Intensity:	30,000 candela
Duration:	40 seconds (minimum)

Introduction

Rocket parachute flares are used to indicate a vessel is in distress. They may not show the exact location they are launched from due to the wind and the angle of launching, but they will attract attention if they are seen. Rocket parachute flares should only be fired if there is a good chance that they will be seen, ie, another vessel has been sighted or inhabited land is nearby.

Rocket parachute flares should be launched pointing skywards, but slightly upwind. They will then re-curve to windward and burst overhead. Rocket parachute flares should ideally be used in pairs because a person seeing one may not be sure what has been seen and ignore it.

Operating a Rocket Parachute Flare

Before operating the rocket parachute flare, the user should read the instructions printed on the flare itself.

Step	Task
1	Take the parachute flare from its storage location and position yourself outside, and on the leeward side, of the vessel/survival craft.
2	Grip the rocket flare in the middle of the tube with the yellow arrow pointing upwards. In this configuration, the grey cap will be at the bottom and the red cap will be at the top.
3	Remove the grey bottom cap revealing the ignition line.
4	With outstretched arms, hold the rocket vertically, away from the body and above the head. Point the rocket flare away from the body and ensure hands remain clear of the top.
5	To fire the rocket, pull down sharply on the ignition line.
6	Grip the flare firmly with both hands and after a two second delay, the flare will fire.

WARNING

If a flare fails to fire, do not look into the top. It may be slow to ignite and could fire without warning. Do not point the flare at another person or directly at an aircraft. Do not use after the expiry date.



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Hand Flares

Line throwing appliance

Manufacturer:	Huahai Marine Signals Mfg. Ltd.
Type:	Red hand held
Model:	JHH-3
No. of sets:	1 (12 each)
Intensity:	15,000 candela
Duration:	60 seconds (minimum)

Hand-held flares are used to indicate the exact position of the distress. They are for use when rescue has arrived in the vicinity and the exact location of the rescue needs to be indicated. They present an extremely bright red light. Hand-held flares should always be held to leeward to ensure the fumes are blown away from the person holding it.

Operating a Red Hand-Held Flare

Before operating a hand-held flare, the user should read the instructions printed on the flare itself.

Step	Task
1	Take a hand-flare from its storage location and move outside onto the leeward side of the vessel/survival craft.
2	Unfold the metal handle downward and hold by this.
3	Remove the red end cap exposing the ignition line.
4	With the flare held by the metal handle away from the body, pull the ignition line to ignite.
5	Hold the flare by the metal handle at arm’s length, in a vertical position and pointed downwind.

WARNING

If a flare fails to fire, do not look into the top. It may be slow to ignite and could fire without warning. Do not point the flare at another person. Do not use after the expiry date.

Red Hand Flare



Manufacturer:	Huahai Marine Safety Equipment Co., Ltd..
Model:	JHH-3
No. of sets:	One line throwing appliance with four projectiles
Propelling distance::	230 metres
Length of throwing line:	270 metres
Canister:	Plastic sea water resistant

Introduction

Every ship has to be provided with a line-throwing appliance which must be capable of throwing a line, with reasonable accuracy, at least 230m in calm weather, utilising a line with a breaking strength of not less than 2kN. The unit should be stored with brief instructions or diagrams clearly illustrating the use of the line-throwing appliance.

The line-throwing appliance is used:

- During rescue operations for establishing connection between the rescue vessel and the ship in distress for towage assistance.
- Evacuation of persons.
- Rigging of breeches buoy.
- When mooring under difficult circumstances.
- In man overboard situations.

Aiming with the Line-Throwing Appliance

The rocket is fired in such a way that the line will fall very close to the person in the water. Aiming with the line-throwing appliance:

Step	Task
1	In calm weather or in following wind or head wind, aim straight at the target. Take into account the direction of the wind (to windward or leeward of the target) and adjust the angle accordingly.
2	In a strong side wind, aim at the windward side of the target as the wind pressure on the line will steer the rocket towards the wind. If the distance to the target is close to the maximum range of the rocket, 230m, aim to windward of the target. The wind pressure will affect the rocket when its speed decreases and draw it to leeward.

The line-throwing device has a rope which is attached to the inside of the canister by a quick-release knot. It has an integral handle/trigger unit and is marked on the casing giving easy reference to the optimum firing angle.

Operation of the Line-Throwing Device

Step	Task
1	Move the line-thrower to a clear area of deck with suitable anchor points and additional ropes.
2	Remove the upper cap.
3	Fasten the line in the canister to a secure point.
4	Remove the safety pin.
5	Hold the handle with the arrow mark aimed at the target firmly and pull the trigger to fire.

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Lifebuoys

SOLAS Chapter III Regulation 7.1 refers to the constructional requirements for lifebuoys, lifebuoy lights, smoke signals and buoyant lifelines. Regulation 32 stipulates the minimum requirements that must be carried by cargo ships and Regulation 7.1 stipulates the type of fittings required.

The lifebuoys are distributed in such a way that they are easily accessible on both sides of the vessel and ready for immediate use, at least one shall be placed in the vicinity of the stern. No lifebuoy is to be permanently secured in position.

Lifebuoy Attachments

- There shall be at least one lifebuoy on each side fitted with a buoyant lifeline.
- At least four of the lifebuoys are to be provided with self-igniting lights.
- At least two of the lifebuoys with self-igniting lights shall also be fitted with self-activating smoke signals, emitting smoke of a highly visible colour and shall be capable of quick release from the navigation bridge.
- Each lifebuoy is to be marked with the ship’s name, port of registration and reflective tape to assist with identification and search and rescue.

This vessel carries a total of 16 lifebuoys, distributed with attachments as follows:

- Lifebuoy with buoyant line (40m) on the upper deck, there are two either side of No.1 cargo tank and two either side of No.2 cargo tank and No.3 cargo tank.
- Lifebuoy with light and smoke; there is one on each bridge wing at the navigation deck level.
- Lifebuoy with a light only; there are two located either side of the upper deck after the accommodation ladder, near the life boat and the stern deck.
- Lifebuoy with a light and a lifeline 40 m; there are two on both sides of the upper deck forward of the accommodation ladder.

Lifebuoy with Self-Igniting Light and Self-Activating Smoke Signal

These units emit dense orange smoke upon releasing the lifebuoy, which pulls the attached smoke float, activating the igniter as the ring falls to the sea. A light will also activated for night vision. The smoke has a duration time of 15 minutes and the light a minimum of 2 hours.

Use Of Lifebuoys

Man Overboard

If man overboard observed:

Step	Task
1	Shout ‘MAN OVERBOARD’ - indicating port or starboard.
2	Throw the nearest lifebuoy overboard - try and maintain visual contact.
3	Raise the alarm and inform the bridge.
4	The officer of the watch will instigate man overboard procedures including releasing a combined light and smoke lifebuoy to assist in marking the area.

In Port or at Anchor

A lifebuoy with a buoyant lifeline should always be at the head of the gangway.

Embarking/Disembarking the Pilot

A lifebuoy equipped with a self-igniting light should always be available near the pilot ladder or gangway. A lifeline fitted with a quoit capable of floating shall also be available.

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2.2 Safety Equipment

2.2.1 Portable Gas Measuring Equipment

2.2.2 Gas Sampling System

2.2.3 Gas Detection System

2.2.4 Fire Detection System

2.2.5 Fire Control and Safety Plans

2.2.6 Self-Contained Air Breathing Apparatus

2.2.7 Emergency Escape Breathing Devices

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2.2.1 PORTABLE GAS MEASURING EQUIPMENT

Introduction

The purpose of the portable gas detection equipment is to provide the vessel’s crew with an independent means of gas testing in remote locations. The portable gas detection equipment on board is both comprehensive and designed to meet the vessel’s needs throughout each stage of operation of carrying LNG. Each instrument is certificated and comes with manufacturer’s operating instructions, recommended spares and test kits. The certificates are to be suitably filed and the monthly tests recorded.

Methane Exposure

Signs and symptoms of methane exposure may include, headaches, heart palpitations, cognitive impairment (poor judgement, memory loss), dizziness, loss of motor coordination and flu-like symptoms (mental uneasiness, lethargy, discomfort). Anybody experiencing such symptoms should leave the area immediately into fresh air. Excessive exposure may require the use of oxygen to offset asphyxiation.

Nitrogen Exposure

Nitrogen gas is a non-toxic, odourless, colourless, tasteless non-flammable gas with 78% (by volume) of the air is nitrogen and oxygen approximately 21%. When nitrogen concentrations increase (eg, when purging equipment) and the oxygen levels drop below 19.5%, rapid suffocation can occur. Inhalation of a nitrogen enriched atmosphere (ie, loss of oxygen) may cause dizziness, drowsiness, nausea, vomiting, excess salivation, diminished mental alertness, loss of consciousness and can lead to death.

Nitrogen gas is an asphyxiant and this means nitrogen will displace oxygen and create oxygen deficient atmospheres without significant physiological effects. Exposure limits are not normally given to ‘simple asphyxiants’ because the limiting factor is the available oxygen. Therefore, nitrogen has no exposure limit. Liquid nitrogen is a colourless, odourless and extremely cold liquid (-190°C). Contact with this liquid or the cold vapours can cause severe frostbite.

Hydrocarbon Exposure

A person usually coughs and chokes after swallowing or inhaling hydrocarbons. A burning sensation can develop in the stomach and the person may vomit. If the lungs are affected, the person continues to cough intensely. Breathing becomes rapid and the skin may become bluish (cyanosis) because of low levels of oxygen in the blood. Sometimes, difficulty breathing does not develop until many hours after the hydrocarbons enter the lungs. Hydrocarbon ingestion also causes neurologic symptoms, including drowsiness, poor coordination, stupor or coma, and seizures.

Carbon Monoxide Exposure

Carbon monoxide is a colourless, odourless, tasteless gas produced as a by-product of incomplete combustion. Carbon monoxide is a chronically toxic gas which bonds to the haemoglobin molecules in red blood cells, and cells contaminated with carbon monoxide are unable to transport oxygen.

Although very high concentrations of carbon monoxide may be acutely toxic, and lead to immediate respiratory arrest or death, it is the long term physiological effects due to chronic exposure at lower levels that take the greatest toll of affected workers. Exposure levels are too low to produce immediate symptoms, but small repeated doses reduce the oxygen carrying capacity of the blood over time to dangerously low levels. This partial impairment of the blood supply may lead over time to serious physiological consequences.

Oxygen Deficiency

In the case of a slight reduced oxygen level, the development of any signs will develop gradually, the symptoms include headaches, fatigue, shortness of breath, a feeling of euphoria and nausea. In severe cases, where the oxygen level drops rapidly, then changes in levels of consciousness, seizures, coma and death can occur.

Toxic Effects of Carbon Monoxide Exposure

Concentration and the Effect

- 35ppm 8 hour time weighted average (TWA).
- 200ppm Ceiling.
- 600ppm Headache, discomfort (60 minutes).
- 2,500ppm Unconsciousness (30 minutes).
- 4,000ppm Fatal within a few minutes.

Time Weighted Average

The time weighted average (TWA) for the exposure to a chemical can be used when both the chemical concentration and time for exposure varies over time. It is thus used as the average exposure to a contaminant to which ship’s crews may be exposed without adverse effect over a period such as in an 8 hour day. They are usually expressed in units of ppm.

Short-Term Exposure Limit

The Short Term Exposure Limit (STEL) is a limit value above which exposure to a chemical substance should not occur and usually relates to a 15 minute reference period. The aim of a STEL is to prevent adverse health effects and other unwanted effects due to peak exposure that may not be controlled by the application of an 8 hour TWA limit. They are usually expressed in units of ppm.

Multi-Gas Detector

The unit can be used for monitoring confined spaces, inerting, purging and pre-entry testing. It has an in-built pump and can be connected to a 30 metre sample tube, as required. The unit is a rugged impact resistant carbon impregnated casing housed in a polypropylene case with an attached neck/shoulder harness.

The two internal batteries are held in place behind a screwed cover and are sealed against dust and water to IP64. The sensors are protected fro dust and water by an integral hydrophobic filter to IP67. The batteries are either two rechargeable 8 hour duration batteries or two alkaline 20 hour batteries. The batteries used must be of the manufacturer’s approved type and must only be charged or replaced in a safe environment.

The unit is controlled from two multi-function soft pushbutton keys operating a menu driven system. There is a central automatic backlit LCD screen with digital readout values on display.

The element to detect the sample consists of a non-dispersive infrared element which is used for measuring the hydrocarbon gases, whilst a galvanic cell is used to measure the oxygen content. The unit can be used for measurement in inert atmospheres, but must not be used in oxygen enriched atmospheres. The unit has two operating modes:

- Confined Space Monitor (CSM) Operation.
- Combustible Gas Indicator (CGI) Operation.

CSM operation is used when entering confined spaces which may contain hazardous gas mixtures. The unit continually samples the atmosphere and emits a confidence beep and/or LED indication (can be disabled). An alarm is generated for low oxygen, high oxygen, flammable gas, high toxic gas, short term TWA (STEL) and long term TWA (LTEL).

CGI is used as a gas indicator drawing samples via a probe where gas is suspected to be present. Alarms are disabled in this mode as the instrument is used for measuring gas levels.

Before use, a visual check should be made of the external condition of the unit, checking for damage to the pushbuttons, display and body. The pump’s flow rate should be checked and the battery voltage checked. If any damage of these items is noticed, fault alerts on the unit, it should not be used until these are corrected.

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Single Gas Monitor

This single gas portable monitor is designed to be continuously worn by an operator without requiring any further input once set into operation. It is clipped to the wearer leaving the operator hands free, with an audible, visual and vibrating alarm to alert to any hazardous changes in atmosphere.

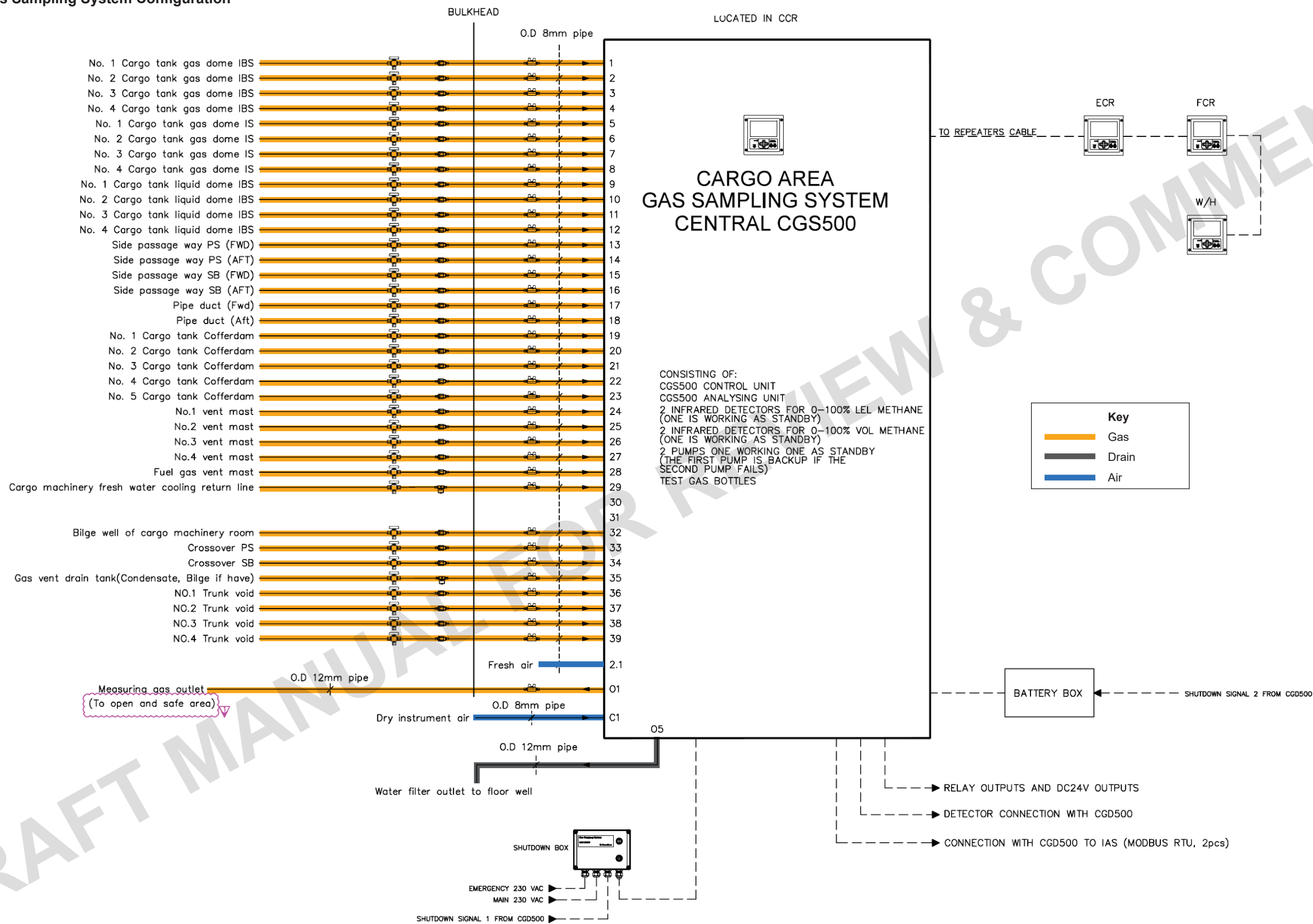
The instrument is a one gas disposable monitor that is operated with a single pushbutton, a direct reading backlit LCD and has a two year life span. It comes with a non-field replaceable lithium-ion battery, filter and sensor already installed and ready for use.

The monitor includes a downloadable data log for twenty-five events and records denoting exposures, calibrations, and gas values. This personal gas detection monitor is designed for monitoring the atmosphere for depletion of oxygen levels.

Before use, a visual check should be made of the external condition of the unit, checking for damage to the pushbutton, display and body. If any damage of these items is noticed, it should not be used until these are corrected.

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Illustration 2.2.2a Gas Sampling System Configuration



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2.2.2 FIXED GAS SAMPLING SYSTEM

Equipment	Manufacturer	Type	No. of sampling points	Detector
Gas Sampling System	C o n s i l i u m Marine AB	Infrared sampling system	39 external	Two Salwico PG11 True IR methane 0-100% LEL Two Salwico PG11 True IR methane 0-100% VOL
Location	Cargo control room Repeater panel: M Gas, Wheelhouse, engine control room, fire control station			

Introduction

The purpose of the fixed gas sampling system is for the detection of methane gas in non-hazardous zones. These include the insulation spaces, cofferdams, liquid domes, vent masts, bosun store, passageways and the inner trunk. In the event of a gas being detected, alarms are activated simultaneously at the sampling unit, the gas sampling repeater panels and the IAS stations.

The fixed gas sampling system is an automatically scanning gas sampling system permanently installed with sampling points. The two gas detectors are common for all sampling points, with two spare sets, which can be used by adjusting the two-way valves from ‘active’ to ‘spare’ as required.

A gas sample is drawn from each sampling point in turn according to the sampling detector in sequence by the transport pump. The sequence is set by the control unit which opens and closes the connected sampling point solenoid valve.

The duration time for opening is dependent upon the distance of the sampling tube and the timing can be adjusted at the control unit to permit a sufficient period to draw a sample from the space to the detector.

Once the timing is set during commissioning, it should require only minor adjustments. The second pump, the analysing pump, draws the sample over the sensors and the sample is expelled to the outside atmosphere.

The sensors in the detector use infrared absorption as a source of measurement of the presence of a gas vapour in the atmosphere and do not require the presence of oxygen. The microprocessor compares the intensity of two specific wavelengths and can determine the value of gas by comparison.

List of Sample Points

No.	Location	Warning	Alarm	Remark
1	No.1 cargo tank IBS - Gas dome	30% LEL	60% LEL	
2	No.2 cargo tank IBS - Gas dome	30% LEL	60% LEL	
3	No.3 cargo tank IBS - Gas dome	30% LEL	60% LEL	
4	No.4 cargo tank IBS - Gas dome	30% LEL	60% LEL	
5	No.1 cargo tank IS - Gas dome	10% Vol	30% Vol	
6	No.2 cargo tank IS - Gas dome	10% Vol	30% Vol	
7	No.3 cargo tank IS - Gas dome	10% Vol	30% Vol	
8	No.4 cargo tank IS - Gas dome	10% Vol	30% Vol	
9	No.1 cargo tank IBS - Liquid dome	30% LEL	60% LEL	
10	No.2 cargo tank IBS - Liquid dome	30% LEL	60% LEL	
11	No.3 cargo tank IBS - Liquid dome	30% LEL	60% LEL	
12	No.4 cargo tank IBS - Liquid dome	30% LEL	60% LEL	
13	Side passage way PS (FWD)	30% LEL	60% LEL	

No.	Location	Warning	Alarm	Remark
14	Side passage way PS (AFT)	30% LEL	60% LEL	
15	Side passage way SB (FWD)	30% LEL	60% LEL	
16	Side passage way SB (AFT)	30% LEL	60% LEL	
17	Pipe duct (FWD)	30% LEL	60% LEL	
18	Pipe duct (AFT)	30% LEL	60% LEL	
19	No.1 cofferdam	30% LEL	60% LEL	
20	No.2 cofferdam	30% LEL	60% LEL	
21	No.3 cofferdam	30% LEL	60% LEL	
22	No.4 cofferdam	30% LEL	60% LEL	
23	No.5 cofferdam	30% LEL	60% LEL	
24	No.1 vent mast	30% LEL	60% LEL	
25	No.2 vent mast	30% LEL	60% LEL	
26	No.3 vent mast	30% LEL	60% LEL	
27	No.4 vent mast	30% LEL	60% LEL	
28	Fuel gas mast	30% LEL	60% LEL	
29	Cargo machinery fresh water cooling return line	30% LEL	60% LEL	* Water filter
30				
31				
32	Bilge well of cargo machinery room	30% LEL	60% LEL	
33	Crossover PS	30% LEL	60% LEL	
34	Crossover SB	30% LEL	60% LEL	
35	Gas vent drain tank	30% LEL	60% LEL	* Water filter
36	No.1 trunk void	30% LEL	60% LEL	
37	No.2 trunk void	30% LEL	60% LEL	
38	No.3 trunk void	30% LEL	60% LEL	
39	No.4 trunk void	30% LEL	60% LEL	
Inside the cabinet is a gas detector which will shut down the gas sampling should this detect methane gas at 30% LEL. * Water filters are provide to avoid the sensor being affected by liquid.				

Operation

The central control panel in the electrical equipment room houses the operator interface to control and monitor the gas detection system, which is also connected to the vessel's IAS. Initial setting and control can be undertaken at the control panel and future monitoring can then be undertaken via the IAS.

The system detects gas, fault, abnormal condition or change in the system status; this is displayed on the control unit panel. The operator can control and check a number of functions in the system. Four access levels are available, where each level is protected by an access code that can restrict unauthorised changes to the equipment status.

The user has to log-in to the system before any operations on high access level can be performed. Without an authorisation code, the user can access permission on the default level.

- Access level 1 is the basic default level, with access to view, local mute and display.
- Access level 2 is for trained personnel level, with access to mute, acknowledge alarms, reset, disablement and calibration.
- Access level 3 is for technical officer level, with level 2 permissions plus full reset capabilities.
- Access level 4 is for authorised service engineers only.

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The system automatically returns to the default access level after 30 minutes of inactivity. The detectors are automatically tested and continuously supervised by the central control unit. In the event of gas being detected, the appropriate output channels are activated and alarm condition is raised.

At preset fixed intervals, the system actively checks that all connected units are in working order and that the contents of the system are according to its configuration. The control system is based upon a menu-driven option, giving the operator various operational functions, checks, changes and disabling active functions as required.

Gas Alarm

A gas alarm is detected in the system. The following information is displayed in the control panel:

- Number of alarms.
- Zone in alarm.
- Type of unit in alarm.
- Time and date of alarm.
- Supplementary text for location.

Mute Gas Alarm

Press the MUTE pushbutton, the access level must be 2 or higher. It silences the internal buzzer and all external alarm devices and mutes the gas alarm indication. The gas indicator stops flashing but remains illuminated until the gas alarm is reset.

Reset a Gas Alarm

Press the RESET pushbutton to reset the current gas alarm. Alarms cannot be reset as long as the device remains in alarm condition.

Disabling

It is possible to disable different devices attached to the system such as zones and detectors. All disabling are shown in the disabling list. Enter access level 2 or higher to make a disabling. Once a new disabling is correctly entered, an acknowledgment is presented on the control panel. The new disabling is also added in the disabling list.

Reconnecting Disabling

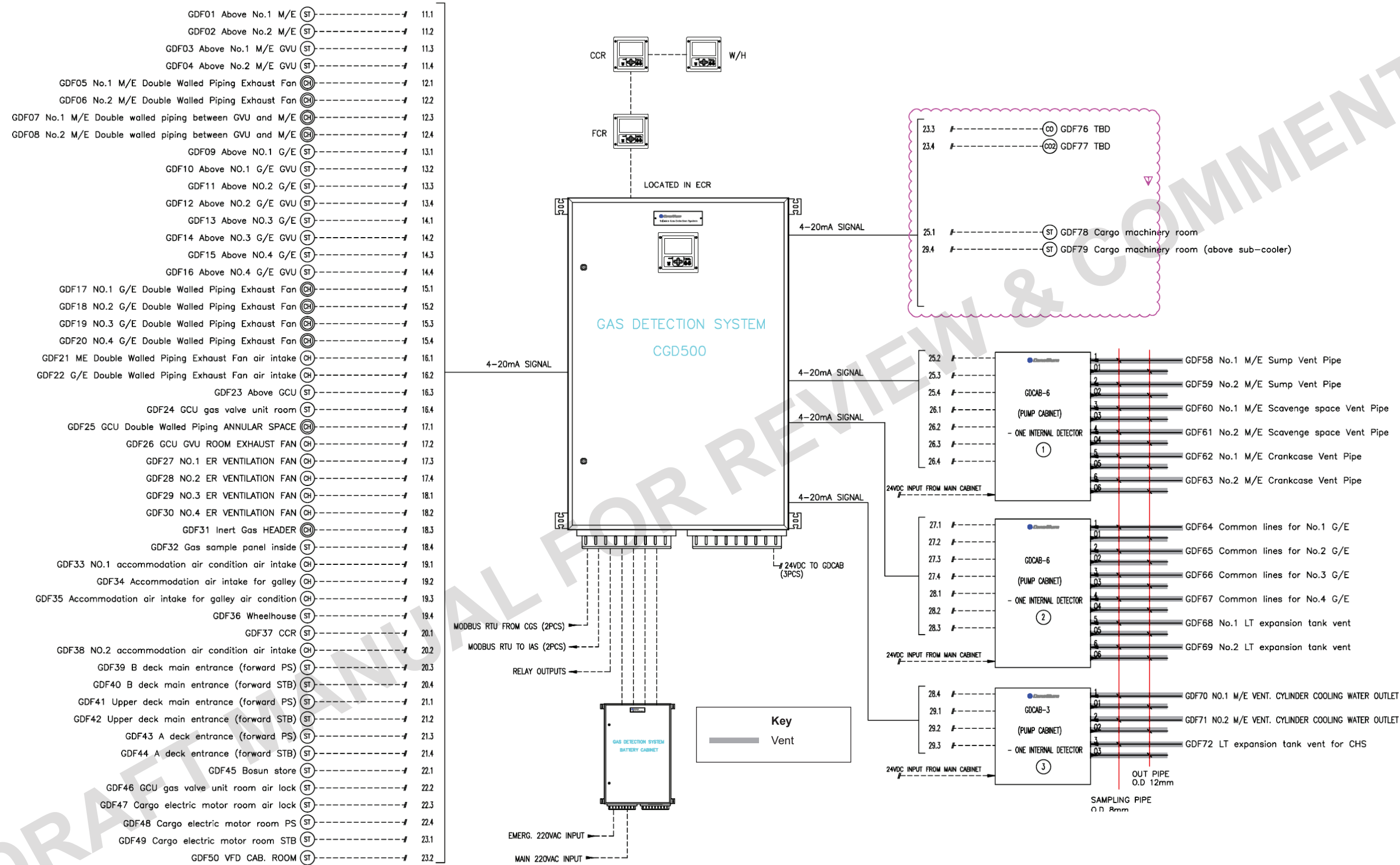
All disabling in the system are presented in the disabling list. Any enabling is made from this list. See the manufacturer's manual for the full operating details.

Automatic Self-Test

The detector provides automatic self-test of its sensing and calibration functions. Malfunction warnings are provided.

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Illustration 2.2.3a Gas Detection System Configuration



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2.2.3 GAS DETECTION SYSTEM

CGD50/500 System: For Engine Room, Other Space and Accommodation Space

Equipment	Manufacturer	Type	Model	Detector
Gas Sampling System	C o n s i l i u m Marine AB	Fixed detectors	GD500	PG11 True IR methane 0-100% LEL PG11 True IR methane 0-100% LEL - Duct mounted ST650 NDIR for methane 0-100% LEL ST450 EC CO, for carbon oxide, 0 - 100 ppm ST650 NDIR CO2 for carbon dioxide, 0 - 5 vol%
Location	Wheelhouse, cargo control room and fire control station			

Introduction

The purpose of the fixed gas detection system is for the detection of methane gas in hazardous zones and the level of oxygen in other zones. The gas detection system consists of individual detector heads located in potentially hazardous areas, which remotely monitor the surrounding atmosphere for the specified gas within the design range. All sensors are explosion-proof rated, specific for their location.

Each detector is connected electrically in loops to a central control station and constant signals are sent with the level of gas at the detector head. Should that level reach a set value, it will activate the control panel into alarm, also on the repeater panels and the connected IAS.

To give greater control and minimise erroneous signals, the gas detectors are specific for that location and their method of measurement is chosen on the basis of the most appropriate type for that situation.

In addition, local gas sampling units are used to monitor crankcase spaces for methane gas, using local sample pump and detector to constantly signal to the central control station with the gas level readings.

Should that level reach a set value, it will activate the control panel into alarm, also on the repeater panels and the connected IAS.

Detector Type

Electrochemical Sensor: ST450 EC CO for Carbon Oxide

The ST450 EC CO uses an electrochemical (EC) cell as the sensing element. Electrochemical sensors consist of electrodes immersed in an electrolyte; when carbon monoxide molecules diffuse into the sensor, they undergo a chemical reaction at the sensing electrode. This reaction generates a small electrical current proportional to the gas concentration. The detector's electronics continuously measure and convert this signal into a calibrated gas concentration value.

Infrared Sensor: ST650 NDIR CH4, Non-dispersive CH4 0-100% LEL

NDIR sensing technology provides high accuracy, long-term stability, and excellent selectivity, as it measures gas concentration based on infrared light absorption at a characteristic wavelength specific to methane. This principle makes the sensor largely immune to sensor poisoning, oxygen deficiency, and humidity variations, ensuring reliable performance in demanding industrial and marine environments.

Infrared Sensor: PG11 CH4 IR, CH4

Consilium PG11 is an explosive-proof rugged optical point gas detector developed for detection of flammable gases he PG11 CH₄ IR is a fixed infrared (IR) gas sensor designed for continuous detection of methane (CH₄) in air. The sensor operates using the infrared absorption principle, measuring methane concentration based on its characteristic absorption of infrared light.

Infrared sensing technology provides high accuracy, long-term stability, and excellent selectivity, and is inherently resistant to sensor poisoning, oxygen deficiency, and humidity effects. These characteristics make the PG11 CH₄ IR sensor well suited for use in industrial and marine environments, particularly in areas where reliable detection of flammable gas leaks is required for safety, alarm activation, and ventilation control.

Operation

The central control panel houses the operator interface to control and monitor the gas detection system, which is also connected to the vessel's IAS. Initial setting and control can be undertaken at the control panel and future monitoring can then be undertaken via the IAS.

The system detects gas, fault, abnormal condition or change in the system status; this is displayed on the control unit panel. The operator can control and check a number of functions in the system. Four access levels are available, where each level is protected by an access code that can restrict unauthorised changes to the equipment status.

The user has to log-in to the system before any operations on high access level can be performed. Without an authorisation code, the user can access permission on the default level.

- Access level 1 is the basic default level, with access to view, local mute and display.
- Access level 2 is for trained personnel level, with access to mute, acknowledge alarms, reset, disablement and calibration.
- Access level 3 is for technical officer level, with level 2 permissions plus full reset capabilities.
- Access level 4 is for authorised service engineers only.

The system automatically returns to the default access level after 30 minutes of inactivity. The detectors are automatically tested and continuously supervised by the central control unit. In the event of gas being detected, the appropriate output channels are activated and alarm condition is raised.

At preset fixed intervals, the system actively checks that all connected units are in working order and that the contents of the system are according to its configuration. The control system is based upon a menu-driven option, giving the operator various operational functions, checks, changes and disabling active functions as required.

Gas Alarm

A gas alarm is detected in the system. The following information is displayed in the control panel:

- Number of alarms.
- Zone in alarm.
- Type of unit in alarm.
- Time and date of alarm.
- Supplementary text for location.

Mute Gas Alarm

Press the MUTE pushbutton, the access level must be 2 or higher. It silences the internal buzzer and all external alarm devices and mutes the gas alarm indication. The gas indicator stops flashing but remains illuminated until the gas alarm is reset.

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Reset a Gas Alarm

Press the RESET pushbutton to reset the current gas alarm. Alarms cannot be reset as long as the device remains in alarm condition.

Disabling

It is possible to disable different devices attached to the system such as zones and detectors. All disabling is shown in the disabling list. Enter access level 2 or higher to make a disabling. Once a new disabling is correctly entered, an acknowledgement is presented on the control panel. The new disabling is also added in the disabling list.

Reconnecting Disabling

All disabling in the system are presented in the disabling list. Any enabling is made from this list. See the manufacturer’s manual for the full operating details.

Automatic Self-Test

The detector provides automatic self-test of its sensing and calibration functions. Malfunction warnings are provided.

Funnel Casing and Accommodation Areas

Detectors: ST650 NDIR CH4, Non-dispersive CH4 0-100% LEL, PG11 CH4 IR, CH4

	Loop	Location	Warning Alarm	High Alarm	Remarks
GDF01	11.1	Above No.1 M/E	30% LEL	60% LEL	ST650 NDIR CH4
GDF02	11.2	Above No.2 M/E	30% LEL	60% LEL	ST650 NDIR CH4
GDF03	11.3	Above No.1 M/E G/V	30% LEL	60% LEL	ST650 NDIR CH4
GDF04	11.4	Above No.2 M/E G/V	30% LEL	60% LEL	ST650 NDIR CH4
GDF05	12.1	No.1 M/E double walled piping exhaust fan	30% LEL	60% LEL	PG11 IR CH4 - Duct mounted
GDF06	12.2	No.2 M/E double walled piping exhaust fan	30% LEL	60% LEL	PG11 IR CH4 - Duct mounted
GDF07	12.3	No.1 M/E double walled piping between G/V and M/E	30% LEL	60% LEL	PG11 IR CH4 - Duct mounted
GDF08	12.4	No.2 M/E double walled piping between G/V and M/E	30% LEL	60% LEL	PG11 IR CH4 - Duct mounted
GDF09	13.1	Above No.1 G/E	30% LEL	60% LEL	ST650 NDIR CH4
GDF10	13.2	Above No.1 G/E G/V	30% LEL	60% LEL	ST650 NDIR CH4
GDF11	13.3	Above No.2 G/E	30% LEL	60% LEL	ST650 NDIR CH4
GDF12	13.4	Above No.2 G/E G/V	30% LEL	60% LEL	ST650 NDIR CH4
GDF13	14.1	Above No.3 G/E	30% LEL	60% LEL	ST650 NDIR CH4
GDF14	14.2	Above No.3 G/E G/V	30% LEL	60% LEL	ST650 NDIR CH4
GDF15	14.3	Above No.4 G/E	30% LEL	60% LEL	ST650 NDIR CH4
GDF16	14.4	Above No.4 G/E G/V	30% LEL	60% LEL	ST650 NDIR CH4
GDF17	15.1	No.1 G/E double walled piping exhaust fan	30% LEL	60% LEL	PG11 IR CH4 - Duct mounted
GDF18	15.2	No.2 G/E double walled piping exhaust fan	30% LEL	60% LEL	PG11 IR CH4 - Duct mounted

	Loop	Location	Warning Alarm	High Alarm	Remarks
GDF19	15.3	No.3 G/E double walled piping exhaust fan	30% LEL	60% LEL	PG11 IR CH4 - Duct mounted
GDF20	15.4	No.4 G/E double walled piping exhaust fan	30% LEL	60% LEL	PG11 IR CH4 - Duct mounted
GDF21	16.1	M/E double walled piping exhaust fan air intake	30% LEL	60% LEL	PG11 IR CH4
GDF22	16.2	M/E double walled piping exhaust fan air intake	30% LEL	60% LEL	PG11 IR CH4
GDF23	16.3	Above GCU	30% LEL	60% LEL	ST650 NDIR CH4
GDF24	16.4	GCU gas valve unit room	30% LEL	60% LEL	ST650 NDIR CH4
GDF25	17.1	GCU double walled piping annular spsce	30% LEL	60% LEL	PG11 IR CH4 - Duct mounted
GDF26	17.2	GCU G/V room exhaust fan	30% LEL	60% LEL	PG11 IR CH4
GDF27	17.3	No.1 engine room ventilation fan	30% LEL	60% LEL	PG11 IR CH4
GDF28	17.4	No.2 engine room ventilation fan	30% LEL	60% LEL	PG11 IR CH4
GDF29	18.1	No.3 engine room ventilation fan	30% LEL	60% LEL	PG11 IR CH4
GDF30	18.2	No.4 engine room ventilation fan	30% LEL	30% LEL	PG11 IR CH4
GDF31	18.3	Inert gas header		60% LEL	PG11 IR CH4 - Duct mounted
GDF32	18.4	Gas sample panel inside	15%LEL	30%LEL trip CGS500	ST650 NDIR CH4
GDF33	19.1	No.1 accommodation air condition air intake	15%LEL	30%LEL trip AC	PG11 IR CH4
GDF34	19.2	Accommodation air intake for galley	15%LEL	30%LEL trip galley fan	PG11 IR CH4
GDF35	19.3	Accommodation air intake for galley air condition	15%LEL	30%LEL trip galley AC	PG11 IR CH4
GDF36	19.4	Wheelhouse	30% LEL	60% LEL	ST650 NDIR CH4
GDF37	20.1	CCR	30% LEL	60% LEL	ST650 NDIR CH4
GDF38	20.2	No.2 accommodation air condition air intake	30% LEL	60% LEL	PG11 IR CH4
GDF39	20.3	B deck main entrance (forward PS)	30% LEL	60% LEL	ST650 NDIR CH4
GDF40	20.4	B deck main entrance (forward STB)	30% LEL	60% LEL	ST650 NDIR CH4
GDF41	21.1	Upper deck main entrance (forward PS)	30% LEL	60% LEL	ST650 NDIR CH4
GDF42	21.2	Upper deck main entrance (forward STB)	30% LEL	60% LEL	ST650 NDIR CH4
GDF43	21.3	A deck main entrance (forward PS)	30% LEL	60% LEL	ST650 NDIR CH4
GDF44	21.4	A deck main entrance (forward STB)	30% LEL	60% LEL	ST650 NDIR CH4
GDF45	22.1	Bosun store	30% LEL	60% LEL	ST650 NDIR CH4
GDF46	22.2	GCU gas valve unit room airlock	30% LEL	60% LEL	ST650 NDIR CH4
GDF47	22.3	Cargo electric motor room airlock	30% LEL	60% LEL	ST650 NDIR CH4
GDF48	22.4	Cargo electric motor room PS	30% LEL	60% LEL	ST650 NDIR CH4
GDF49	23.1	Cargo electric motor room STB	30% LEL	60% LEL	ST650 NDIR CH4
GDF50	23.2	VFD cabinet room	30% LEL	60% LEL	ST650 NDIR CH4
GDF76	23.3	TBD	30% LEL	60% LEL	ST450 EC CO
GDF77	23.4	TBD	30% LEL	60% LEL	ST650 NDIR CO2
GDF78	25.1	Cargo machinery room	30% LEL	60% LEL	ST650 NDIR CH4

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	Loop	Location	Warning Alarm	High Alarm	Remarks
GDF79	29.4	Cargo machinery room (above sub-cooler)	30% LEL	60% LEL	ST650 NDIR CH4
GDF73	26.4	Local sample cabinet No.1 inside	30% LEL	30%LEL trip SAL-GDCAB-1	ST650 NDIR CH4
GDF74	28.3	Local sample cabinet No.2 inside	30% LEL	30%LEL trip SAL-GDCAB-2	ST650 NDIR CH4
GDF75	29.3	Local sample cabinet No.3 inside	30% LEL	30%LEL trip SAL-GDCAB-3	ST650 NDIR CH4
GDF58	25.2	No.1 M/E sump vent pipe	30% LEL	60% LEL	PG11 IR CH4
GDF59	25.3	No.2 M/E sump vent pipe	30% LEL	60% LEL	PG11 IR CH4
GDF60	25.5	No.1 M/E scavenge space vent pipe	30% LEL	60% LEL	Band B True IR CH4
GDF61	26.1	No.2 M/E scavenge space vent pipe	30% LEL	60% LEL	Band B True IR CH4
GDF62	26.2	No.1 M/E crankcase vent pipe	30% LEL	60% LEL	Band B True IR CH4
GDF63	26.3	No.2 M/E crankcase vent pipe	30% LEL	60% LEL	Band B True IR CH4
GDF64	27.1	Common lines for No.1 G/E	30% LEL	60% LEL	Band B True IR CH4
GDF65	27.2	Common lines for No.2 G/E	30% LEL	60% LEL	Band B True IR CH4
GDF66	27.3	Common lines for No.3 G/E	30% LEL	60% LEL	Band B True IR CH4
GDF67	27.4	Common lines for No.4 G/E	30% LEL	60% LEL	Band B True IR CH4
GDF68	28.1	No.1 expansion tank vent	30% LEL	60% LEL	PG11 IR CH4
GDF69	28.2	No.2 expansion tank vent	30% LEL	60% LEL	PG11 IR CH4
GDF70	28.4	No.1 M/E vent cylinder cooling water outlet	30% LEL	60% LEL	PG11 IR CH4
GDF71	29.1	No.2 M/E vent cylinder cooling water outlet	30% LEL	60% LEL	PG11 IR CH4
GDF72	29.2	LT expansion tank vent for CHS	30% LEL	60% LEL	PG11 IR CH4

CGD50/500 System: for O₂ detection and Freons detection

Introduction

The purpose of the fixed gas detection system is for the detection of the level of oxygen in the spaces near nitrogen generators, inert gas generator, LNG subcooler and level of refrigerant gas R407c near the inert gas system water chiller unit. The gas detection system consists of individual detector heads located in potentially hazardous areas, which remotely monitor the surrounding atmosphere for the specified gas within the design range. Sensors in LNG subcooler area are explosion-proof rated, specific for their location.

Each detector is connected electrically in loops to a central control station and constant signals are sent with the level of gas at the detector head. Should that level reach a set value, it will activate the control panel into alarm, also on the repeater panels and the connected IAS.

To give greater control and minimise erroneous signals, the gas detectors are specific for that location and their method of measurement is chosen on the basis of the most appropriate type for that situation.

Detector Type

Electrochemical Sensor: ST450 EC for Oxygen

The ST450 EC is a robust, industrial-grade fixed gas detector that uses an electrochemical sensor to continuously monitor the concentration of oxygen (O2) in ambient air. It is designed for reliable and accurate detection in marine, industrial, confined space, and hazardous area applications.

The ST450 EC employs an electrochemical sensing element, which typically consists of two or three electrodes immersed in an electrolyte. When oxygen interacts with the sensor electrodes, an electrochemical reaction occurs, producing an electrical signal proportional to the oxygen concentration. This signal is processed by the detector

electronics to calculate and display the oxygen level. Electrochemical sensors are highly selective, provide stable long-term performance and are less susceptible to cross-sensitivity with other gases compared with some alternative technologies.

The detector is housed in a robust enclosure with ATEX and IECEx approvals for use in explosive atmospheres without requiring additional zener barriers. It is suitable for installation in Zone 1/2 and Zone 21/22 hazardous areas.

A replaceable sensor module allows the oxygen sensing element to be changed at the end of its service life without replacing the entire detector. The unit incorporates a graphic display with backlight, showing real-time measured values and status indications. A capacitive key interface enables calibration, alarm set point adjustment, and navigation through diagnostics and menu functions; access to these settings is secured with a password to prevent unauthorised changes.

Semiconductor Sensor: ST350-SCF for refrigerants R407c / R410a

The heart of the detector is a replaceable BUSENS D SCF gas sensor. The SCF sensor is suitable for the detection of synthetic refrigerants (H)CFKs and HFKs. These are well known freons such as R134a, R407c and the newer type R1234yf.

The sensor used in the ST350-SCF is a semiconductor (metal oxide) sensor (MOS) that responds to changes in gas concentration by detecting variations in electrical resistance on its active sensing surface. This sensing principle offers a fast response and reliable detection of refrigerant gases in the air, making it suitable for environments where continuous monitoring is required for safety and system protection.

The ST350-SCF output signals typically include a 4-20 mA analogue output and optionally RS-485 Modbus RTU communication, allowing integration with building management systems, alarm panels or control systems. Status indicators including LEDs provide local visual feedback for normal operation, alarm conditions or system faults.

The unit is certified for use in potentially explosive atmospheres (ATEX / IECEx) and designed with appropriate ingress protection, making it suitable for installation in industrial and marine environments where leak detection of refrigerants such as R-407C and R-410A is critical for safety and regulatory compliance.

	Loop	Location	High Level Alarm	Remarks
GDF51	11.1	No.1 generator space	19% Vol	ST450 EC O2
GDF52	11.2	No.2 generator space	19% Vol	ST450 EC O2
GDF53	11.3	No.1 IGG space	19% Vol	ST450 EC O2
GDF54	11.4	No.2 IGG space	19% Vol	ST450 EC O2
GDF55	12.1	No.1 subcooler space	19% Vol	ST450 EC O2
GDF56	12.2	No.1 subcooler space	19% Vol	ST450 EC O2
GDF57	12.3	Inert gas system water chiller unit	1,000 ppm	ST350, R407c

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ENGINE ROOM LOCAL FIXED GAS SAMPLING

Introduction

The purpose of the local fixed gas sampling system is for the detection of methane gas in engine internal spaces. In the event of a gas being detected, alarms are activated simultaneously at the sampling unit and the Integrated Automation System (IAS) stations.

The local fixed gas sampling system is an automatic gas sampling system, permanently installed with sampling points. The gas detectors each have one sampling line and a separate sampling pump on separate intake lines and exhaust lines which vent to outside the engine room.

After the detector, the sample pump passes through a reducing nipple and a flame trap to an exit line outside the engine room.

A shut-off valve is fitted to completely isolate the sampling line immediately after the take-off point. To ensure no oil or water contamination is carried over and contaminate the detector, the sampling line is fitted with a particle/water filter, a coalescing/oil filter, a flame trap and a stop valve with purging, all located downstream of the detector.

The engine room cabinet internals are fitted with a gas detector unit to monitor for gas leaks within the cabinet.

The sensors in the detector use infrared absorption as a source of measurement for the presence of a gas vapour in the atmosphere and do not require the presence of oxygen. The microprocessor compares the intensity of two specific wavelengths and can determine the value of gas by comparison.

The Salwico True IR Band B gas detector is a fixed infrared (IR) flame and gas detection device designed for reliable detection of hydrocarbon gas clouds and fire-related infrared radiation in industrial and marine environments. Using infrared optical sensing technology, the detector provides fast and stable response while being highly resistant to false alarms caused by sunlight, hot surfaces, or electrical interference.

Sampling Points for No.1 Local Gas Sampling Cabinet for M/E

	Loop	Location	Warning Alarm	High Alarm	Remarks
GDF58	25.2	No.1 M/E sump vent pipe	30% LEL	60% LEL	PG11 IR CH4
GDF59	25.3	No.2 M/E sump vent pipe	30% LEL	60% LEL	PG11 IR CH4
GDF60	25.5	No.1 M/E scavenge space vent pipe	30% LEL	60% LEL	Band B True IR CH4
GDF61	26.1	No.2 M/E scavenge space vent pipe	30% LEL	60% LEL	Band B True IR CH4
GDF62	26.2	No.1 M/E crankcase vent pipe	90% LEL	95% LEL	Band B True IR CH4
GDF63	26.3	No.2 M/E crankcase vent pipe	90% LEL	95% LEL	Band B True IR CH4

* If the system detects more than 30% LEL inside the cabinet the system automatically shuts down.

Sampling Points for No.2 Local Gas Sampling Cabinet for G/E & LT & HT

	Loop	Location	Warning Alarm	High Alarm	Remarks
GDF64	27.1	Common lines for No.1 G/E	90% LEL	95% LEL	Band B True IR CH4
GDF65	27.2	Common lines for No.2 G/E	90% LEL	95% LEL	Band B True IR CH4
GDF66	27.3	Common lines for No.3 G/E	90% LEL	95% LEL	Band B True IR CH4
GDF67	27.4	Common lines for No.4 G/E	90% LEL	95% LEL	Band B True IR CH4
GDF68	28.1	No.1 expansion tank vent	30% LEL	60% LEL	PG11 IR CH4
GDF69	28.2	No.2 expansion tank vent	30% LEL	60% LEL	PG11 IR CH4

* If the system detects more than 30% LEL inside the cabinet the system automatically shuts down.

Sampling Points for No.3 Local Gas Sampling Cabinet for M/E & LT

	Loop	Location	Warning Alarm	High Alarm	Remarks
GDF70	28.4	No.1 M/E vent cylinder cooling water outlet	30% LEL	60% LEL	PG11 IR CH4
GDF71	29.1	No.2 M/E vent cylinder cooling water outlet	30% LEL	60% LEL	PG11 IR CH4
GDF72	29.2	LT expansion tank vent for CHS	30% LEL	60% LEL	PG11 IR CH4

* If the system detects more than 30% LEL inside the cabinet the system automatically shuts down.

Diagram illustrating the Fire Alarm System (FAS) configuration, showing the connection between the Fire Alarm Central Panel (NAV. EQIPT. RM.) and various Repeater Panels (Fire Control Station, Cargo Control Room, Wheelhouse Console, ECC) and the Control Panel.

The system components and connections are as follows:

- Fire Alarm Central Panel (NAV. EQIPT. RM.):** Labeled "FIRE DETECTION SYSTEM SALVICO CARGO". It receives power from "BACKBONE BUS EXTERNAL CHANNEL A + 24VDC" (FA-1 4x2x0.75) and "BACKBONE BUS EXTERNAL CHANNEL B + 24VDC" (FA-1A 4x2x0.75). It outputs "LOOPS", "I/O & RELAY OUTPUTS", "MAINS 220 VAC", and "EMERGENCY 220 VAC".
- Control Panel (Wheelhouse Console):** Receives power from the central panel (FA-3A 4x2x0.75) and outputs to the VDR (FA-VDR 2x2x0.75) and IAS (FAP-3FPD 2x2x0.75).
- Repeater Panels:**
 - REPEATER PANEL FIRE CONTROL STATION:** Receives power from the central panel (FA-4 4x2x0.75) and outputs to the ECC (FA-5A 4x2x0.75).
 - REPEATER PANEL CARGO CONTROL ROOM:** Receives power from the central panel (FA-3 4x2x0.75) and outputs to the VDR (FA-VDR 2x2x0.75) and IAS (FAP-3FPD 2x2x0.75).
 - REPEATER PANEL ECC:** Receives power from the Fire Control Station (FA-5 4x2x0.75) and outputs to the VDR (FA-VDR 2x2x0.75) and IAS (FAP-3FPD 2x2x0.75).

Legend for cable types and sizes:

- FA-1 4x2x0.75
- FA-3 4x2x0.75
- FA-4 4x2x0.75
- FA-5 4x2x0.75
- FA-3A 4x2x0.75
- FA-5A 4x2x0.75
- FA-VDR 2x2x0.75
- FAP-3FPD 2x2x0.75
- FA-1A 4x2x0.75

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2.2.4 FIRE DETECTION SYSTEM

Manufacturer:	Consilium
Model:	Salwico Cargo System

CD-PH Detectors

CD-PH+IP-ADAPT+IP-BASE	
Detector type:	Addressable smoke& heat detector with lockable base for wet space, IP44
No. of sets:	72
Sensor type:	Optical light scattering thermistor
Temperature range:	-40°C to +70°C
Nominal voltage:	24V DC
CD-PH Ex ia+IP-ADAPT+IP-BASE	
Detector type:	IS type addressable smoke& heat detector with lockable base for hazardous space, IP44
No. of sets:	20
Sensor type:	Optical light scattering thermistor
Temperature range:	-40°C to +70°C
Nominal voltage:	24V DC
CD-PH+IP-ADAPT+IP-BASE	
Detector type:	Addressable smoke detector with lockable base for wet space, IP44
No. of sets:	153
Sensor type:	Optical light scattering thermistor
Temperature range:	-40°C to +70°C
Nominal voltage:	24V DC
CD-PH+CD-S+CD-B	
Detector type:	Addressable smoke detector with lockable base and alarm sound for dry space, IP22
No. of sets:	43
Sensor type:	Optical light scattering thermistor
Temperature range:	-40°C to +70°C
Nominal voltage:	24V DC
CD-PH+CD-B	
Detector type:	Addressable smoke detector with lockable base for dry space, IP22
No. of sets:	54
Sensor type:	Optical light scattering thermistor
Temperature range:	-40°C to +70°C
Nominal voltage:	24V DC
CD-PH+CD-B	
Detector type:	Addressable heat detector 54°C with lockable base for dry space, IP22
No. of sets:	2
Sensor type:	Optical light scattering thermistor
Temperature range:	-40°C to +70°C
Nominal voltage:	24V DC

EV-H2-CS Detectors

EV-H2-CS+IP-ADAPT SCI+IP-BASE	
Detector type:	Addressable heat detector 84°C with lockable base for wet space, IP44
No. of sets:	14
Sensor type:	Heat thermistor fixed at 84°C + 0.8°C/minute rate of rise
Temperature range:	-25°C to +80°C
Nominal voltage:	24V DC

Manual Call Points

CD-MCP WP Ex ia	
MCP type:	IS type addressable manual call point for hazardous space, IP66
No. of sets:	4
MCP-A SCI WP	
MCP type:	Addressable manual call point for wet space, IP67
No. of sets:	33
MCP-A SCI	
MCP type:	Addressable manual call point for dry space, IP23
No. of sets:	24

The Salwico Cargo Fire Detection System is a computerised, fully addressable analogue fire alarm system with analogue detectors. The power supplies are 220V AC from the main switchboard and 220V AC from the emergency switchboard. The unit also contains an emergency 7.2Ah battery in the event that the other supplies fail.

Equipped with a range of detectors, the system can respond to smoke, UV, flame and heat generated by a fire.

The system comprises:

- 6 loops for all zones.
- 7.2Ah battery/24V DC.
- 2.3Ah battery/24V DC.
- One repeater panels (M4.3), console-mounted in Fire Control Station.
- Two control/repeater panels (M4.3), console-mounted in the wheelhouse and CCR.
- 358 detector heads of various types.

The fire alarm and detection system is a computerised fully addressable fire alarm system with analogue detectors. The operating panel, control unit and power supply are contained in a central cabinet located in the Navigation Equipment Room. Control panel is located in the wheelhouse. This panel allows the ship's staff to monitor alarms and scroll through the alarms in the queue list, to perform any disconnections or reconnections. Additionally, three repeater panels are located in Fire Control Station, Cargo Control Room and Engine Control Console.

There are six detector loops connected in the system with a battery back-up provided in the event of a power failure.

The detection system consists of a wide range of detectors and sensors to suit different needs and conditions and includes detectors for different parameters such as smoke, heat and flame. Manual call points, short-circuit isolators and a timer are also connected into the loops where required. A fault in the system or a false alarm is detected immediately, since the function of the detectors and other installed loop units are automatically and continuously tested. The system can also identify defective detectors in each loop and can be monitored via the IAS.

Each main control unit contains a USB interface, allowing for the upload and download of data and to save a configuration file.

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The system is supplied from the 220V AC Ship General UPS cabinet, with a back-up supply from the emergency 220V AC feeder panel. Both the main and emergency supplies are fed into the changeover module. The output from the changeover module is sent to the power supply unit (PSU) module. The supply voltage is then rectified and changed down to a 24V DC 5A signal which is sent to the charger and used to charge an emergency battery supply. There are two control panels, one is mounted on the bridge at the rear side of the radio table and one in the cargo control room.

The system is connected to the ship's PA system for sounding the alarm over the public address after a set time delay of 2 minutes. There is also a control signal from the fire alarm control system which will trip the accommodation fans, and release the stairwell magnetic door holding open units.

An output from the fire detection system is directed to the IAS monitoring and control system and the Voyage Data Recorder (VDR) which will record all alarm activation.

System Loops and Zones

There are six detector loops around the vessel to monitor selected areas. Loops 1 to 3 cover the accommodation, the cargo & forward areas, loop No.6 covers the emergency generator room, electric motor room, while No.4 and No.5 cover the engine room. Each loop is broken down into several zones.

Fire Detector

The vessel is fitted with an automatic fire detection system designed to give early warning of fire by detecting heat and/or smoke in different areas of the ship. All detectors are connected to the fire alarm system and continuously monitor the surrounding conditions. When abnormal heat or smoke is detected, an alarm is sent to the fire control panel, and the location of the detector is identified.

Heat Detectors

Heat detectors are used in areas where smoke may be present during normal operations (such as machinery spaces).

- They detect high temperature or rapid temperature increase
- An alarm is activated when the temperature exceeds a set limit
- A red light (LED) on the detector will illuminate in alarm condition
- The alarm remains active until it is reset from the control panel

Smoke and Heat Detectors (Combined Detectors)

These detectors are installed in areas such as accommodation and control spaces.

- They detect smoke, heat, or both
- Smoke is detected at an early stage of fire development
- Sensitivity is adjusted automatically to reduce false alarms caused by dust
- A red LED indicates alarm condition

Detectors in Hazardous Areas

Special detectors are installed in hazardous areas (e.g. gas-related spaces).

- These are designed to be intrinsically safe
- They operate in the same way as standard detectors but are safe for explosive atmospheres

System Operation

- Each detector has its own address, allowing the exact location of the alarm to be shown

- The system continuously checks the condition of all detectors
- In case of fault or alarm, the control panel will indicate the affected detector

Additional Features

- Some detectors are connected to remote indicators (e.g. outside cabins)
- The system includes protection against electrical faults to maintain operation

Crew Actions

1. Check the location indicated on the fire panel
2. Proceed to the area and verify the situation
3. Report to the bridge or duty officer
4. Take action according to emergency procedures

Maintenance

- Keep detectors clean and unobstructed
- Do not paint or cover detectors
- Report any damage or faults immediately

The fire detection system provides early warning of fire by monitoring heat and smoke, allowing quick response by the crew and improving onboard safety.

Gas Detection Panel



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Main Control Panel

The main control panel is used to monitor and control the functions of the fire detection system. The main control panel is divided into three sections:

- System indicators.
- Operational controls.
- Alphanumeric keypad.

Control/Repeater Panel M 4.3 – Front View Description

The Control/Repeater Panel M 4.3 is equipped with an LCD display and multifunction control keys, providing local operation and indication for the fire detection and alarm system.

Main features (front view):

Step	Task
1	LED Status Indicators (left vertical row) – Display system status such as power on, fault, alarm, disablement, and test conditions.
2	USB Port (bottom left) – For configuration upload/download and firmware updates.
3	Fire and Navigation Keys (left button group) – Four programmable pushbuttons with integrated LED indicators for system acknowledgement, menu navigation, and quick actions.
4	LCD Display (centre) – High-contrast graphical display showing system menus, active alarms, faults, and status messages.
5	Function Keys (below display) – • Acknowledge/Scroll – Navigates through messages or confirms actions. • System Reset – Clears alarm or fault conditions after resolution. • Silence Buzzer – Mutes the local panel buzzer.
6	Navigation Keys (right cluster) – Arrow keys for menu navigation and value adjustment.
7	Numeric/Soft Keypad (right button group) – Allows direct numeric entry or soft key functions depending on the active menu

System Indicators

POWER: Steady green light indicates the correct power supply to the panel.

SYSTEM FAULT: Flashing yellow light indicates a main process fault and a serious system error.

TEST: Steady yellow light indicates that at least one zone has been manually set to test mode.

WARNING: Steady yellow light indicates that at least one warning is present.

ZONE:

- Steady yellow light indicates that at least one zone or fire detector is disabled.
- Flashing yellow light indicates that at least one zone or fire detector is in fault.

ALARM DEVICE:

- Steady yellow light indicates that the output from at least one alarm device has been disabled.
- Flashing yellow light indicates that the output from at least one alarm device has developed a fault.

ALARM DELAY OFF: Steady yellow light indicates that the alarm delay function has been disabled.

A USB connection is provided for flash memory to load or save system configuration files.

Operational Controls

FIRE: This indicates the presence of a fire alarm and provides access to the fire alarm list.

- Flashing red light indicates an unacknowledged fire alarm is present.
- Steady red light indicates that all fire alarms have been muted.

PRE-ALARM: This indicates the presence of a pre-alarm and provides access to the pre-alarm list.

- Flashing orange light indicates that an unacknowledged pre-alarm is present.
- Steady orange light indicates that all pre-alarms have been muted.

FAULT: This indicates the presence of a fault and provides access to the fault list.

- Flashing yellow light indicates that an unacknowledged fault is present.
- Steady yellow light indicates that all faults have been muted.

DISABLEMENTS: A steady yellow light indicates the presence of a disabled function in the system and provides access to the disablement menu.

SHORTCUTS: This can be customised to provide access to a short-cut list specified by the operator.

STATUS: This provides the operator with quick access to the system status summary.

MENU: This provides the operator with quick access to the main menu screen and access to all system functions.

System Operation

Operator Access

In order to prevent any unauthorised changes to the system, an operator must log into the system with the correct access code before any vital operation can be performed. Without the correct access code, the user can only view fire and fault alarms and mute the local buzzer. There are three different access levels:

- Operator Access (Level 2B) - This is the default access level. The system will automatically return to this level after 30 minutes of inactivity. With 2B access, the operator can mute the local buzzer, view fire and fault alarms, access the main menu, reset and mute alarms and disable parts of the system.
- Power User Access (Level 3) - This level can be accessed by entering the correct code via the main menu. This carries the same permissions as 2B plus the ability to make changes to the system configuration.
- Service Access (Level 4) - This level is only allowed to qualified service engineers.

To change the user access level:

Step	Task
1	Press Menu.
2	Select option 4 LOGIN and select user.
3	Enter the correct access code for the user level.

Pre-Alarms

The pre-alarm function provides early alarms to fire conditions such as a smouldering fire. The indication PRE-ALARM illuminates on activation of a pre-alarm. The priority level of a pre-alarm is lower than that of a fire alarm.

It is not possible to reset a pre-alarm. The operator can only mute the alarm. The pre-alarm will remain in the pre-alarm list until the fire condition is under the level for pre-alarm or over the level required for a fire alarm.

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Fire Alarms

Upon the activation of a fire alarm, the following information is displayed:

- Number of alarms.
- Zone in alarm.
- Type of unit in alarm and address of unit.
- Supplementary text.

Press OK for further details.

If a fire alarm sounds:

Step	Task
1	Press MUTE (🔇) to silence the internal buzzer and all external alarm devices. The FIRE indicator will stop flashing but will remain illuminated until the alarm is reset.
2	Press OK for further details.

To reset a fire alarm, press reset (🔄).

Note: An alarm cannot be reset as long as the sensor remains in alarm condition.

If multiple fire alarms activate, the (🔊) indicator will activate. The first and last alarms will be displayed on the control panel. Press either multiple alarm (🔊) or the arrow keys to scroll through the alarms.

Fault Alarms

If a fault condition occurs, the following information will be displayed on the main control panel:

- Number of detected faults.
- Type of fault.
- Identification of faulty unit.
- Supplementary text.

Press OK for further details.

If a fault alarm activates:

Step	Task
1	Press MUTE (🔇) to silence the internal buzzer.
2	Press OK for further details.

To reset a fault from the fault list:

Step	Task
1	Press MENU.
2	Select 1 FAULT ALARMS.
3	Select 1 FAULT LIST.
4	Select the fault from the list to be reset, then press reset (🔄).

To reset all faults from the fault list:

Step	Task
1	Press MENU.

Step	Task
2	Select 1 FAULT ALARMS.
3	Select 3 RESET ALL FAULTS.
4	Press OK.

Disablements

An operator can disable parts of the system such as zones, loops, detectors, manual call points, etc. Once an item has been disabled it is added to the disablements list. The disablement indicator (🔇) will illuminate when at least one item is on the list and remain illuminated until all items have been removed from the disablements list.

When disabling a part of the system, there are four different types of disablement:

- Permanent - The device is permanently disabled until it is re-enabled by the operator.
- Timer - The device is disabled for a certain time period as entered by the operator in the disablement timer (hh:mm). The maximum possible time is 23 hours and 59 minutes.
- Clock - The device is disabled immediately and is reconnected at the time entered by the operator. The maximum possible time is 23 hours and 59 minutes.
- Periodic - The device is disabled for certain time periods on certain days of the week.

When a new disablement is entered, an acknowledgement is shown on the control panel.

Disabling Zones

Step	Task
1	Press the MENU button.
2	Select 3 DISABLEMENTS.
3	Select 1 NEW DISABLEMENT.
4	Select 1 ZONES.
5	Select the zone number.
6	Choose the type of disablement required (permanent, timer, etc). If the type of disablement chosen requires a time to be entered, the operator will then be required to enter it here.

Disabling Addressable Detectors and Manual Call Points

Step	Task
1	Press the MENU button.
2	Select 3 DISABLEMENTS.
3	Select 1 NEW DISABLEMENT.
4	Select 2 DETECTORS & MANUAL CALL POINTS.
5	Select zone number, detector types, disablement type and enter a disablement time.

Reconnecting a Disabled Item

Step	Task
1	Press the MENU button.
2	Select 3 DISABLEMENTS.
3	Select 2 DISABLEMENT LIST.
4	Choose the disablement from the list.
5	Press reset (🔄).

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Reconnecting All Disablements

Step	Task
1	Press the MENU button.
2	Select 3 DISABLEMENTS.
3	Select 5 REMOVE ALL DISABLEMENTS.
4	Press the OK button to remove all disablements.

System Testing

When a zone is in test mode, external alarm devices and controls are not activated in case of fire. This setting can be deactivated and so should be checked before testing the system.

In test mode, the alarm level is lowered in order to provide a quick response to the detector test equipment. The LED indicator on the loop unit under test will begin to blink. The units in alarm will automatically reset once the heat or smoke concentration falls below the alarm limit or if the manual call point is reset. The zone that has been set to test mode will automatically return to normal operation after two hours.

Placing a Zone in Test Mode

In order to place a zone into test mode, the procedure is as follows:

Step	Task
1	Press MENU.
2	Select 3 DISABLEMENTS.
3	Select 1 NEW DISABLEMENT.
4	Select 6 TEST MODE.
5	Select the required zone and press OK.

The disablements, test and zone/unit indicators will illuminate to confirm that they are now under test mode. Once under test mode, the detectors and manual call points in the fire zone can be tested during the following two hours.

Note: The zone that has been set to test mode will automatically return to normal operation after two hours.

To view the alarms generated whilst testing the system:

Step	Task
1	Press MENU.
2	Select 7 HISTORY.
3	Select 1 FIRE HISTORY LIST.

Reset Test Mode

Once testing has been completed, the zone should be returned to normal operation.

Step	Task
1	Check that no detectors or manual call points are in alarm condition.
2	Press MENU.
3	Select 3 DISABLEMENTS.
4	Select 2 DISABLEMENT LIST.
5	Select the desired zone and reset (⏮).

Test Control Panel Display

This function allows an operator to test the display and all indicators on the main control panel. To test the display, the procedure is:

Step	Task
1	Press MENU.
2	Select 5 SETTINGS.
3	Select 4 TEST DISPLAY.

Repeater Control Panel

The repeater panel monitors functions in the fire detection system. The screen display allows the operator to view any fire alarms, faults and disablements. The panel comprises the following in addition to the screen:

- Power lamp - this is a steady green lamp showing that the panel power source is being supplied.
- Local mute key - this is used to silence the local buzzer alarm.
- Navigation keys (next/previous) - this allows the operator to scroll to the next item in the selected list.
- Day mode key - this key is used to increase the screen display brightness.
- Night mode key - this key is used to decrease the screen display brightness.
- Fire - Fault - Dis key - this key is used to select the fire alarm, fault alarm and disablements.
- Lamp test - pressing the day and night keys together will illuminate the panel indicators and display an on-screen test card. If this test fails, it is an indication that there is a fault on the panel.

Test Control Panel Display

This function allows an operator to test the display and all indicators on the main control panel. To test the display, the procedure is:

Step	Task
1	Press MENU.
2	Select 5 SETTINGS.
3	Select 4 TEST DISPLAY.

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2.2.5 FIRE CONTROL AND SAFETY PLANS

Illustration 2.2.5a Fire Control and Safety Plan

SYMBOL	DESCRIPTION	SUNK DECK	STEERING GEAR ROOM	BOSUNSTORE	FWO PUMP ROOM	E.R. FLOOR	E.R. 4TH-FLAT	E.R. 3RD-FLAT	E.R. 2ND-FLAT	PASSAGEWAY	MAIN DECK TRUNK DECK	A DECK	B DECK	C DECK	D DECK	NAV. DECK	COMP. DECK	SUM																			
	Horn fire alarm 火警铃								1	4	9	2	6	3	2			27	 Lifejacket 救生衣								2	8	8	9	16	13	2		58		
	Space or group of spaces monitored by flame detector(s) of fixed local application water fire-extinguishing system 由火焰探测器监控的固定水喷淋保护的处所或处所群						1	7	2		1		1					12	 Lifebuoy with light and smoke 带灯和烟雾信号的救生圈													2		2			
	Space or group of spaces monitored by heat detector(s) 由感温探测器监控的处所或处所群					1	1	1	3		4	3	1	2				16	 LIFEBOUY WITH LIGHT&LIFELINE 带灯和救生索救生圈									2						2			
	Space or group of spaces monitored by smoke detector(s) of fixed local application water fire-extinguishing system 由感烟探测器监控的固定水喷淋保护的处所或处所群						1	7	2		1		1					12	 Lifebuoy with light 带灯救生圈	2								4						6			
	Fuel pump(s) remote shut-off 燃油泵遥控关闭								2		1	2				1		6	 Lifebuoy with line 带索救生圈									2						2			
	Lube oil pump(s) remote shut-off 滑油泵遥控关闭								2		1	2				1		6	 Lifebuoy 救生圈									4						4			
	Remote control for emergency fire pump or fire pump supplied by the emergency source of power 应急消防泵或由应急电源供电的消防泵的遥控控制				1							1				1		3	 20P 20人救生筏									4						4			
	Emergency fire pump 应急消防泵(220m3x1.0Mpa)				1													1	 6P 救生筏(6人)									1						1			
	Remote control for fire pump(s) 消防泵遥控控制					2						2				2		6	 Lifeboat/rescue boat 救生艇/救助艇									1						1			
	Ventilation remote control or shut off for Accommodation and service spaces 起居处所和服务处所的通风遥控控制或关闭								1		1					1		3	 Lifeboat 救生艇									1						1			
	Ventilation remote control or shut off for Machinery spaces 机器处所的通风遥控控制或关闭								2		1	2				1		6	 Fire and Safety Plan 防火与救生安全图									2						2			
	TWO-WAY VHF RADIO TELEPHONE 双项VHF对讲机															3		3	 B-class hinged self-closing fire door B 级自闭式铰链防火门									3	9	4	3			19			
	Emergency Position-indicating Radio Beacon(EPRIB) 应急无线电示位标															1		1	 A-class hinged self-closing fire door A 级自闭式铰链防火门					2	2	2	12	8	9	4	3	2	3		47		
	Emergency telephone 应急电话		2			2			1		3		1	2	1			12	 B-class hinged fire door B 级铰链防火门										1	8	18	13			40		
	Search And Rescue Transponder 搜索和救助应答器											1				1		2	 A-class hinged fire door A 级铰链防火门								2	2	6	3	4			17			
	Safety Equipment 安全装备										3							3	 Secondary escape route 次脱险路线																		
	Fireman's outfit & locker 消防员装备及箱		1								4	2						7	 Primary escape route 主脱险路线																		
	Muster Station 集合站											2						2	 B-class division B 级分隔																		
	Stretcher 担架										1							1	 A-class division A 级分隔																		
	Embarkation ladder or alternative approved device 登乘梯											4						4	SYMBOL	DESCRIPTION	SUNK DECK	STEERING GEAR ROOM	BOSUNSTORE	FWO PUMP ROOM	E.R. FLOOR	E.R. 4TH-FLAT	E.R. 3RD-FLAT	E.R. 2ND-FLAT	PASSAGEWAY	MAIN DECK TRUNK DECK	A DECK	B DECK	C DECK	D DECK	NAV. DECK	COMP. DECK	SUM
	Line throwing appliance 抛绳设备															1		1																			
	Rocket parachute flares 火箭降落伞火焰信号															12		12																			
	Emergency escape breathing device (EEBD) 应急逃生呼吸装置					2	2	2	4			1	1		1			13																			
	Emergency escape breathing device (EEBD) /Training 应急逃生呼吸装置/训练用															1		1																			
	Immersion suit 救生服								2		8		9	16	13	10		58																			

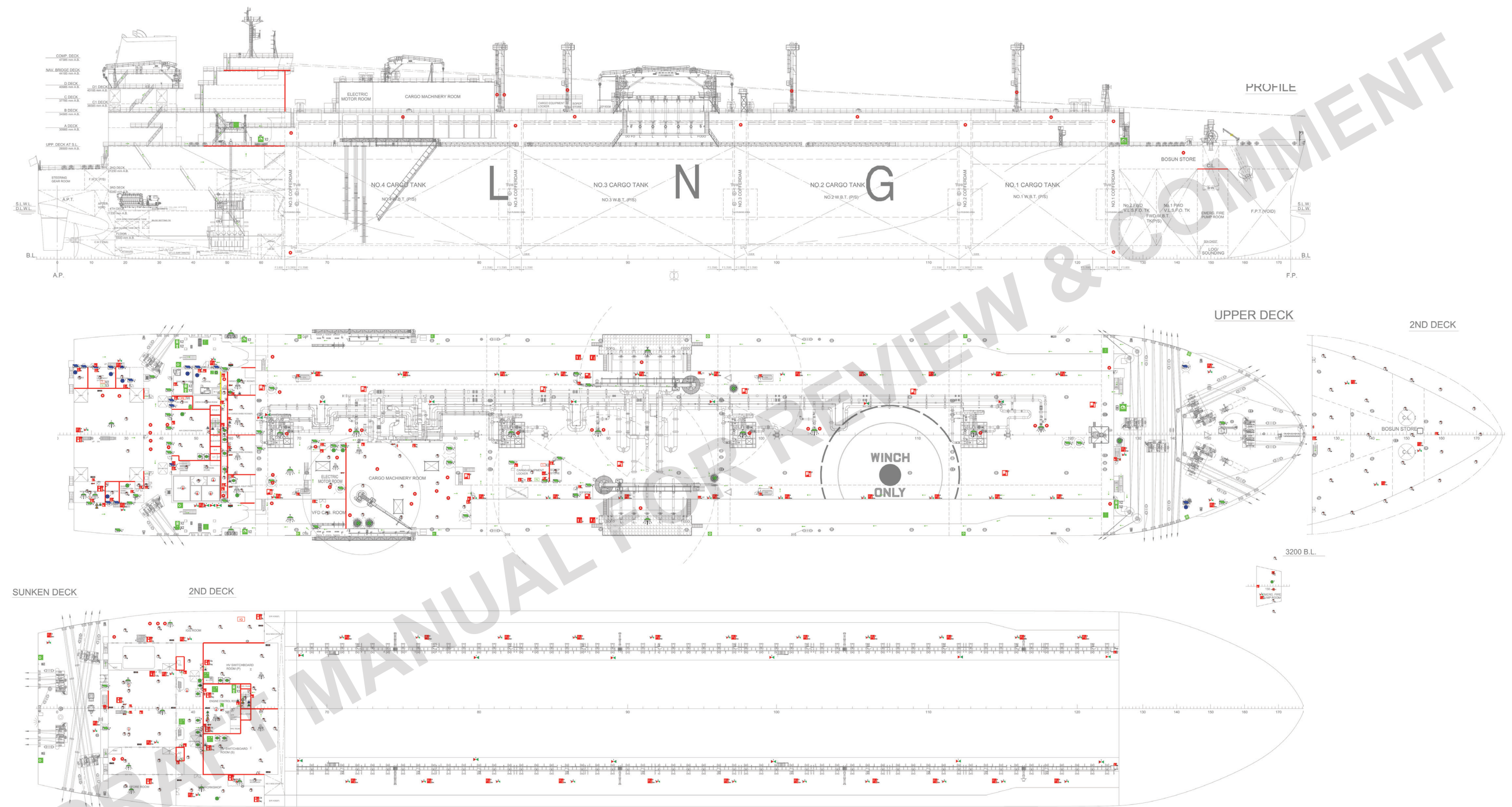
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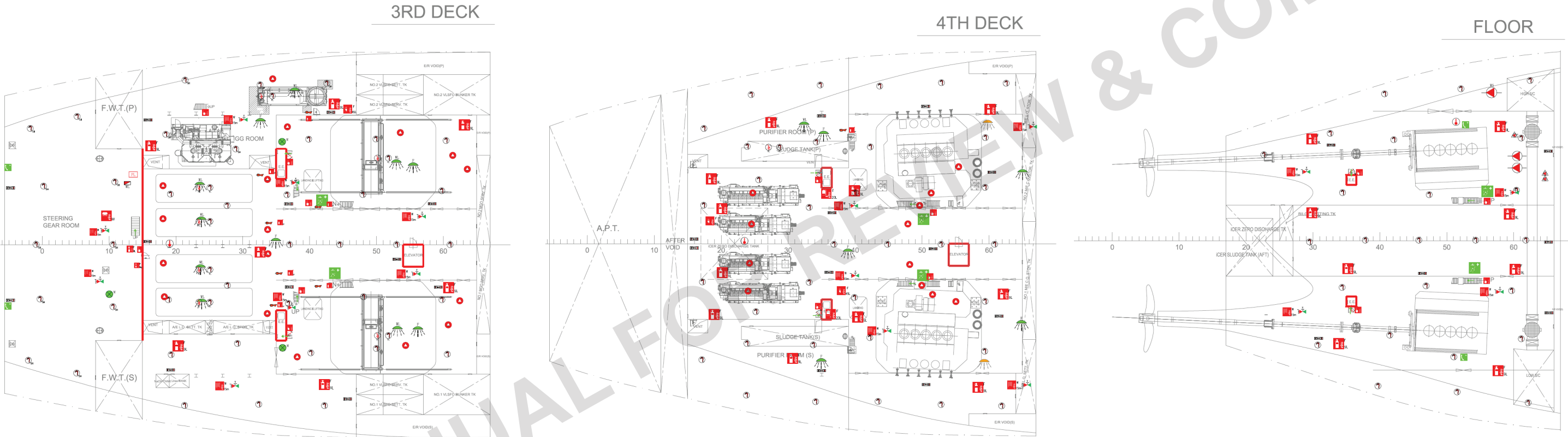
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Illustration 2.2.5c Fire Control and Safety Plan



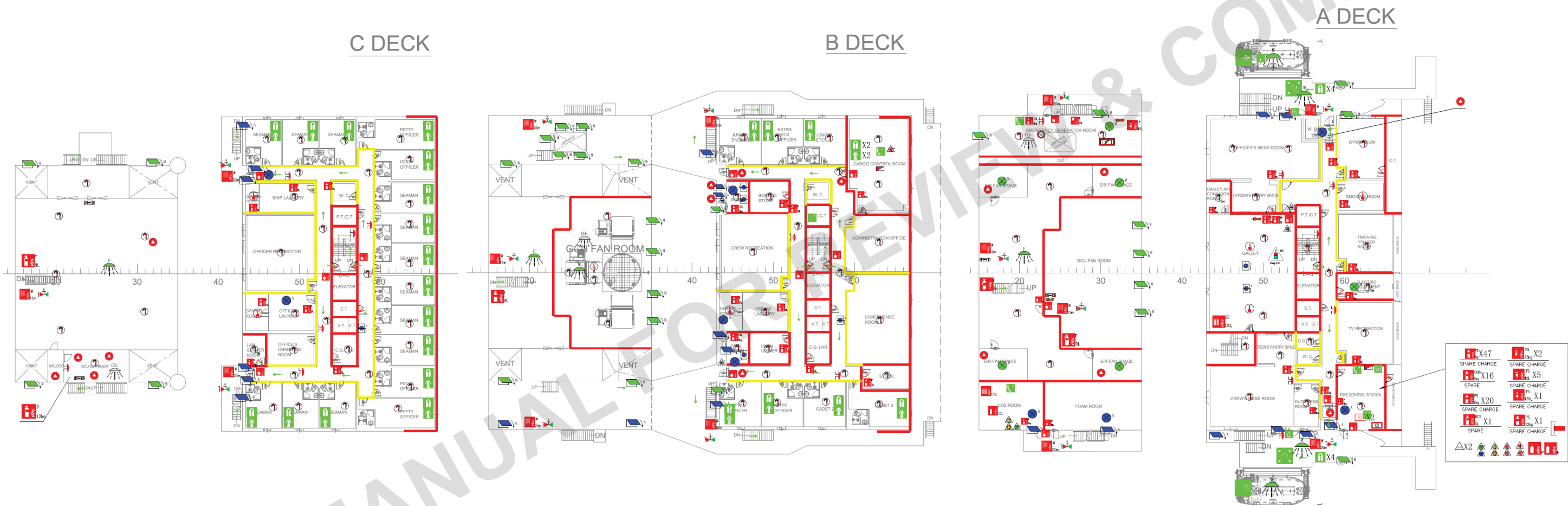
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Illustration 2.2.5d Fire Control and Safety Plan



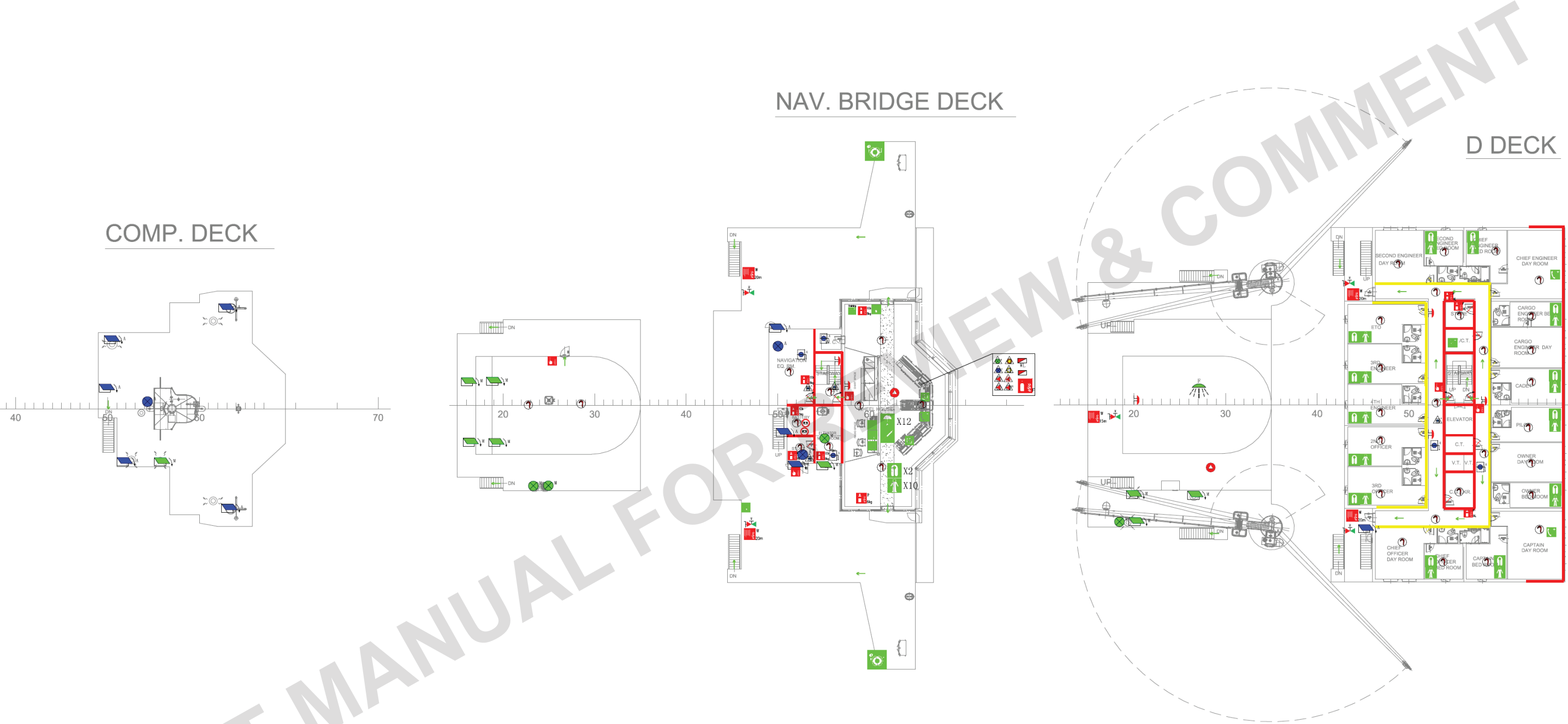
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Illustration 2.2.5e Fire Control and Safety Plan



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Illustration 2.2.5f Fire Control and Safety Plan



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2.2.6 SELF-CONTAINED AIR BREATHING APPARATUS

General Description

The Breathing Air Compressor is a high-pressure, electrically driven compressor unit designed to supply clean, breathable air for filling SCBA cylinders and other emergency breathing equipment onboard.

The installed unit is a COLTRI EASY STANDARD model, powered by a 380 V three-phase electric motor operating at 60 Hz. The compressor is compact, skid-mounted, and designed for reliable marine operation.

The system compresses atmospheric air through a multi-stage process and delivers it at high pressure through a triple filtration system, ensuring that the air quality is suitable for breathing applications.



Main Technical Data

- Manufacturer / Model: COLTRI – EASY STANDARD
- Power supply: 380 V, 3-phase, 60 Hz
- Motor power: 4.0 kW
- Free air delivery: approx. 235 L/min
- Working pressure: 150 to 350 bar
- Filling capability:
 - Approx. 6–8 minutes to fill a 10 L cylinder to 200 bar
- Weight: approx. 95 kg
- Noise level: approx. 80.7 dB
- Dimensions: approx. 100 × 46 × 73 cm
- Certification: DNV

Construction and Components

The compressor consists of the following main components:

- Electric motor driving the compressor block
- Multi-stage compressor unit for progressive air compression
- Intermediate and final coolers (stainless steel) to reduce air temperature
- Condensate separation system to remove moisture and oil
- Triple filtration system ensuring breathable air quality
- Adjustable safety valve (150–350 bar range)
- Pressure gauges and control fittings
- Two filling connections with valves and two 1.2 m high-pressure hoses
- Steel frame with protective coating, including handle and installation base

Operating Principle

The compressor draws in atmospheric air and compresses it in several stages to the required high pressure. During compression:

- Air is cooled between stages to improve efficiency
- Moisture and oil are removed in separators
- Final filtration removes remaining contaminants

The compressed air is then delivered through filling hoses into SCBA cylinders.

Cylinder Filling Arrangement

The compressor is equipped to fill cylinders directly:

- Two filling connections allow simultaneous filling of two cylinders
- Standard cylinders of 6–8 L capacity (steel or composite) can be used
- Filling pressure is adjustable up to the required working pressure

Air Quality and Filtration

The compressor includes a triple filtration system, which removes:

- Moisture
- Oil vapour
- Particulate contaminants

This ensures that the compressed air is suitable for breathing purposes and meets applicable safety requirements.

Safety Features

The compressor is fitted with the following safety features:

- Adjustable safety valve to prevent overpressure
- Cooling system to control air temperature
- Pressure gauges for monitoring inlet and outlet pressure
- Robust steel construction for durability

Additionally, when used with a filling cabinet (if provided):

- Explosion-protected enclosure
- Mechanical locking system during filling

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- Automatic stop when opening the cabinet door
- Pressure regulation for safe cylinder filling

Operational Use

The breathing air compressor is used onboard for:

- Filling SCBA cylinders for fire fighting teams
- Supporting emergency response operations
- Providing breathable air supply for safety equipment

During operation:

- Air intake shall be from a clean, uncontaminated location
- Pressure and temperature shall be monitored
- Cylinders shall be filled in accordance with safety procedures

Summary

The COLTRI EASY STANDARD breathing air compressor provides a reliable source of high-pressure breathable air. It combines multi-stage compression, effective cooling, and triple-stage filtration to ensure safe and efficient filling of breathing apparatus cylinders onboard.

Procedure for Operation of the Breathing Air Compressor

Step	Task
1	Check the compressor oil level and top up if required using the manufacturer-approved Coltri compressor oil.
2	Open the condensate drain valves on the separators/filters and blow through to ensure they are clear.
3	Ensure all drain valves are closed prior to starting the compressor.
4	Ensure the air intake area is clean and free from contaminants.
5	Connect the air cylinders to the filling connections using the provided high-pressure hoses. Ensure all connections are secure.
6	Open the cylinder valves slowly.
7	Start the electric motor and allow the compressor to run.
8	Monitor compressor operation, including pressure and temperature, and check for abnormal noise or vibration.
9	During operation, periodically open the condensate drain valves (every 15–30 minutes, depending on humidity) to remove accumulated moisture and oil.
10	Continue filling until the required cylinder pressure is reached.
11	Upon completion of filling: • Stop the compressor • Close the cylinder valves • Slowly vent pressure from the filling line • Disconnect the cylinders
12	Open condensate drain valves briefly to release any remaining condensate.
13	Record compressor running hours in the log.
14	Replace the filter cartridge (triple filtration system) in accordance with the manufacturer’s maintenance instructions.

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2.2.7 EMERGENCY ESCAPE BREATHING DEVICES

Manufacturer:	Jiangsu Huayan Marine Equipment CO.,Ltd.
Type:	Self-contained compressed oxygen
Model:	TH15B
No. of sets:	4
Duration:	≥15 minutes

Introduction

The vessel carries 4 Emergency Escape Breathing Devices (EEBDs) and one set for training purpose.

Chapter II-2 of the SOLAS Amendments prescribe the following:

- 4.3Emergency escape breathing devices.
- 4.3.1On all ships, within the machinery spaces, emergency escape breathing devices shall be situated ready for use at easily visible places, which can be reached quickly and easily at any time in the event of fire. The location of emergency escape breathing devices shall take into account the layout of the machinery space and the number of persons normally working in the spaces.
- 4.3.2The number and location of the devices shall be indicated in the Fire Control Plan required in Regulation 15.2.4.
- 4.3.3Emergency escape breathing devices shall comply with the Fire Safety Systems Code.

The Emergency Escape Breathing Device (EEBD) TH15B is a portable breathing apparatus designed to provide a short-term supply of breathable air to enable personnel to escape from hazardous atmospheres.



The unit consists of a compressed air cylinder, a pressure regulator, and a hood with an integrated breathing system, all contained within a compact carrying bag for quick access.

The air cylinder is manufactured from alloy steel with a capacity of approximately 3.0 litres and is charged to a working pressure of 210 bar. When activated, the unit supplies breathable air for a minimum duration of 15 minutes, depending on the user’s breathing rate.

The EEBD is of the hood type, allowing it to be worn quickly without the need for face seal adjustments. The hood fully encloses the head and provides a continuous flow of air, helping to protect the wearer from smoke and toxic gases during escape.

The device is lightweight (approximately 6.5 kg or less) and compact, making it suitable for installation at strategic locations throughout the vessel. It is typically stored in a protective carrying case and is ready for immediate use.

The EEBD complies with relevant international standards, including SOLAS 1974 (as amended), IMO FSS Code, EN 1146, and ISO 23269-1, and is certified accordingly (MED approved).

The unit is intended for escape purposes only and shall not be used for fire fighting or rescue operations.

Donning Procedure – Emergency Escape Breathing Device (EEBD)

The EEBD is provided for emergency escape and supplies breathable air for a minimum of 15 minutes.

1. Removal and Opening

Step	Task
1	Remove the EEBD from its storage position.
2	Open the carrying bag and take out the cylinder and hood assembly.

2. Activation of Air Supply

Step	Task
1	Open the cylinder valve fully by turning it anticlockwise.
2	Confirm that air is flowing to the hood.

3. Donning of Hood

Step	Task
1	Remove the hood from the bag and place it over the head.
2	Ensure that: - The hood fully covers the head - The neck seal is properly positioned and not twisted - Hair and clothing do not interfere with the seal

4. Breathing Check

Step	Task
1	Breathe normally and confirm a continuous air supply.
2	Hold the cylinder or carry it using the shoulder strap while moving.

5. Escape

Step	Task
1	Proceed immediately to the nearest safe escape route.
2	Keep calm and avoid unnecessary exertion to conserve air.

6. After Use

Step	Task
1	Remove the hood only when in a safe atmosphere.
2	Close the cylinder valve if required.
3	Do not reuse the unit once activated.
4	Report use for replacement.

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Maintenance

Monthly

The anti-tamper tag should be checked to ensure it is intact and the gauge reading compared to the previous check to ensure that it remains full.

Annually

In addition to the monthly test check, the carrying case, hood and exhalation valve should be checked annually.

WARNING
EEBD is to be used only for lifesaving purposes and not for fire fighting as the SCBA.

- 3.1 Fire Fighting Systems and Procedures
 - 3.1.1 Fire And Deck Main Systems
 - 3.1.2 Water Spray System
 - 3.1.3 High Expansion Foam Fire Fighting System
 - 3.1.4 Engine Room Local Fire Fighting (Water Mist) System
 - 3.1.5 CO2 Fire Extinguishing System
 - 3.1.6 Dry Powder System
 - 3.1.7 Emergency Shut-Off Valve And Fan Damper System
 - 3.1.8 General Fire Safety Practices
 - 3.1.9 Fire Fighting Techniques And Procedures

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3.1 FIRE FIGHTING SYSTEMS AND PROCEDURES

3.1.1 FIRE AND DECK MAIN SYSTEMS

Fire and Water Curtain Pump

Manufacturer:	Shinko Ind., Ltd.
Type:	Vertical centrifugal self-priming
Model:	SVW200MEJ
No. of sets:	1
Capacity:	180m³/h at 10.0 bar
Motor:	440V, 110kW at 1,800 rpm

Emergency Fire Pump

Manufacturer:	Shinko Ind., Ltd.
Type:	Vertical centrifugal self-priming
Model:	KV300KS
No. of sets:	1
Capacity:	725m³/h at 10.0 bar
Motor:	440V, 500kW at 1,800 rpm

Fire Line Pressurising Pump (Jockey Pump)

Manufacturer:	Shinko Ind., Ltd.
Type:	Vertical centrifugal self-priming
Model:	SHQ50M
No. of sets:	1
Capacity:	2.0m³/h at 10.0 bar
Motor:	440V, 5.5kW at 3,600 rpm

Bilge, Fire and General Service Pump

Manufacturer:	Shinko Ind., Ltd.
Type:	Vertical centrifugal self-priming
Model:	SVW200MEJ
No. of sets:	2
Capacity:	150/260m³/h at 10.0/3.5 bar
Motor:	440V, 110kW at 1,800 rpm

Introduction

The sea water fire main and deck wash system is maintained under a constant pressure by a 2.0m³ hydrophore unit, and its associated fire line pressuring pump, the pump is supplied from the emergency switchboard. The hydrophore unit is used to maintain the system pressure and accommodate and replace any small losses in the system. By this means, the fire main and deck wash system is maintained under operational pressure to supply water to any of the hydrants. The fire pump is supplied from the main switchboard(s) group panel LGSP-3

The bilge, fire and general service pumps, plus the fire line pressuring pump, are located in the engine room with both pumps supplied from the main switchboards group panel LGSP-3 for pump No.1, and pump No.2. The emergency fire pump which is supplied from the emergency switchboard is located in the forward pump room which is designated as an enclosed space, therefore the space must be ventilated and the atmosphere tested before the space can be entered. Pumps in the engine room for fire fighting duties take suction from the main sea water crossover line. The emergency fire pump has its own designated sea chest in the forward space.

The supply of fire water from the fire pump/ line pressurising pump is supplied to the accommodation and deck hydrants via the engine room fire water riser and isolating valve (EAV301). Prior to splitting to the port and starboard risers and associated isolating valves to the deck ring main system installed valve EAV302 (N.O.). The deck fire water ring main system comprises of the following section transverse and longitudinal isolating valves enabling the fire main to remain in service in the event of a section failure:

- Accommodation piping system isolating valve EAV303 (N.O.) sited aft of the accommodation.
- Port isolating valve (EWV69) sited on the upper deck port side of the accommodation.
- Starboard isolating valve (EWV47) sited on the upper deck starboard side of the accommodation.
- Port longitudinal isolating valve (EWV68) sited in the passageway (P) adjacent No.4 trunk.
- Starboard longitudinal isolating valve (EWV46) sited in the passageway (S) adjacent No.4 trunk.
- Port longitudinal isolating valves (EWV67/66) sited in the passageway (P) adjacent No.3 trunk.
- Starboard longitudinal isolating valves (EWV45/44) sited in the passageway (S) adjacent No.3 trunk.
- Port longitudinal isolating valve EWW65) sited in the passageway (P) adjacent No.2 trunk.
- Starboard longitudinal isolating valve (EWV43) sited in the passageway (S) adjacent No.2 trunk.
- Port longitudinal isolating valve (EWV64) sited in the passageway (P) adjacent No.1 trunk.
- Starboard longitudinal isolating valve (EWV42) sited in the passageway (S) adjacent No.1 trunk.

The deck fire water supplies various hydrants and fire water to the following utilities:

- Port and starboard deluge to the respective cargo manifolds.
- Fire water supply to the port and starboard eductor driving lines in the forward bosuns store.
- Port and starboard anchor washing.
- Fire water to the port and starboard bilge eductors for passageways

Firewater and Line Pressurising Pump Operation

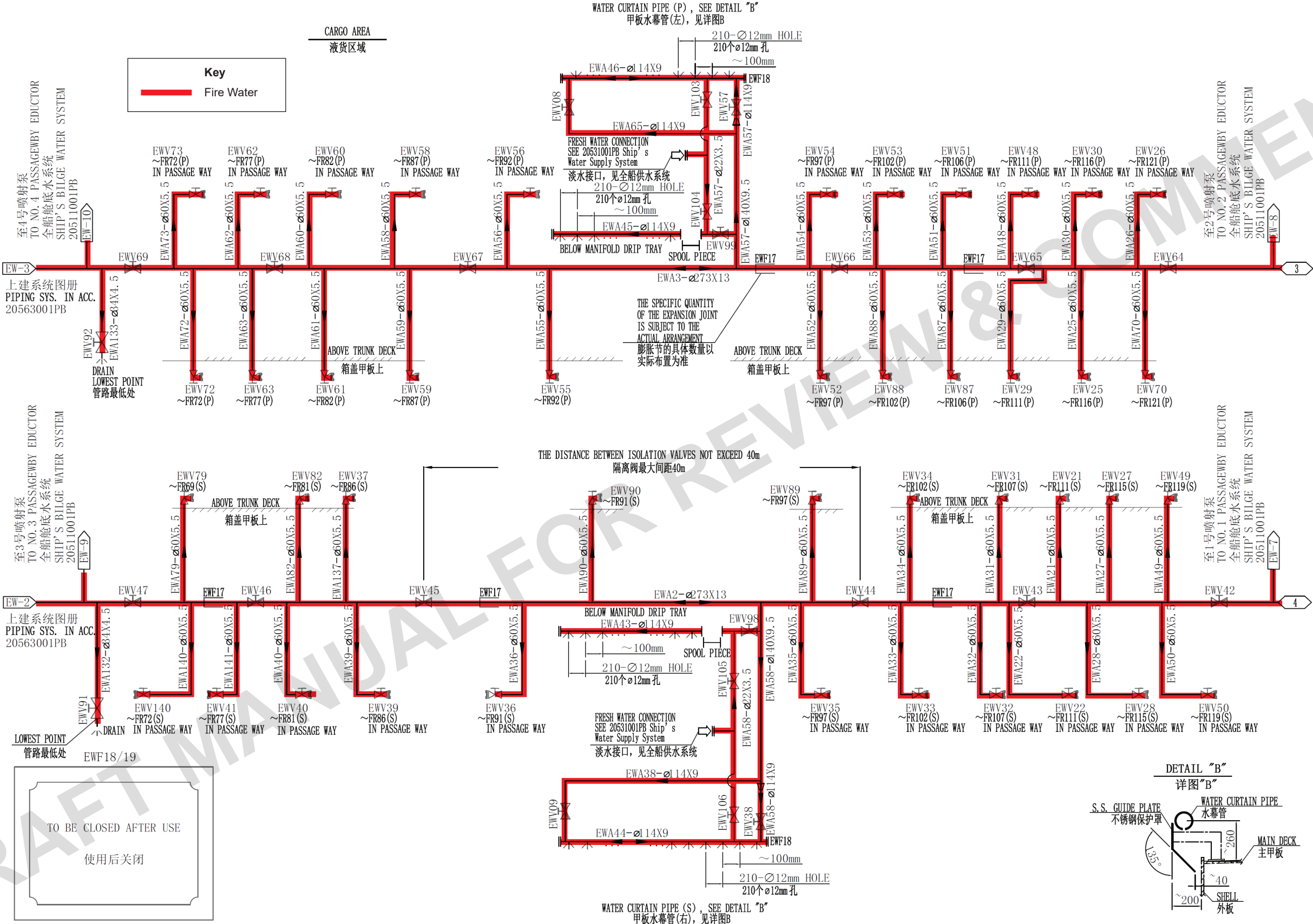
The firewater pressure within the ring-main system is constantly maintained automatically by a firewater line pressuring pump, taking suction from the seawater systems transverse manifold and delivery the firewater to the fire main. When the fire water pressure falls to a pre-determined pressure of 10.0 bar initiated by pressure transmitter (WMF200), the line pressuring pump starts with the pump stopping when the pressure increases to 11.5 bar.

In event of a fire incident, the opening of the fire hydrant will cause a rapid water pressure fall in the fire line pressuring pumps discharge to the fire main, where the line pressurising pump cannot compensate, and the lower water limit of the pressure falls to 9.0 bar activating a low pressure alarm. The low pressure of 8.0 bar starts the fire pump and provides the required high flow for fire suppression.

It should be noted the emergency fire pump can be started manually via the IAS. In addition, the fire water pump can if required be operated manually from their local start (start/ stop) from the bridge and from the IAS. Its should be noted that No.1 fire, bilge and GS pump inlet valve (BMV58) and outlet valve (BMV03) are ‘normally open’.

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Illustration 3.1.1b Ship's Water Fire System (ii)



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Note: It is essential that all fire hoses are stored correctly and that the nozzles are located with the hoses. All items must be maintained in serviceable condition at all times.

For routine wash deck requirements, a bilge, fire and general service pump will be started manually via the IAS, with the fire main then being bled through one or more of the hydrants or hawse pipes. When any of the pumps which can be used to supply the fire main and deck wash system are running, at least one of the hydrants must remain open at all times to ensure there is always a flow of water through the pump’s casing to prevent overheating.

Pressurising the Fire Main Using the Pressurising Pump and Hydrophore Unit

It has been assumed in the following description that the fire main has already been flooded before starting the hydrophore unit.

Step	Task
1	Ensure that the sea water suction main is flooded with either the high or low sea suction valve open.
2	Open the fire line pressuring pump suction valve (WMV205) from the transverse sea water main, and the pump discharge valve (WMV204). Vent the sea water hydrophore tank, and using the pressuring pump in manual control, fill the hydrophore unit until the water level is approximately 75% in the gauge glass.
3	Connect the compressed air supply hose to the hydrophore and open the valve to the compressed air connection and pressurise the accumulator tank to a pressure of approximately 12 bar by slowly opening the hydrophore compressed air inlet valve. On completion close the valve.
4	Set the line pressurising pump to AUTO on the IAS screen. Open (slowly) the hydrophore unit outlet valve WMV201 when the pressure in the hydrophore unit reduces to 10 bar, the pump will start, the pump will automatically stop when the pressure reaches 11.5 bar. Water/air can be drained/vented as necessary to establish the correct working level of approximately 50%. Open a high deck level hydrant to remove any entrapped air within the fire main system until the line pressuring pump/ hydrophore pressure stabilises.
5	The sea water fire main hydrophore unit is now operational. The pump will now operate and start and stop automatically on system pressure and maintain the pressure in the fire hydrant system.

The fire line pressurising pump will now maintain the fire main pressure between the upper and lower limits. No.1 bilge, fire and general service pump is normally left with their suction and discharge valves open, the pressure in the fire main developed by the pressurising pump will be retained, because the bilge, fire and general service pumps have been fitted with a non-return valve on their discharge sides.

Note: Valves which are normally open or closed on any pump must be maintained in a good working condition. These valves must be lubricated and should be tested at regular intervals to ensure they operate correctly. After testing, the valve must be returned to its normal open or closed position and a check made to ensure the valve is correctly set.

Preparation for Operation of the Fire Hydrant System

Prior the start-up procedure all hydrant valves are assumed to be closed.

Step	Task
1	Set the suction and discharge valves of the designated emergency fire pump for fire main duties. Confirm the suction and discharge valves of the emergency fire pump for supplying sea water to the fire main.
2	All of the manual intermediate section isolating valves along the fire main on the main deck, plus the section isolating valves for the aft deck area, accommodation (port and starboard) and the forward deck area, must be open.
3	Ensure all the pressure gauge and instrumentation valves are open.
4	Confirm the control air is available to both Nos.1 and 2 fire, bilge and GS pumps

Step	Task
5	Align the system valves accordingly.

No.1 Bilge, Fire and General Service Pumps

The bilge, fire and general service pumps will normally be set for suction from the sea water crossover main and discharge to the fire main with the valves set as in the following table.

Pump Starting Locations

The two bilge, fire and general service pumps have their electrical supply from the main switchboard and can be started from the following locations:

- Locally.
- Fire control station/foam room.
- Wheelhouse.
- ECR and CCR via the IAS workstations.

Although the bilge, fire and general service pumps can be started and stopped locally, or manually from the IAS screen, it is normal practice to set one pump to automatic standby cut-in via the IAS screen so that a pump will cut-in automatically if the fire main pressure falls to 6 bar.

Emergency Fire Pump

The emergency fire pump is located at the base of the forward pump room, and is powered by a electric drive motor. A vertical self-priming centrifugal pump which can be started locally or remotely from the fire control room, the bridge and the IAS workstations in the ECR and CCR. The power supply is taken from the emergency switchboard.

The emergency fire pump has its own sea chest and associated sea water suction valve EWV01 which is locked open, so when the pump is started, it will supply water to the fire main immediately. The pump non-return discharge valve EWV81 is also locked in the open position. The discharge from the emergency fire pump links into the fire main. Before starting the emergency fire pump, its vacuum pump should be started first, approximately 10 seconds later the emergency fire pump should be started.

Note: The emergency fire pump can be started locally. If it is to be started remotely, the MODE CONTROL local selector switch.

Operation of the Fire Main

The outlets on the fire main are opened as required and the line isolating valves on the fire main are normally maintained in the open position so that hydrants will receive water immediately they are opened. Fire hydrant valves should be test opened at least once per month to ensure that they are operational.

Fire mains are located in the port and starboard side passages on the forward deck with line valves for isolation being set at predetermined intervals.

The fire main valves are as in the following tables:

Engine Room Fire Fighting Hydrants

Description	Hydrant Number
Engine room floor hydrants	WMV206, WMV207, WMV208, WMV209
Engine room 4th deck hydrants	WMV210, WMV211, WMV212, WMV213, WMV214, WMV215

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Description	Hydrant Number	International Shore Connections
Engine room 3rd deck hydrants	WMV216, WMV217, WMV218, WMV219, WMV220, WMV221, WMV222	The fire main system has been supplied with an international shore connection that will allow shore side fire appliances to connect onto the ship's fire main. The international shore connection is kept in the fire control station.
Engine room 2nd deck hydrants	WMV223, WMV224, WMV225, WMV226	
Engine room upper deck hydrants	WMV227	
Engine room casing, A deck level	WMV228, WMV229	
Engine room casing, B deck level	WMV230	
Engine room casing, C deck level	WMV231	
Engine room casing, D deck level	WMV232	

Engine Casing External Fire Fighting Hydrants

Description	Hydrant Number
Sunk deck fire hydrants	EWV19, EWV96
Steering gear room fire hydrants	EWV20, EWV101
Out of engine casing, upper deck fire hydrants	EWV10, EWV11
Out of engine casing, A deck fire hydrants	EWV12, EWV13
Out of engine casing, B deck fire hydrants	EWV14, EWV15

Accommodation Block Fire Hydrants

Accommodation	Hydrant Number
Upper deck hydrants	EAV308, EAV309
A deck hydrants	EAV310, EAV311
B deck hydrants	EAV312, EAV313
C deck hydrants	EAV314, EAV315
D deck hydrants	EAV316, EAV3179
Navigation bridge hydrants	EAV318, EAV319

Note: The hydrant BF-90A fire water line has a boss with pressure gauge and stop valve. In addition a similar arrangement is available at the forward hydrant (BF-05F) on the focsle deck.

Main Deck Fire Main Section Isolating Valves

For full details in the 'Main Deck Fire Main Section Isolating Valves' reference the listing in the introduction.

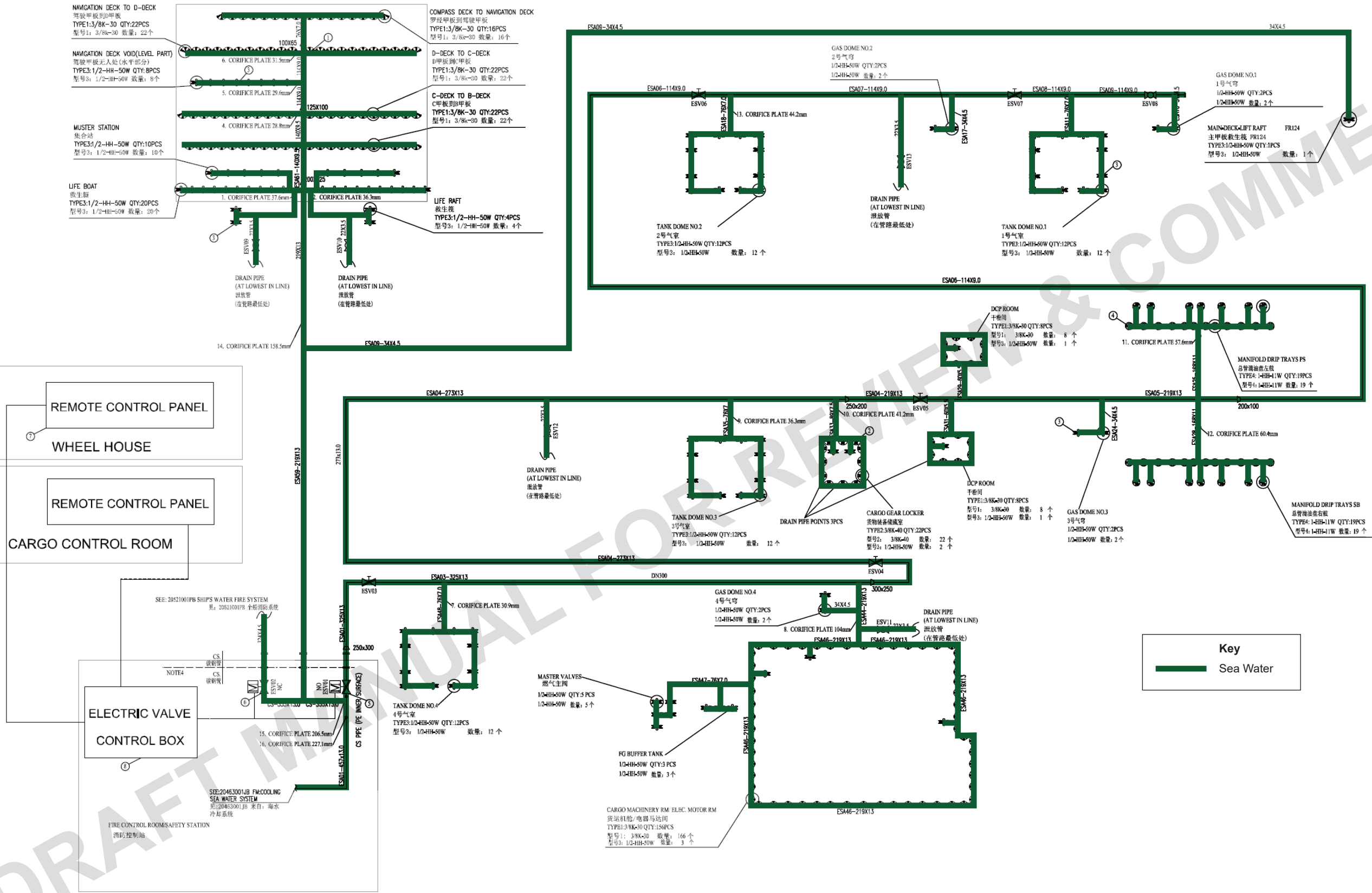
Note: If the vessel is trading in very cold climatic conditions where it is possible that the deck main could freeze, it will be necessary to drain the fire main down, and leave it in a dry condition and close the isolation valves.

Water Spray System

In an emergency, the bilge, fire and general service pumps operating on the fire main system can be cross-connected to the water spray system via crossover (NC) valve, although their flow rate is quite small in comparison to the spray pump. This line is primarily used to flush the water spray line with fresh water via No.1 bilge, fire and general service pump drawing from fresh water tank.

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Illustration 3.1.2a Deck Sea Water Spray System



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3.1.2 WATER SPRAY SYSTEM

Water Spray Pump

Manufacturer:	Shinko Industries Ltd
Type:	Electric, motor-driven, vertical centrifugal
Model:	KV350KEJ
No. of sets:	1
Capacity:	1,600 m³/h at 1.0 MPa at 32°C
Power output:	700 kW
Rating:	6,600V, 60Hz at 1,800 rpm
Spray rating:	10 litres/min for horizontal protected surfaces
Spray rating:	4 litres/min for vertical protected surfaces
Location:	Engine room lower floor

System Hydraulic Characteristics

The deck spray system hydraulic design is based on the following parameters:

Design inlet pressure at nozzle	4 bar
System design pressure	16 bar
Minimum required flow	1202.7 m³/h
Hydraulic calculation basis	10 bar fire main pressure

The purpose of the water spray system is to provide cooling and fire protection for the vessel on or near the trunk deck. The water spray system can be divided into two sections, allowing selective operation depending on the location of the fire hazard.

This arrangement allows cooling protection to be applied to the affected areas without unnecessary operation of the entire system.

Section 1 – Accommodation and Upper Deck Protection

Section, which covers the accommodation front wall area and a further 3m of those sides and the lifeboat and liferaft muster station facing the cargo area and forward life raft area.

Protected areas include:

- Compass deck and navigation deck
- Accommodation decks
- Muster stations
- Lifeboat embarkation areas
- Liferaft stations

Cooling of these areas ensures protection of the accommodation structure and maintains safe evacuation routes during an emergency.

Section 2 – Cargo Deck and LNG Equipment Protection

Section provides cooling protection for cargo handling equipment and LNG transfer systems.

Protected areas include:

- Cargo manifolds
- Cargo pipelines

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- Tank domes
- Gas domes
- Cargo machinery room bulkheads
- Cargo gear locker
- Fuel gas buffer tank area

These areas require higher spray densities due to the presence of LNG equipment and potential fire hazards.

The water spray system is supplied from a single dedicated pump and because only one pump has been fitted, provision has been made for cross-connection to the water spray system with the fire main system if the water spray pump is not available.

It should be noted that the capacity of the bilge, fire and general service pumps are considerably lower than that of the water spray pump, therefore the primary emergency supply would be provided by the emergency fire pump which is rated at 725 m³/h at 1.45 MPa.

The trunk deck spray line can be isolated by isolation block valves along its length, and each branch line leading to the spray nozzles can be further isolated by isolation valves. Each spray line is fitted with a series of spray nozzles designed to give a spray pattern and the number required has been calculated at the design stage to meet the required needs.

Pump Starting Locations

The water spray pump has its electrical supply from the main switchboard and can be started from the following positions:

- Locally.
- Fire control station.
- Wheelhouse.
- Engine control room via the IAS.

The water spray pump takes its suction from the sea via the engine room crossover main. The water spray pump discharges sea water through a single 457mm diameter pipeline via a sea water spray filter arrangement and onto the water spray system onto the deck.

To maintain the water spray system in the standby condition, the suction valv WCV79 and discharge valve WSV81 of the water spray pump and all the manually operated isolating valves on the groups are normally left in the open position, except the drain valves.

After use, a flushing facility is fitted to the sea water spray filter for fresh water rinsing, using a dedicated fresh water supply line from port side fresh water tank with check valve WSV83; this has a designated wash rinsing cycle which is manually started and stopped. Washing drains lead bilge well, this is not used for water spray line rinsing.

The bilge, fire and general service pumps are used for fresh water rinsing and flushing the lines and nozzles with fresh water following a test or operation of the system. To facilitate this, the pump suctions have a connection to the fresh water tanks. Rinsing the system after use will reduce the amount of internal corrosion to the lines and prevent the nozzles from becoming clogged with saline crystals.

Two valves, ESV01 & ESV02 located in Fire Control Station are operated remotely from the fire safety control box in the FCS on upper deck, or remotely from Cargo Control Room and Wheel House. Section 2 sea water supply valve ESV01 is normally open and valve connecting fire system with deck spray system ESV02 - normally closed.

After use, conduct a full fresh water rinse of all lines and nozzles and drain down through the drain valves. The water spray pump filter should be correctly backflushed with a fresh water rinse cycle.

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Spray Nozzle Arrangement

The spray system uses several types of spray nozzles selected according to the required spray capacity.

Nozzle Type	Quantity
3/8K-30	264 pcs
3/8K-40	22 pcs
1/2-HH-50W	111 pcs
1-HH-11W	38 pcs

These nozzles provide uniform spray coverage over the protected areas.

Spray Nozzle Description

3/8K-30

Used primarily for structural cooling of bulkheads and deck structures where moderate spray density is required.

Typical installation locations:

- Accommodation bulkheads.
- Cargo machinery room bulkheads.
- Deck structural surfaces.

3/8K-40

Provides increased discharge capacity compared with the K-30 nozzle and is used for larger structural areas requiring improved cooling coverage.

1/2-HH-50W

Medium-to-high capacity nozzle installed in areas requiring increased spray density such as:

- Tank domes.
- Gas domes.
- Safety critical equipment areas.
- Muster stations.

1-HH-11W

High-capacity spray nozzle installed in the most critical cargo handling areas including:

- Cargo manifold stations.
- LNG transfer pipelines.
- Cargo valve clusters.

Spray Nozzle Materials

The spray nozzles are manufactured from stainless steel SUS316, providing high resistance to corrosion and ensuring reliable spray performance in the marine environment.

Section Activation Philosophy

The spray system is designed so that each section can be activated independently depending on the location of the fire hazard.

Section 1 Activation

Section 1 is activated when protection of the accommodation structure or evacuation areas is required.

Typical situations include:

- Cargo deck fire affecting accommodation structures.
- Radiant heat exposure to muster stations.
- Protection of lifeboat embarkation areas.
- Protection of liferaft embarkation areas including forward liferaft.

Section 2 Activation

Section 2 is activated when fire or excessive heat occurs in cargo handling areas including:

- Cargo manifold stations.
- LNG pipelines.
- Tank domes.
- Cargo valve stations.

Simultaneous Operation

During major emergency situations both sections may be activated simultaneously.

The installed spray pump capacity is sufficient to maintain the required spray density across all zones.

Zone Coverage Table

The spray system provides coverage for the following zones.

Section 1 Zones

Zone	Nozzles	Protected Area
Compass deck to Navigation deck	16	70 m²
Navigation deck to D deck	22	100 m²
Navigation deck void (port)	8	22 m²
D deck to C deck	22	100 m²
C deck to B deck	22	100 m²
Muster station (port)	5	12.6 m²
Muster station (starboard)	5	12.6 m²
Lifeboat area (port)	10	28 m²
Lifeboat area (starboard)	10	28 m²

Section 2 Zones

Zone	Nozzles	Protected Area
Cargo machinery room forward bulkhead	30	140 m²
Cargo machinery room port bulkhead	52	229 m²
Cargo machinery room starboard bulkhead	54	229 m²
Cargo machinery room aft bulkhead	30	140 m²
Cargo gear locker	22	142 m²
Tank dome No.4	12	33 m²
Tank dome No.3	12	33 m²

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Zone	Nozzles	Protected Area
Tank dome No.2	12	33 m²
Tank dome No.1	12	33 m²
Manifold area (port)	19	140.5 m²
Manifold area (starboard)	19	140.5 m²

System Operation

To activate the deck spray system:

1. Start the Deck Water Spray Pump.
2. Confirm pump discharge pressure.
3. Confirm water discharge from spray nozzles.

Emergency Supply from the Fire Main to the Water Spray System

In the event that the water spray pump is out of service, the supply can be provided by the emergency fire pump, plus either of the bilge fire and general service pumps, but at a lower designed flow rate of the water spray pump. The water supply is carried out by opening the crossover isolating valve ESV02 from the fire main onto the water spray delivery line.

Inspection and Maintenance

Routine inspection should include:

- Checking spray nozzles for blockage or damage.
- Verifying nozzle alignment.
- Inspecting spray piping for corrosion.
- Checking valve operation.
- Periodic testing of the spray pump.

Nozzles showing irregular spray pattern should be cleaned or replaced.

Safety Considerations

When the spray system is operating:

- Visibility on the cargo deck may be reduced.
- Deck surfaces may become slippery.
- Water accumulation may occur on deck.

Personnel should exercise caution when moving in spray protected areas.

SCHEMATIC DIAGRAM

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3.1.3 HIGH EXPANSION FOAM FIRE FIGHTING SYSTEM

Technical Specification

The High Expansion Foam Fire Fighting System is a fixed total flooding fire-extinguishing system designed for protection of enclosed machinery and auxiliary spaces onboard.

- System Type: High Expansion Foam (Total Flooding)
- Standards:
 - SOLAS Chapter II-2.
 - FSS Code (Chapter 3.1).
 - IMO MSC/Circ. 127.1
 - EN 13565-1.
- Foam Type: Synthetic high-expansion foam concentrate.
- Mixing Ratio: 2%.
- Foam Expansion Ratio: approx. 600–645:1.
- Foam Generator Types:
 - HG-25 (primary type used.)
 - Capacity: approx. 58–60 m³/min per unit.
 - Working pressure: approx. 6 bar.
- Total Installed Foam Generators: 138 (HG-25 type).
- Maximum Simultaneous Operation: 114 generators.
- Foam Tank Capacity: approx. 6300 litres.
- Foam Pump Capacity: approx. 12.3 m³/h.
- Fire Water Capacity: approx. 600–635 m³/h.
- System Pressure: approx. 9.3 bar at foam skid inlet.
- System Capacity:
 - Filling of largest protected space within approx. 10 minutes .
 - Foam supply for minimum 30 minutes or 5× space volume.

General Description

The High Expansion Foam System is intended for rapid extinguishing of fires in enclosed spaces by generating large volumes of foam that fill the protected space.

The extinguishing effect is achieved by:

- Smothering: displacement of oxygen from the fire zone
- Cooling: heat absorption by foam moisture
- Re-ignition prevention: maintaining a foam blanket over hot surfaces

Compared to CO₂ systems, the foam system is less hazardous to personnel during initial discharge and provides both cooling and inerting effects.

Protected Spaces

The system provides protection to the following spaces:

Main Machinery Spaces (Engine Room – Multi-Level)

- D Deck.
- C Deck.
- B Deck.
- A Deck.
- Upper Deck to A Deck trunk.
- 2nd Deck.
- 3rd Deck.
- 4th Deck.
- Above Floor (engine top area).
- Below Floor (tank top / bilge area).

Auxiliary Spaces

- Purifier Room (Port and Starboard).
- Engine Room Workshop.
- Electrical Workshop.
- Incinerator Room.
- IGG Room (3rd Deck).
- IGG Room (2nd Deck).

All spaces are arranged in separate foam zones, allowing selective operation.

System Arrangement

The system consists of the following main components:

- Foam generators located in protected spaces.
- Foam concentrate storage tank.
- Foam concentrate pump.
- Foam proportioner (2% mixing unit).
- Fire water supply (main or emergency fire pump.)
- Main foam distribution piping (DN200).
- Branch distribution lines (DN150 / DN80).
- Section control valves (MOV type).
- Main control panel (MCP).
- Alarm and shutdown system.

Foam generators are installed at ceiling level to ensure top-down filling of spaces.

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Principle of Operation

Foam Generation

Foam is produced by mixing water and foam concentrate and passing the solution through foam generators equipped with water-driven rotating blades, which draw in air and create high expansion foam.

Foam Expansion and Flooding

The foam expands up to 600 times its original volume and:

- Rapidly fills the protected space.
- Covers all equipment and surfaces.
- Displaces oxygen and suppresses combustion.

Total Flooding Concept

The system is designed to:

- Fill the entire protected space within a defined time
- Maintain foam coverage to prevent re-ignition
- Provide sufficient foam volume for extended fire fighting duration

System Operation

Automatic Operation

Upon activation:

Step	Task
1	Alarm Activation • Audible and visual alarms activated • Personnel warned to evacuate
2	Ventilation Shutdown • Fans in protected space are stopped
3	Water System Start • Sea water supply valve opens • Fire water pump starts
4	Zone Selection • Section valve for selected space opens
5	Foam System Activation • After approx. 120 seconds: • Foam pump suction valve opens. • Foam concentrate pump starts. • Foam solution is generated.
6	Foam Discharge • Foam generators discharge foam into space • Space is progressively filled

Manual Operation

The system may be manually operated by:

- Activating alarm.
- Stopping ventilation.
- Opening water supply.
- Opening section valve.
- Starting fire water pump.
- Starting foam pump.

Zonal and Time-Sharing Operation

- The system operates on a zonal basis.
- Only the selected space is flooded at one time.
- Foam generators are arranged by deck and space.
- Time-sharing logic ensures optimal distribution of foam generators.

Not all generators operate simultaneously, ensuring system efficiency and capacity management.

Safety Features

The system includes:

- Pre-discharge alarms.
- Ventilation shutdown interlocks.
- Sequential start logic (water before foam).
- Control panel monitoring and indication.
- Electrical and valve interlocks.

All critical equipment is located outside protected spaces to ensure operation during fire.

Operational Considerations

- Personnel must evacuate before foam discharge.
- Visibility will be reduced during operation.
- Foam accumulates from top to bottom.
- Re-entry only after atmosphere is confirmed safe.

Inspection and Testing

Routine checks include:

- Control panel functionality.
- Alarm system operation.
- Valve and pump operation.
- Foam proportioning accuracy.
- Generator condition.

Testing is normally conducted without actual foam discharge.

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Advantages of the System

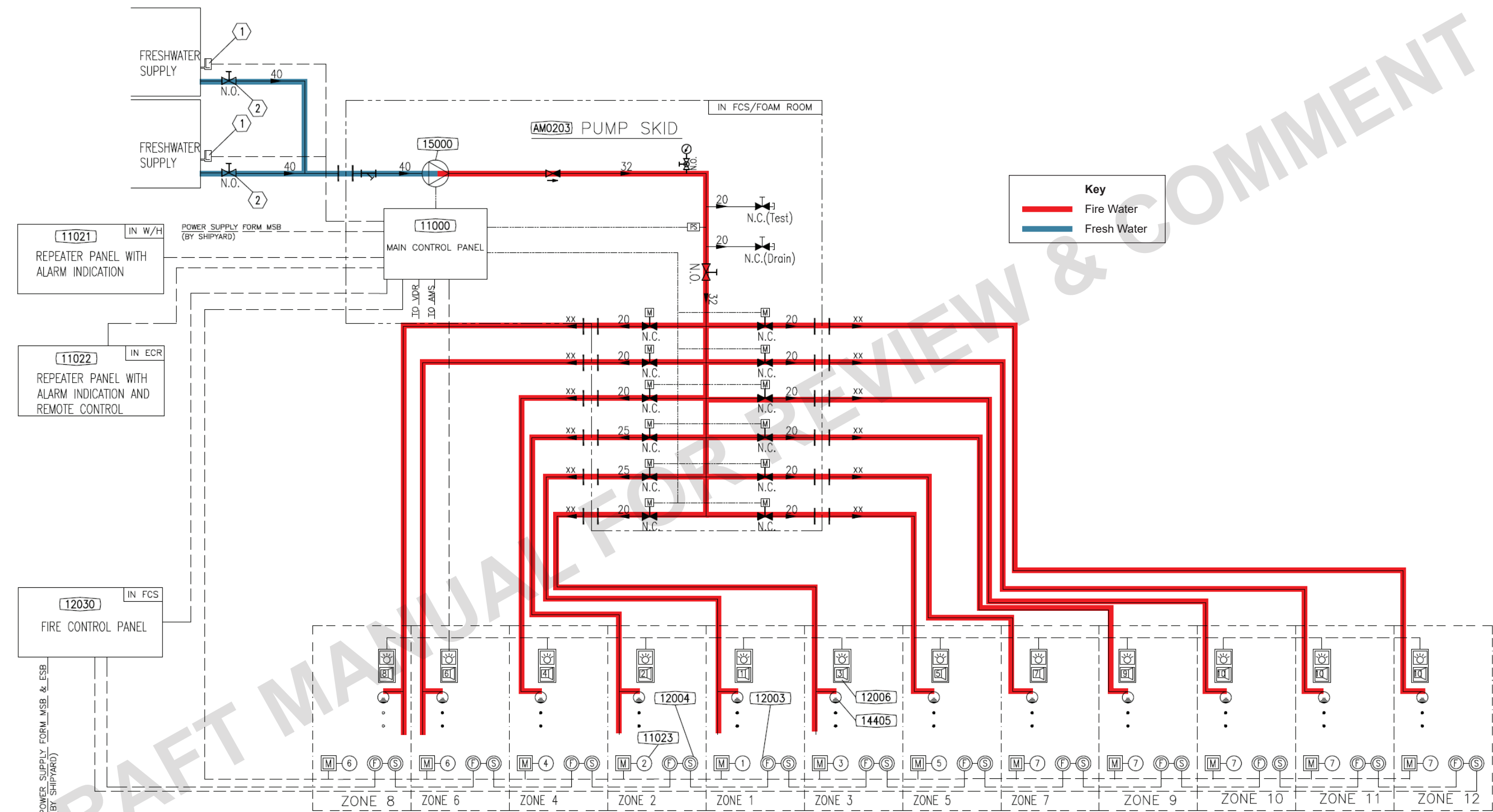
- Rapid fire suppression in large enclosed spaces.
- Cooling and smothering effect.
- Reduced risk compared to CO₂ systems.
- Effective for multi-level machinery spaces.

Summary

The High Expansion Foam Fire Fighting System provides a comprehensive and efficient fire protection solution for enclosed machinery and auxiliary spaces. By combining high expansion foam generation with zonal control and automated operation, the system ensures rapid fire suppression while maintaining safety for personnel and vessel integrity.

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Illustration 3.1.4a Fixed Water-Based Local Application Fire Fighting System



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3.1.4 ENGINE ROOM LOCAL FIRE FIGHTING (WATER MIST) SYSTEM

Technical Specification

The Water Mist System is a fixed local application fire fighting system designed to protect high-risk equipment in machinery spaces by applying fine water spray directly onto fire sources.

- System Type: Fixed Water-Based Local Application System (Water Mist)
- Standards:
 - SOLAS Chapter II-2.
 - IMO MSC/Circ. 913.
 - IMO MSC/Circ. 1276.
 - IMO MSC/Circ. 1387.
 - Classification Society (DNV) requirements.
- Extinguishing Medium: Fresh water (fine mist).
- Discharge Density: approx. 1.11 mm/min.
- Operating Pressure: approx. 4–8 bar at nozzles.
- Water Supply Requirement:
 - Main Engine section: approx. 5.4 m³/h @ 8 bar.
- Minimum Operating Duration: 20 minutes.
- Minimum Water Volume Required: approx. 1.8 m³.
- Nozzle Type: Water mist spray nozzles (GW M5 type).
- Nozzle Flow: approx. 10 l/min per nozzle.
- Coverage per Nozzle: approx. 2.25 m².

General Description

The Water Mist System is designed to provide localised fire protection for high-risk equipment in machinery spaces, such as engines, boilers, and fuel systems.

The system extinguishes fire by:

- Cooling: rapid heat absorption due to fine water droplets.
- Oxygen displacement: local steam generation reduces oxygen concentration.
- Radiant heat attenuation: prevents fire spread to adjacent equipment.

Unlike total flooding systems, this system applies water mist directly onto specific equipment, minimising water damage while ensuring effective fire suppression.

Protected Areas and Equipment

The system is arranged in independent sections, each protecting a specific piece of equipment.

Protected Sections (as per design data):

1. Main Engine No.1 – 9 nozzles
2. Main Engine No.2 – 9 nozzles
3. Generator Engine No.1 – 4 nozzles
4. Generator Engine No.2 – 4 nozzles
5. Generator Engine No.3 – 4 nozzles

6. Generator Engine No.4 – 4 nozzles.
7. Oil-Fired Boiler No.1 – 1 nozzle.
8. Oil-Fired Boiler No.2 – 1 nozzle.
9. Incinerator – 1 nozzle.
10. Fuel Oil Purifiers (Port) – 3 nozzles.
11. IGS Burner Unit – 1 nozzle.
12. GCU Burner – 1 nozzle.

Each section operates independently and is activated only when required.

Note: Each section operates independently and is activated only when required.

System Arrangement

Based on the system schematic, the Water Mist System consists of:

- Pump skid unit (main water mist pump system).
- Sectional distribution piping network.
- Water mist nozzles installed above protected equipment.
- Section control valves for each zone.
- Pressure switches and monitoring devices.
- Main Control Panel (MCP).
- Remote Release Panel (RRP).
- Release Push Buttons (RPB).
- Alarm and indication system.

The system piping is typically stainless steel or copper, designed for pressures up to 16 bar.

The diagram shows a central pump skid distributing water to multiple independent zones, each equipped with control valves and nozzles.

Principle of Operation

Water Mist Generation

Water is delivered under pressure through specially designed nozzles, producing a fine mist consisting of very small droplets.

This mist:

- Rapidly absorbs heat.
- Evaporates into steam.
- Displaces oxygen locally.

Local Application Concept

- The system is not a total flooding system
- Water mist is applied directly onto:
 - Engine surfaces.
 - Fuel systems.
 - High-risk machinery.

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This ensures targeted fire suppression with minimal impact on surrounding equipment.

System Operation

Automatic Operation

The system is activated automatically by the fire detection system:

Step	Task
1	Fire Detection - Activation of one detector → pre-alarm - Activation of two detectors → system release
2	Alarm Activation Audible and visual alarms activated
3	System Start - Pump starts automatically - Section valve for affected zone opens
4	Water Mist Discharge - Water is discharged through nozzles - Mist is applied directly to protected equipment

The system operates automatically once fire conditions are confirmed.

Manual Operation

Manual activation can be carried out:

- From Main Control Panel (MCP).
- From Remote Release Panel (RRP)
- From local Release Push Buttons (RPB).

This allows activation in case of detector failure or emergency.

System Shutdown

- System can be stopped manually from MCP.
- Section valves close.
- Pump stop.

Zonal Arrangement

- The system is divided into 12 independent sections.
- Each section corresponds to specific equipmen.t
- Only the affected section is activated.

This ensures:

- Efficient water use.
- Targeted fire fighting.
- Reduced system load.

Safety Features

The system includes:

- Automatic detection-based activation.
- Pre-alarm warning system.
- Manual override capability.
- Section isolation valves.
- Control panel monitoring.
- Audible and visual alarms (CCR and navigation bridge).

Installation Requirements

- Piping must be securely supported.
- Maximum spacing between pipe supports defined (as per table).
- Nozzles must not be painted or obstructed.
- Drain lines installed for system testin.g
- Air blowing required after installation to prevent clogging.

Inspection and Testing

Routine Checks

- Daily:
 - Alarm system check.
 - Panel indication.
- Monthly:
 - Pump operation test.
 - Valve functionality.
 - Alarm verification.
- Every 6 months:
 - Functional system testing.
- Annually:
 - Pump overhaul.
 - System inspection.

All tests must be recorded in the logbook.

Operational Considerations

- System is intended for localised protection only.
- Personnel may remain in space during discharge.
- Water mist reduces visibility locally.
- Minimal water damage compared to sprinkler systems.

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Advantages of the System

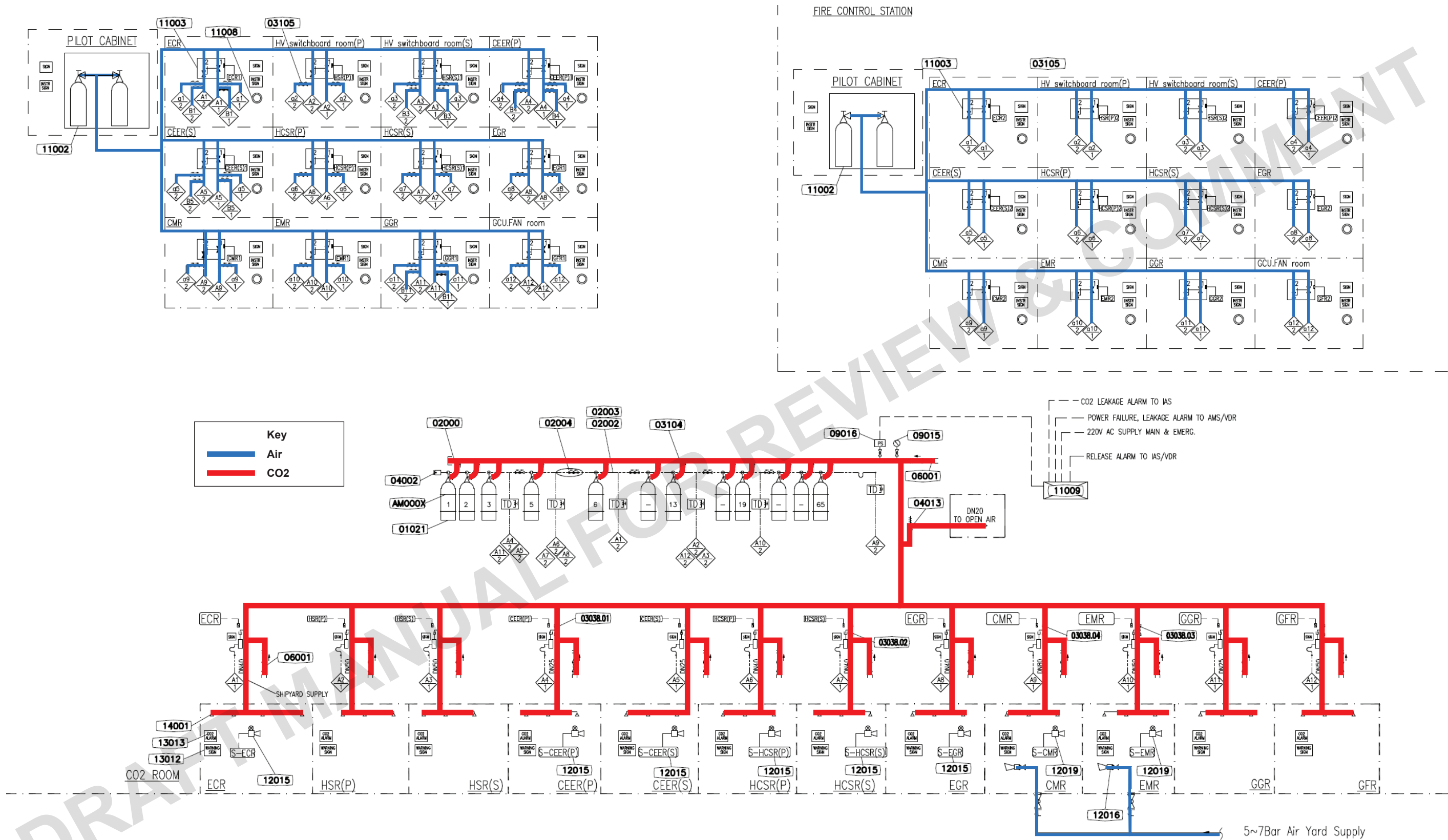
- Rapid fire suppression at source.
- Reduced water consumption.
- Minimal damage to equipment.
- Safe for occupied machinery spaces.
- Automatic and manual operation capability.

Summary

The Water Mist Fire Fighting System provides efficient local fire protection for critical machinery by delivering fine water mist directly onto high-risk equipment. The system operates automatically or manually, ensuring rapid response, minimal water usage, and effective fire suppression in machinery spaces.

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Illustration 3.1.5a CO2 Fire Extinguishing System



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3.1.5 CO2 FIRE EXTINGUISHING SYSTEM

Manufacturer:	VTI Fire Products Co., Ltd
Type:	High pressure
Capacity:	65 cylinders 45kg + 1 spare cylinder

Introduction

Carbon dioxide acts to reduce the oxygen content of the atmosphere in the region of a fire, and in doing so, extinguishes the fire. The oxygen content must be reduced sufficiently and it is essential that the correct amount of CO₂ is injected into the compartment in which the fire has occurred. The free volume of a protected compartment is determined, and from this, the amount of CO₂ required to give a 40% volume (45% for cargo areas) can be determined. CO₂ is stored in gas cylinders and the number of cylinders required to protect a particular compartment can be determined. It is essential that all of the cylinders designated for a compartment are released together to achieve the necessary concentration to extinguish the fire, and regulations require that the gas discharge has to be completed within two minutes.

Prior to the release of the CO₂, all ventilation must be stopped and dampers closed to ensure that the CO₂ remains in the compartment and that the concentration is maintained. On this vessel, this action is achieved during the pre-discharge sequence which is followed prior to the release of the gas. CO₂ is heavier than air and will tend to move to the bottom of the compartment, thus the injection nozzles may be placed above equipment and positioned to give the required spread of gas upon release. Any fire in a bilge area will be tackled by the descending gas.

Carbon dioxide does not react with other materials or leave deposits, nor does it conduct electricity. These factors make it suitable for electrical fires and fires in small compartments containing equipment such as switchgear consisting of many small parts. After a fire in a compartment, there is no residue to be removed before the equipment may be returned to service; all that is required is adequate ventilation.

Even in low concentrations, CO₂ will result in asphyxiation, therefore, any compartment into which CO₂ has been injected must be thoroughly ventilated before personnel are allowed to enter. The atmosphere of a compartment must be proved safe before entry without the aid of breathing apparatus is allowed. Even at this stage, pockets of gas may remain in places such as bilge wells and this should be remembered prior to entering such enclosed spaces.

Company procedure regarding the release of CO₂ and the subsequent re-entry into a compartment following the release of CO₂ must be read and understood by all personnel, not just those with the authority to release CO2 in the event of a fire.

WARNING
Liquid or recently evaporated CO ₂ may cause low temperature burns when in contact with the skin. In such cases the affected area should be thoroughly irrigated with clean water and afterwards dressed by a trained person.

WARNING
Upon activation of the CO ₂ alarm, all personnel must evacuate the space immediately.

System Description

Areas Protected

The CO₂ system consists of a central bank of cylinders arranged in four double rows which are installed in a designated compartment in the ship (CO₂ room) in order to protect the following areas:

- Engine room area - covers the three areas below which can also flooded by their own release.
- Engine control room.

- HV switchboard room (S).
- HV main switchboard room (P).
- GCU GVT room.
- GCU fan room.
- HV cargo switchboard room (S).
- HV cargo switchboard room (P).
- Cargo electric equipment room (S).
- Cargo electric equipment room (P).
- Emergency diesel generator room.
- Cargo machinery room.
- Electric motor room.

The CO₂ room is located on the starboard aft side of the engine casing on ‘A’ deck, refer to illustration 7.6.3b for the configuration and identification numbers for the cylinders and the protected spaces they as assigned to cover.

Central Bank CO2 System

The central CO₂ bank is located in the CO₂ room and consists of 65 cylinders, each containing a 45kg CO₂ charge.

These cylinders are connected to discharge nozzles within the protected spaces via cylinder manifolds and distribution pipework. A pressure gauge, pressure relief valve and pressure switch are fitted to the main CO₂ manifold.

The system is designed to discharge the required number of cylinders into each space as indicated in the following table:

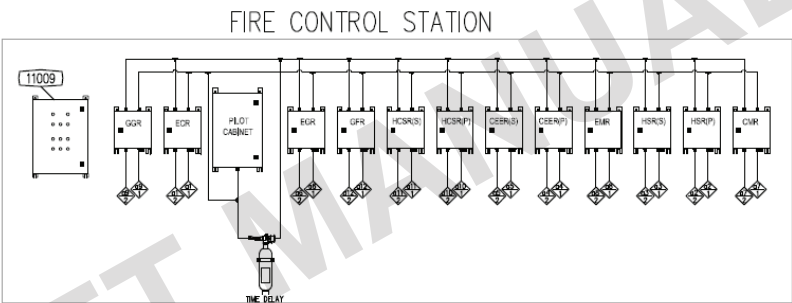
Protected Space Engine Room Area	CO ₂ Cylinders Required
HV switchboard room (P)	13
HV switchboard room (S)	13
Engine control room	6
GCU GVT room	2
GCU fan room	12

Protected Space Cargo Area	CO ₂ Cylinders Required
Cargo machinery room	65
Electric motor room	19

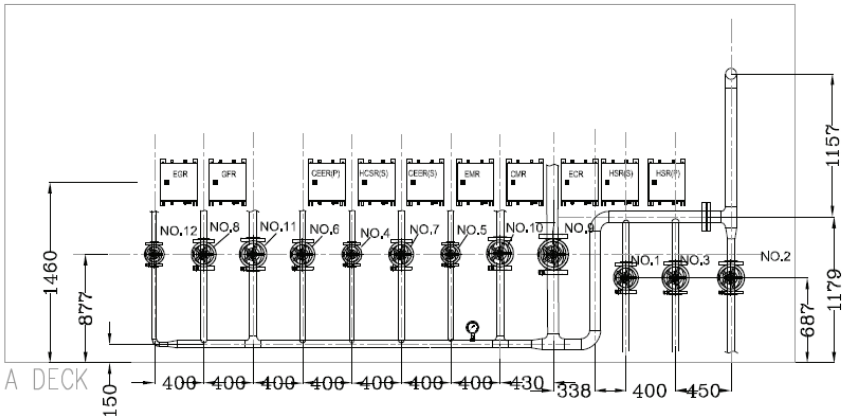
Protected Space Accommodation	CO ₂ Cylinders Required
HV cargo switchboard room (P)	5
HV cargo switchboard room (S)	5
Cargo electric equipment room (P)	3
Cargo electric equipment room (S)	3
Emergency diesel generator room	4

The largest protected space consists of the cargo machinery room which requires the total release of all 65 CO₂ cylinders for complete protection. In order to allow the other protected spaces to be release separately from the main gang of cylinders, sets of non-return valves in the pilot gas line are fitted which will accurately control the release of the designated cylinders for that compartment/space.

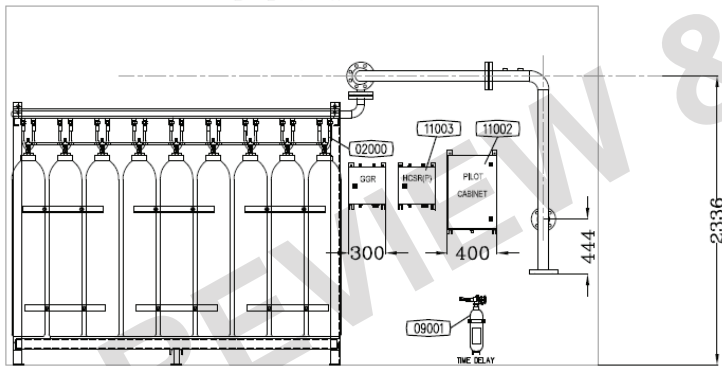
A DECK



A-A VIEW

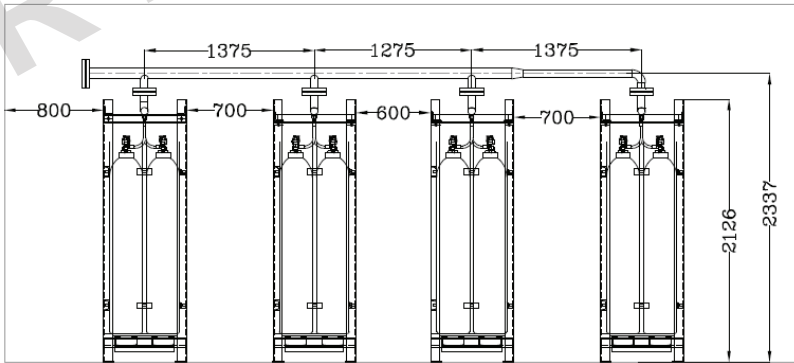


B-B VIEW



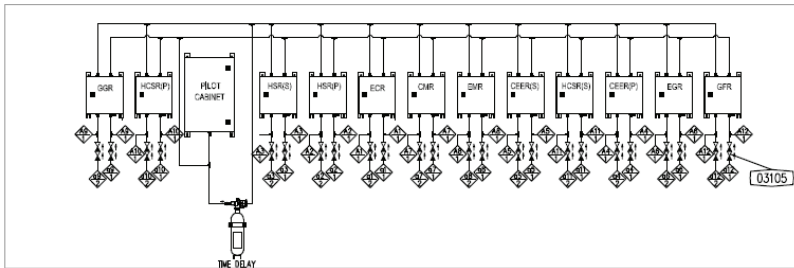
A DECK

C-C VIEW



A DECK

ACTUTATION PIPE REFERENCE



NO.1	ENGINE CONTROL ROOM
NO.2	HV SWITCHBOARD ROOM(P)
NO.3	HV SWITCHBOARD ROOM(S)
NO.4	CARGO ELECTRIC EQUIPMENT ROOM(P)
NO.5	CARGO ELECTRIC EQUIPMENT ROOM(S)
NO.6	HV CARGO SWITCHBOARD ROOM(P)
NO.7	HV CARGO SWITCHBOARD ROOM(S)
NO.8	EMERGENCY GENERATOR ROOM
NO.9	CARGO MACHINERY ROOM
NO.10	ELECTRIC MOTOR ROOM
NO.11	GCU FAN ROOM
NO.12	GCU GVT ROOM

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It must be appreciated that if any CO₂ cylinders are released to protect other compartments, there will be insufficient CO₂ available to protect the cargo machinery room. Therefore, if any CO₂ cylinders are released, they must be replenished at the earliest opportunity.

Pilot cylinders are used for activating the release of the main CO₂ cylinders and for opening the distribution valves. Two pilot cylinders are located in the control cabinets in the CO₂ room and the fire control station for the protected spaces.

Distribution Valves

A pilot gas operated distribution valve for each protected space is installed into the CO₂ discharge pipework between the CO₂ manifold and the discharge nozzles to direct the CO₂ gas into the protected space where the fire is located, the distribution valve manifold is located on the outboard bulkhead in the CO₂ room. The CO₂ discharge activation can be carried out from the fire control station, or the CO₂ room. The the distribution valve is opened automatically by the pilot gas when the CO₂ release system for a particular space is activated. For the engine room area, the distribution valves for the two main switchboard rooms plus the valve for the ECR will be opened.

A manually operated anti-piracy valve is installed and located in the pipeline downstream of the distribution valve for the ECR, in normal operations this valve is maintained in the open position, a safety pin is fitted to secure it in the open position. The valve is situated in the ECR inside a locked cabinet which has an inspection glass in the cabinet door in order that its open/closed position can be checked, a limit switch is installed which will activate an alarm through the IAS if the door is opened, an electric horn and rotating light in the ECR will also be activated. Additionally, the valve handle is fitted with two limit switches which will give indication in the IAS if the valve is not fully in the open position, one limit switch acts as a spare if the in-service switch fails. The valve would only be closed if the ship’s company had been mustered into the ECR which acts as the citadel in the event of a piracy attack.

Control Valve Cabinet Positions

Discharge of the CO₂ is manually operated from control valve cabinets which are located in the CO₂ room, and in the fire control station. These cabinets contain control valves which direct the pilot gas to open the appropriate distribution valve and to release the designated gang of CO₂ cylinders.

At each release location there are detailed instructions for the release of CO₂ into the protected space. A key box is located at each release cabinet and this is used to open the cabinet. Upon opening of the release box an audible and visual alarm is activated in the protected space, additionally the ventilation system for that space is stopped. The Master’s designated operator MUST NOT release CO₂ into a protected space until satisfied that there are no personnel in the protected space.

Alarms are initiated via relays from switches mounted on the control valve cabinet doors, control cabinet door and on the control valves and on main CO₂ release valves.

A pneumatic time delay unit is fitted in the pilot gas line to gang release line from the control cabinet in the fire control station and CO₂ room, it has a default minimum setting of 30 seconds. The time delay unit allows time for anyone in the protected space to evacuate immediately. The time delay unit is fitted with a bypass line and manual valve in order that the pilot gas can be directed to the gang of cylinders if the time delay unit fails to operate.

The delivery valves in the CO₂ pilot release control cabinet are positioned so that the cabinet door cannot be closed with the valves in the open position.

Operating Procedure from the CO₂ Room or Fire Control Station

On discovering a fire in one of the protected spaces, and after all initial attempts to control it have failed, the Master will take the decision to release the CO₂. It is assumed that the manual anti-piracy isolating valve for the ECR is open and the time delay unit bypass valve is closed.

- On mustering, a head count of all ship’s staff is taken. All ship’s staff must be accounted for before any further action is taken.
- Activate all remote machinery stops from the cabinet in the fire control station.

- Close all ventilation dampers and tank valves from the fire control station.

Releasing the CO₂

Step	Task
1	In either the CO ₂ room or fire control station break the acrylic panel on the key box located adjacent to the control cabinet and remove the key.
2	Unlock and open the control valve cabinet for the protected space, an audible alarm will sound in the protected space and the ventilation fans will trip.
3	Open the control valve No.1 (rotate vertically - left hand valve) for the distribution valve, followed by valve No.2 (pull the valve handle to the horizontal - right hand valve) for the gang release.
4	Unlock and open the control cabinet door.
5	Open one of the pilot cylinder valves, check the pilot pressure is above 3MPa, if not, open the other pilot cylinder valve.
6	Open pilot valve No.1 (rotate vertically - left hand valve), followed by pilot valve No.2 (pull the valve handle to the horizontal - right hand valve). The pilot gas will be directed to the time delay unit for the gang release, after the delay period has expired the pilot gas will release the designated bank of cylinders.
7	Allow sufficient time for the fire to be extinguished, then close the pilot cylinder valves.
8	When the pilot pressure gauge within the control box is zero, close both pilot isolation valves.

Emergency Operation

Should the pilot CO₂ cylinders fail to open after 40 seconds, the main CO₂ cylinders for the protected compartment from either of the remote release station or local position, the cylinders must be opened manually. The main line valve to the protected compartment must also be operated manually.

The CO₂ cylinders for each protected compartment are marked in the CO₂ room to allow for ease of recognition which cylinders must be manually opened for the protected compartment. The procedure for opening the main cylinders is as follows; these should be opened by operation of the manual release lever on top of each CO₂ cylinder as follows:

Step	Task
1	Proceed to the CO ₂ room and open the distribution valve leading to the protected space by operating the lever for the room.
2	At the cylinder bank, remove the safety pin of the valve actuator and pull up on the operating lever. The CO ₂ in that cylinder is now discharged. The designated master cylinder for each associated space should be activated. Or if necessary repeat the opening procedure for the designated cylinder for the affected space have been discharged. The illustration 7.6.3b outlines the CO ₂ cylinder associations and master cylinders for the identifies the protected spaces.

WARNING

During a CO₂ discharge, the storage cylinders can generate large static charges. In addition, the storage cylinders will ice-up, freezing anything in contact with them. Do not touch the cylinders during or immediately after discharge.

CAUTION

It must be remembered that CO₂ gas should be delivered to a protected space within a period of two minutes for effective fire fighting action. Any manual release of storage bottles must therefore be carried out as swiftly as possible in order to comply with this time restriction.

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Time Before Discharge

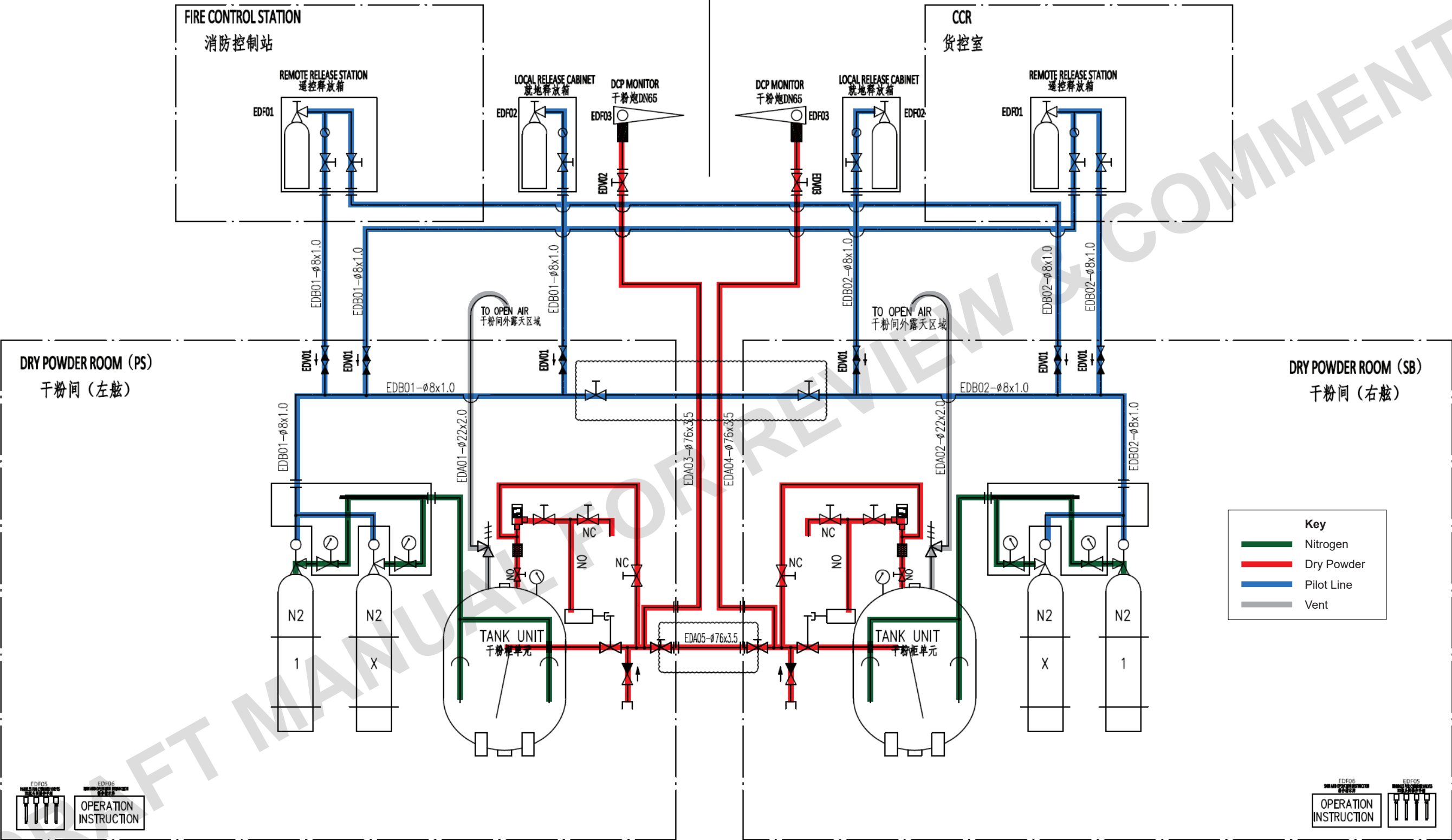
Thirty seconds after the initiation of audible alarm, the CO₂ will be released.

After Discharge

- Allow sufficient time for the CO₂ to extinguish the fire.
- Do not enter the space until it is absolutely established that the fire is out and cannot re-ignite.
- Do not enter the space until it has been thoroughly ventilated.
- Persons entering the space must wear breathing apparatus until the atmosphere has been checked and verified that the oxygen content is 21%.

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Illustration 3.1.6a Dry Powder Fire Extinguishing System



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3.1.6 DRY POWDER SYSTEM

Dry Powder System

Manufacturer:	VTI Fire Products Co., Ltd
Extinguishing Medium	BC type dry chemical powder
No. of sets:	2 independent tank units
System pressure:	1.35MPa at 35°C
No.1 tank (S) capacity:	1,230 litres at 1,130kg powder
No.1 tank (S) nitrogen:	6 cylinders x 67.5 L
No.2 tank (P) capacity:	1,230 litres at 1,130kg powder
No.2 tank (P) nitrogen:	6 cylinders x 67.5 L
Hand hoses:	11 sets at 33m length
Monitor stations:	1 port and 1 starboard manifold

Introduction

The dry powder fire-fighting system provides immediate fire-fighting capability in the event of a fire occurring in the cargo manifold area, LNG transfer equipment or trunk deck cargo piping.

Dry chemical powder is particularly effective against flammable gas fires, such as LNG vapour fires, as it interrupts the combustion chain reaction and rapidly extinguishes the flame.

The system consists of two independent dry powder storage units, located on the port and starboard sides of the vessel. Each unit contains dry chemical powder stored in a pressurised tank and nitrogen cylinders used as the propellant gas.

Dry powder is distributed through fixed pipelines to:

- cargo manifold monitors, and
- hand-held hose stations located along the cargo manifold deck.

The port and starboard powder systems normally operate independently to ensure system redundancy.

Dry Chemical Powder

The dry chemical powder used in the dry powder fire-fighting system is an extinguishing medium consisting of very fine particles of potassium and sodium bicarbonate treated with additives to improve flow characteristics and prevent moisture absorption, packing and caking.

Dry chemical powders used in LNG cargo fire-fighting systems are specifically designed for flammable gas and liquid fires, where the powder acts by interrupting the chemical chain reaction of combustion and rapidly suppressing the flame.

Only dry chemical powders based on potassium salts shall be used in this system.

The type of dry chemical powder installed in the system shall not be changed unless testing has been carried out to verify performance to the satisfaction of the Administration. Different dry chemical powders shall not be mixed, as this may adversely affect extinguishing performance and flow characteristics.

The dry powder system installed on board utilises potassium bicarbonate powder (70%), which has been tested in accordance with:

Chemical Composition of Dry Powder

Potassium hydrogen carbonate	70%
Sodium bicarbonate	13%
Calcium carbonate	13%
Silicone oil	2%
Other additives	2%

The additives included in the powder composition are intended to improve powder dispersion, prevent moisture absorption and ensure consistent discharge performance during system operation.

Safe Handling

When handling dry chemical powder, appropriate precautions shall be taken to avoid exposure.

Personnel involved in handling or refilling operations shall:

- Wear appropriate protective clothing and gloves;
- Avoid inhalation of powder dust;
- Avoid contact with eyes and skin;
- Ensure that handling operations are carried out in a well-ventilated area.

During handling or transfer of powder containers, care shall be taken to avoid damage to packaging, as moisture ingress may affect the performance of the powder.

Storage Requirements

Dry chemical powder shall be stored under the following conditions:

- In a cool and dry location;
- In tightly closed containers;
- In a well-ventilated storage area;
- Protected from heat sources, ignition sources and open flames;
- Protected from moisture.

Dry chemical powder shall not be stored near substances that may cause chemical reaction or degradation.

Incompatible Materials

The powder should be kept away from:

- Acids.
- Strong oxidising agents.

Exposure to moisture must be avoided as this may lead to powder compaction and reduced discharge performance.

Storage Area Safety

The storage location for dry chemical powder shall be provided with:

- Appropriate fire-fighting equipment; and
- Equipment for emergency leakage control and clean-up.

Periodic inspection of stored powder containers shall be carried out to ensure that containers remain intact and free from moisture contamination.

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System Arrangement

Each dry powder unit consists of the following main components:

- Dry powder storage tank.
- Nitrogen cylinder banks.
- Pilot cylinders for remote activation.
- Powder discharge valves.
- Powder distribution piping.
- Cargo manifold monitors.
- Hose stations.

When activated, nitrogen gas enters the powder tank, fluidising the powder and pressurising the tank. The powder is then discharged through the piping network to the selected monitor or hose station.

Dry Powder Distribution Network

The dry powder piping network runs along the cargo manifold deck and connects both powder storage tanks to all monitors and hose stations.

The distribution system ensures:

- Full fire-fighting coverage of cargo transfer areas.
- Redundancy through two independent powder tanks.
- The ability to operate monitors or hose stations from either side of the vessel.

Where installed, crossover valves allow the powder systems to support each other during emergencies.

Dry Powder Monitors

Two dry powder monitors are installed at the cargo manifold area.

These monitors are arranged to cover:

- LNG liquid manifolds.
- Vapour return manifolds.
- Cargo transfer valves.
- Trunk deck cargo piping.

The monitors are normally directed towards the cargo manifold area and locked in position to allow immediate use.

The monitors can be activated:

- Locally at the monitor station; or
- Remotely from release cabinets located in the Cargo Control Room (CCR) and Fire Control Station (FCS).

Dry Powder Hose Stations

Hand-held hose stations provide local fire-fighting capability along the cargo manifold deck.

Each hose cabinet contains:

- Dry powder hose reel.
- Discharge nozzle.
- Pilot nitrogen cylinder.
- Powder supply control valve.

Opening the pilot nitrogen cylinder activates the main nitrogen cylinders connected to the powder tank, pressurising the tank and allowing powder discharge.

Hose Station Coverage

Hose Station No.	Location	Supplied From
No.1	Starboard manifold deck forward	Starboard powder unit
No.2	Port manifold deck forward	Port powder unit
No.3	Starboard manifold deck mid-forward	Starboard powder unit
No.4	Port manifold deck mid-forward	Port powder unit
No.5	Starboard manifold deck centre	Starboard powder unit
No.6	Port manifold deck centre	Port powder unit
No.7	Starboard manifold deck mid-aft	Starboard powder unit
No.8	Port manifold deck mid-aft	Port powder unit
No.9	Starboard manifold deck aft	Starboard powder unit
No.10	Port manifold deck aft	Port powder unit

This arrangement ensures that the entire cargo manifold area can be reached by at least one hose station.

Monitor Coverage

Monitor	Location	Supplied From
Port Monitor	Port side manifold deck	LNG liquid and vapour manifolds
Starboard Monitor	Starboard manifold deck	LNG liquid and vapour manifolds

The monitors are arranged to provide wide-area coverage of the cargo transfer equipment.

Operation of Dry Powder Hose System

Step	Task
1	Open the cabinet door at the selected hose station.
2	Remove the brake and fully extend the hose.
3	Open the nitrogen pilot cylinder valve inside the cabinet.
4	Open the powder supply ball valve.
5	The powder tank will automatically pressurise.
6	Aim the nozzle at the base of the fire.
7	Open the nozzle discharge valve.

Operation of Dry Powder Monitors

The monitors may be activated locally or remotely.

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Remote Operation

Step	Task
1	Open the release cabinet in the CCR or FCS.
2	Open the pilot nitrogen cylinder valve.
3	Open the monitor discharge valve.
4	Powder discharge will begin automatically.

Local Operation

If remote operation fails, the monitor may be activated locally at the powder storage unit.

After Use of the System

After the system has been used:

Step	Task
1	Close the powder discharge valve.
2	Close the nitrogen supply valve.
3	Vent pressure from the powder tank.
4	Vent pressure from nitrogen supply lines.
5	reset all valves to normal positions.
6	Refill the powder tank.
7	Recharge nitrogen cylinders.

Powder Agitation

Dry powder may compact during long periods of inactivity. The powder tank is therefore equipped with a nitrogen agitation system to fluidise the powder periodically.

Agitation shall be performed according to the manufacturer’s maintenance instructions.

Safety Precautions

- Wear full fire-fighting protective equipment when operating the system.
- Ensure hoses are fully extended before discharge.
- Be prepared for strong recoil forces when opening the nozzle.
- Maintain stable footing when operating hose units.
- Avoid directing powder discharge directly into strong wind conditions.

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3.1.7 EMERGENCY SHUT-OFF VALVE AND FAN DAMPER SYSTEM

General Description

The Emergency Shut-Off Valve (Quick-Closing Valve – QCV) and Ventilation Damper System is provided to rapidly isolate fuel and lubricating oil sources and stop air supply to machinery spaces in the event of fire.

The system is designed to:

- Prevent the supply of fuel oil and lubricating oil to a fire.
- Reduce fire intensity by eliminating combustible liquid flow.
- Isolate engine room ventilation to limit oxygen supply.
- Provide remote and emergency operation from safe locations.

The system complies with SOLAS requirements for remote closing of oil tank outlet valves and closure of ventilation openings.

System Arrangement

Quick-Closing Valve Control System

The quick-closing valves are operated by a centralised pneumatic control system, consisting of:

- Quick-closing valve control box.
- Air distribution manifold with multiple outlet lines.
- Individual air lines to each valve actuator.
- Local manual operating handles.
- Air inlet and outlet connections.

The maker’s drawing shows that the control box distributes compressed air to multiple QCV actuators via dedicated outlet connections, allowing grouped or individual operation of valves.

Air Supply System

- Air Source: Service / control air system
- Air Bottle (Reservoir):
 - o Dedicated air bottle installed for emergency operation.
 - o Equipped with pressure gauge, safety valve and drain arrangement.
- Operating Principle: Pneumatic actuation via air pressure release.

The air system ensures fail-safe operation, allowing valves and dampers to close even during loss of power.

Grouping of Quick-Closing Valves

The system is divided into functional groups:

- Main Engine / Generator Groups (Port and Starboard).
- Auxiliary Equipment Group (e.g. gearboxes, incinerator.)
- Ventilation Damper Group

Each group is supplied via separate air control lines from the control box, enabling selective or grouped shutdown.

Quick-Closing Valves (QCVs)

Function

Quick-closing valves are fitted on:

- Fuel oil tank outlet lines.
- Lubricating oil tank outlet lines.

Their purpose is to:

- Immediately stop oil flow in case of fire.
- Prevent fuel supply to machinery or fire source.

Operating Principle

The QCVs are air-operated, spring-assisted shut-off valves.

Normal Condition

- Valves are held in open position.
- Air system is charged and ready.

Activation (Emergency Closure)

- Activation is carried out via control box or remote release .
- Air pressure is applied or released (depending on system design).
- Actuator mechanism operates and:
 - Releases the valve locking mechanism.
 - Valve closes rapidly under spring force or gravity

The control box shown in the maker’s drawing distributes air to multiple actuators simultaneously, ensuring rapid closure of all valves within the selected group.

Resetting of Valves

After activation:

- Air supply is restored or vented as required.
- Actuator pistons are reset.
- Valve mechanism is manually re-engaged.
- Valves are reopened manually.

Important Note

Some tanks (e.g. certain service or daily tanks) may not be fitted with QCVs.

- These valves shall be kept normally closed
- Opened only when required for operation.

Ventilation Damper System

Function

The ventilation damper system is designed to:

- Stop air supply to machinery spaces.
- Prevent spread of fire and smoke.
- Support effectiveness of fire extinguishing systems.

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Arrangement

The system includes:

- Air-operated ventilation dampers.
- Control air supply from the same pneumatic system.
- Grouped control lines for different ventilation zone.s

Dampers are fitted on:

- Engine room ventilation fans.
- Funnel ventilation systems.
- Machinery space intake and exhaust ducts.
- Auxiliary spaces (e.g. purifier room, control room, converter rooms).

Operating Principle

Normal Condition

- Dampers are held open by air pressure.

Emergency Condition

- Upon activation (fire detection system or manual release):
 - Air supply is vented.
 - Actuator pressure is lost.
 - Dampers close automatically under spring force.

This ensures fail-safe closure in case of fire or loss of air supply.

System Operation

Automatic Operation

Upon fire detection:

- Signal sent from Fire & Gas System.
- Activation of:
 - Quick-closing valves (fuel isolation) .
 - Ventilation dampers (air isolation).
- Air control system actuates valves and dampers simultaneously.

Manual Operation

The system may be activated manually from:

- Fire Control Station.
- Local release stations.
- Control box (manual levers/valves).

Safety Features

- Centralised control from safe location.
- Pneumatic operation independent of electrical power.
- Fail-safe closure of valves and dampers.

- Grouped and individual control capability.
- Pressure monitoring of air system.
- Safety valve fitted on air reservoir.

Operational Considerations

- System must always be charged and ready.
- Air pressure in reservoir must be maintained.
- Regular testing of:
 - QCV operation.
 - Damper closure.
 - Control box functionality.
- After activation:
 - System must be manually reset.
 - Valves and dampers verified in correct position.

Summary

The Emergency Shut-Off Valve and Ventilation Damper System provides a critical safety function by:

- Rapidly isolating fuel and lubricating oil supply.
- Cutting off air supply to fire zones.
- Supporting overall fire containment strategy.

The system ensures fast, reliable, and fail-safe response through pneumatic actuation and centralised control.

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3.1.8 GENERAL FIRE SAFETY PRACTICES

Introduction

There are three requirements necessary to enable a fire to exist:

- 1) A combustible substance (fuel).
- 2) A supply of oxygen, in the form of air.
- 3) A source of heat and/or ignition.

Fire Triangle

Fires will spread once they have started by direct flame transfer or by any one or all of the following ways:

- **Conduction:** Heat can be conducted through the steel structure of the vessel from one space to another.
- **Radiation:** Heat from a fire can raise the temperature and ignite substances in the vicinity.
- **Convection:** When gases and liquids are heated, they expand and rise therefore the heat from a fire will rise and could cause substances above to be set alight.

Housekeeping to Prevent Fires

General housekeeping is the first line of defence in ensuring a safe environment on board. To ensure good housekeeping the following must be observed:

General Areas

- At the end of each work task, clear away left over materials into segregated containers appropriate for disposal.
- Clear away tools and equipment and store in the appropriate location.
- Do not overfill bins.
- For oil spills, where possible use appropriate oil absorption material rather than rags or other wipes.
- Corridors and fire escape routes are to be kept clear at all times.
- Ensure that flammable liquids such as paint, paint thinners, other solvents and glues are stored in a designated lockable store or cupboard and that the method of storage is secure.
- Never hang cloths over cookers or heaters to dry.

Machinery Spaces

- Ensure safety equipment such as fuel shut-off valves, overflow alarms and heat sensors are kept in working order.
- Store other combustible materials such as timber and rag boxes in designated areas away from locations where there is a heightened risk of ignition such as machinery spaces.
- Dirty waste, rags, sawdust and other rubbish, especially if contaminated with oil, are dangerous if left lying around. They must be disposed of immediately in the correct bin to eliminate any possibility of spontaneous combustion or accidental ignition.
- Combustible materials, such as wood, paints, spirits and tins of oil should not be kept in machinery spaces and must be stored in areas designated for the storage of that material only.
- If rags are used, dispose of them in a manner consistent with hazardous flammable liquids.
- Do not use rags as drip collectors.
- Machinery spaces should be kept clean and tidy at all times. Oil leaks and spills should be rectified and cleaned immediately. Oil-soaked lagging should be replaced. Oily waste should be disposed of correctly.

Electrical Areas

- Carry out earth test and inspections on wiring normally out of sight.
- Electrical appliances should be regularly inspected for defects.
- When not in use disconnect equipment.
- Never leave soldering irons unattended.

Laundries

- Always ensure lint filters in laundry equipment are clean before and after use. Do not allow lint to build up.
- Never leave hot irons unattended.

Hot Work

- Hot work must be undertaken in accordance with a Permit to Work system with no exceptions.
- Equipment should only ever be used by personnel trained and authorised to do so.
- Any oxy-acetylene hoses should be checked for leaks and twists and that they are attached securely.
- Check surrounding, and adjoining areas above and below for combustible materials.
- Any openings that sparks could fly through should be closed.
- Appropriate fire extinguishers should be immediately available to personnel carrying out the hot work.
- Personnel should be posted to carry out a fire watch in areas not visible to personnel carrying out the hot work ie, in adjacent tanks, spaces and decks above and below.
- On completion of hot work being carried out continue to monitor the area for several hours to ensure no hot surfaces are left unattended.

Smoking

- Extinguish cigarettes and matches properly in ashtrays and disposal facilities.
- Never smoke in bed.
- Only smoke in designated smoking areas.
- Adhere to no-smoking areas.
- Do not smoke when moving around the ship.

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3.1.9 FIRE FIGHTING TECHNIQUES AND PROCEDURES

Fire Extinguishing Techniques

Fire fighting techniques use one, or a combination, of the below methods to extinguish a fire:

- Removal of Fuel (Starvation).
- Removal of Heat (Cooling).
- Removal of Air (Smothering).

There are therefore, three aims which are the basic principles of fire extinguishing and they are starvation, cooling and smothering. When tackling a fire, these three principles should be borne in mind, as should use of the ship's structure and installations.

The closing of watertight doors and hatches, valves in oil supply lines and flaps in ventilation systems, are an important factor in supplementing the effects of fire fighting appliances. In a major fire, a change of ship's course can be beneficial in preventing flames from being fanned and spread.

Removal of Fuel (Starvation)

This cannot be completely achieved in practice, but a reduction in the intensity of the fire can be achieved in many instances as follows:

- Shut off the oil supply if the fire is being fed from a fuel leak or piece of faulty equipment.
- Pump out all free liquids if the fire is located in bilges or on tank tops.
- Remove all burning material by manual means (if possible).
- Apply boundary starvation; ie, remove fuel from adjacent areas, thus forming a fire break.

Removal of Heat (Cooling)

The obvious cooling agent is water, which has a high capacity to absorb heat and is usually available in plentiful quantities. The initial requirement for ignition is the heating of a fuel to its flashpoint. Therefore, if a combustible has been heated to this temperature, cooling will reduce or stop vaporisation and the fire will be controlled. Additionally, entry of a fire-affected space can usually be made shortly after the fire has been extinguished by means of water.

Although water cools the flames and may quench them, the primary object is to cool the fuel and inhibit the creation of vapour. If water extinguishes the flames and the fuel remains hot, it will continue to vaporise and will fill the compartment with unburnt vapour. The conditions are then right for a back-flash. The main gaseous product of most fires is carbon dioxide, which does not support life or combustion. Carbon monoxide is a deadly odourless gas, produced when combustion is incomplete, which is usually the case when a fire exists in an enclosed compartment. When air reaches the carbon monoxide formed in this situation, it may burn violently to form carbon dioxide. This is the explanation of back-flash, of which fire fighters must be warned. This can occur spontaneously if the vapours are above ignition temperature, or by the application of external heat in the form of a spark or flame. A back-flash can erupt with sufficient magnitude to damage a steel compartment, with subsequent injury or loss of life.

It is essential to cool the combustibles below their flashpoint and to reduce the heat in the structure and masses in the vicinity of the fire, thus eliminating as far as possible the risk of re-ignition.

Removal of Air (Smothering)

Various techniques can be employed and are divided into three categories:

- Smothering the fires by using non-combustible materials such as sand or fire blankets etc.
- Envelopment with an inert vapour or gas, including chemical and mechanical foam compounds.
- Total sealing of the affected compartment.

Locating a Fire

The way in which any fire is tackled with fire fighting equipment will depend on the type of equipment, the fire fighting media and the type of fire. In the case of a small fire, it would be advantageous to tackle the fire as quickly as possible with portable equipment before it has a chance to grow larger. Other fires may be too large to tackle with portable equipment when discovered. However, some fire fighting techniques are universal and apply to all equipment. Some general rules are as follows:

- Always raise the alarm before attempting to tackle any fire. Use the manual call points provided, by breaking the glass cover, use a radio or phone to alert the bridge.
- After assessing the severity of the fire after raising the alarm, protect yourself from the fire using objects within the space where possible. If the fire becomes beyond your capabilities to tackle, retreat and attempt to contain the fire as best as possible until help arrives.
- Test fire fighting equipment prior to entry into a space containing a fire to ensure it is working correctly.
- Be aware of potential flash-over whilst entering a space containing fire. Remain low at all times and always have some means of communicating with personnel outside the space.
- Do not leave access doors unattended and open behind you for any reason.
- Be prepared to ask for assistance from external sources at the earliest opportunity. Shore fire fighting organisations should be contacted as soon as possible if in port.

Containing a Fire Area

The closing of doors will prevent the spread of fire and smoke, this includes all fire doors and watertight doors.

Considerations to be made when isolating a fire area:

- Closing doors as soon as possible to stop the spread of fire and smoke, taking into consideration that certain areas may need to remain open long enough to facilitate the evacuation of large numbers of people.
- The opening or closing of doors during an incident will affect the overall status of the ventilation system and should be closely monitored.
- Fire door flaps should be used as far as possible for hoses.

Operational Fire Fighting

Pre-Entry Stage

Prior to sending a fire team into a fire area, the following **MUST** be performed:

- Communications checked.
- Entry point and staging area established.
- Fire equipment checked and tested.
- Fire teams briefed.
- Main and back-up hoses and nozzles rigged and charged.
- Fire team No.2 on standby.
- Support teams at the staging area.
- Watertight doors and sliding fire doors in local control.
- Electrical isolation completed.
- Fire zone evacuated and secured.

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The following must be considered:

- Fire size and type.
- Hazards in the vicinity of the fire area.
- Actions taken by the first on scene.
- Fixed fire fighting systems activated.
- Fire control plans reviewed.
- Ventilation status.
- Boundary cooling commenced.
- CCTV reviewed.

Fire Group Preparedness

Having donned and checked equipment, the fire teams will proceed to the staging area and wait to be called forward by the On Scene Command (OSC) at the entry point.

The OSC will brief the fire teams prior to entry with the following:

- Fire location, type and development.
- Search plan and/or pattern and additional exit options.
- Known dangers.
- Any missing persons.

Breathing Apparatus Operational Guidance

When using Breathing Apparatus (BA) the following should be observed:

- The BA controller must monitor the fire team's cylinder pressures from entry to exit of the fire area.
- The fire team must verify BA operation by going 'on air' prior to entering the space.
- The fire team must not make an entry with less than 80% cylinder pressure.
- Individual fire team members must constantly monitor their equipment and advise the OSC if there are any issues.
- If any fire team member has a safety-related problem, the team must immediately withdraw from the fire area.
- Fire teams must maintain regular communications.
- Fire teams must carry out cylinder pressure gauge checks:
 - Before entry to the fire area.
 - At regular intervals.
 - Before attacking the fire.
 - Before changing level (eg, at a ladder or stairs).
 - Before moving into another space (at doors).
 - Before increased workload (eg, casualty rescue).
 - If conditions change (eg, the fire intensifies).
- Fire team members must maintain physical contact at all times in restricted visibility; or in clear visibility, must be in sight of each other.
- During an entry, a back-up fire team must always be dressed and on standby, but not on air.
- Spare BA cylinders should be located at the staging area.
- The fire team will be de-briefed by the OSC after exit from the fire area.

Door Opening Procedure

The temperature of the door surface should be tested prior to opening.

Fire teams must:

- Look around the edges of the door for signs of heat and smoke.
- Observe any distress to the door (eg, paint discolouration, burning).
- Apply a short burst of water onto the top of the door.
- If steam forms from the door, it is too hot to proceed and the team should withdraw, advising the OSC.

If the door opens TOWARD the fire team:

- Move the team onto the HINGE SIDE.
- Use the door as a protective shield.
- The fire team leader is to brace (push) their lower leg against the door.
- All of the team are to lower to the crouch position.
- The fire hose nozzle is passed to the second team member.
- Communicate a clear warning to the team prior to opening the door.
- Open the door approximately 10cm for 10 seconds, if a backdraft/rush of air is felt, close the door immediately.
- If safe, look around the door into the space to assess the conditions, inform the OSC.
- The fire team leader takes back the nozzle and leads the team through the door.

If the door opens AWAY from the fire team:

- Move the team onto the HANDLE SIDE.
- Use the bulkhead on the handle side as protection.
- All team members are to lower to the crouch position.
- The fire team leader holds the handle to maintain control of the door.
- The fire hose nozzle is passed to the second team member.
- Shout a clear warning to the team prior to opening the door.
- Open the door approximately 10cm for 10 seconds, if a backdraft/rush of air is felt, close the door immediately.
- If safe, look around the door into the space to assess the conditions, inform the OSC.
- The fire team leader takes back the nozzle and leads the team through the door.

WARNING

Fire doors should always be CLOSED after a fire team withdraws to stop the spread of smoke and fire.

Finding the Fire

Upon entry into area, follow the instructions and the route provided in the pre-entry briefing to try to locate the fire.

WARNING

Fire team must not be committed into a fire area unless a second fire team is standing by.

First enter the space where the fire is expected to be, using the search pattern given during the brief.

There are two methods of route search:

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1. Left Hand Search

- a) The fire team keeps contact/proximity with a bulkhead on the left hand side while progressing towards the incident, holding the hose with the right hand.
- b) When the team is withdrawing, a right hand progress will bring the team back to the entry point.

2. Right Hand Search

- a) The fire team keeps contact/proximity with a bulkhead on the right hand side while progressing towards the incident, holding the hose with the left hand.
- b) When the team is withdrawing, left hand progress will bring the team back to the entry point.

Direct and Indirect Routes

For example, a right hand search is carried out as follows:

- The leading team member (No.1) will use the back of the right hand (slightly clenched) in an upward/downward, slightly arcing movement in constant contact with the bulkhead. This movement assists in identifying door handles and other protruding items.
- The nozzle will be held in No.1's left hand and should be in water-wall mode (the nozzle is ready for protection/defence while, if accidentally knocked on and this will give a radial jet reaction).
- Protection from walking into an overhead obstruction is reduced by sweeping the left-hand (holding the hose nozzle) in a vertical movement from head height to lower chest.
- The deck area in front is swept with the outside foot (ie, leg furthest away from a fixed bulkhead). Weight is to be maintained on the inside leg (the leg nearest the bulkhead) and feet are to be in contact with the deck at all times.
- When changing deck level (stairs or ladders), No.1 goes first (defending the entry) before calling for the rest of the group and equipment.

Left hand and right hand search may be carried out in two different ways:

1. Direct Route:
Proceed directly towards the incident or a casualty with one behind the other in single file following the fire team leader. This is used when a general search is not required in order to search quickly.
2. Indirect Route:
Constantly maintain contact with each other in a line at a 90° angle to a bulkhead in order to maximise the search area. This is used in larger open spaces where the location of the incident may be unknown or uncertain.

If visibility is poor, thermal imaging should be utilised (if available).

There may be times when the fire cannot be detected for the following reasons:

- Inside a void space.
- Behind internal linings such as walls, ceilings, etc.
- Obstacles and blockages.
- Faulty detection system or equipment.
- Dangerous to enter.

Heat Monitoring

Regular steam and heat checks are required whilst fire fighting to ensure a safe working environment.

Entry Heat Check

The following method can be used to heat check upon entry into a fire area and should be carried out by a fire team who are fully dressed in protective clothing and wearing BA:

- The check would normally be done by any fire team member, initially in a low crouched-down position, at the entry point to the fire area.
- Before entry, observe any distress to the door (eg, paint discolouration, burning) or excess steam when applying water as part of the door opening procedure.
- After entry, observe if water is falling as droplets when spraying the deckhead, or if steam is being generated instead.
- The fire team members will need to consider their own physical condition regarding the heat.
- If a fire team member is feeling distressed, they must inform their fire team leader.

Steam Check

Whilst fighting a fire the fire team should:

- Monitor the steam/heat barrier.
- If necessary, STOP applying water if it becomes uncomfortably hot.
- Crouch down into cooler air.
- Assess the heat level.
- When the steam/heat barrier rises and it is safe to do so, stand up and continue fire fighting.
- Repeat the checks as necessary.

WARNING

If any member of a fire team at any time experiences heat conditions which are uncomfortably hot, they must withdraw from the fire area immediately, with or without completing a heat check.

Search and Recovery

When there is a suspected or confirmed missing person, a fire team must be deployed. If within a smoke filled area, this must be a fire team fully dressed and equipped with BA. Fire fighting should not be compromised, however, rescue and lifesaving takes priority over fire fighting. Ventilation and lighting must be reviewed for any area where a missing person is located or suspected of being located. It may be appropriate to increase smoke extraction from, or improve lighting for that space to aid rescue.

Casualty Recovery

If, during a search, a casualty is located, the fire team leader should:

- Inform the OSC who in turn will inform the Command and Control.
- Place an Emergency Escape Breathing Device (EEBD) on the casualty (if required).
- Perform a group gauge check (if wearing BA).
- Nominate two suitable fire team members to recover the casualty.
- Command and Control will direct the medical team to have a stretcher waiting at the exit point.

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Identifying a Fire

Class of Fire	Description
Class A	Fires in ordinary combustible materials, such as wood, cloth, paper, rubber, and many plastics.
Class B	Fires in flammable liquids, combustible liquids, petroleum greases, tars, oils, oil-based paints, solvents, lacquers, alcohols, and flammable gases
Class C	Fires that involve energised electrical equipment.
Class D	Fires in combustible metals, such as magnesium, titanium, zirconium, sodium, lithium, and potassium.
Class K	Fires in cooking appliances that involve combustible cooking media (vegetable or animal oils and fats).

If a fire is uncontrollable or too large to tackle, the fire must be contained inside the fire area by closing down fire doors and performing boundary cooling/starvation.

WARNING
When using fire hoses to tackle fire on the upper decks, attention should be paid to the stability of the vessel. The addition of water high up in the vessel, coupled with the inherent free surface effect, will have a negative effect on the vessel’s stability and in extreme cases could cause the vessel to capsize.

Withdrawal

After completing an attack of a fire, the fire team should exit using a reversed direct route, ensuring that all doors are closed during the withdrawal. No.1 leads the group out, carrying the nozzle to maintain team protection. When changing deck level (stairs or ladders), No.1 goes last (defending the exit). Once all the team members are on the same level, No.1 resumes the lead position.

The fire team must be debriefed upon exit and proceed to the staging area to recover, replenish and prepare for re-entry if necessary.

Post Incident Actions

The following actions must be taken in the post-incident phase:

- OSC or nominated person is to regularly assess the affected area:
 - Note the risks and the possibility of re-ignition of the fire.
 - Observe areas that may contain pockets of toxic gas or are oxygen deficient.
 - Enter the affected area(s) using BA and multi-gas detectors until the atmosphere is confirmed safe.
- Post a fire watch and have appropriate fire fighting equipment at hand.
- Carefully manage ventilation for the removal of smoke and steam, in order to avoid its spread.
- All fire equipment and fire stations are to be returned to operational readiness.
- Restore power, propulsion and services as soon as possible.
- Ensure the welfare of the response teams, with particular regard to BA wearer hydration and fatigue.

3.2 Emergency Procedures - Engine

- 3.2.1 Flooding in the engine room**
- 3.2.2 Emergency Steering**
- 3.2.3 Emergency Fire Pump**
- 3.2.4 Fire in the Engine Room**
- 3.2.5 Electrical Power Failure**
- 3.2.6 Electrical safety**
- 3.2.7 High Voltage Systems Isolation and Permit To Work System**

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3.2 EMERGENCY PROCEDURES - ENGINE

3.2.2 EMERGENCY STEERING

3.2.1 FLOODING IN THE ENGINE ROOM

Authors note: Section in progress.

Engine Room Flooding

Flooding in the engine room can occur due to a defect in the hull structure, possibly due to grounding, berthing or collision damage, or, more likely, due to a defect in the sea water pipeline system.

Measures to Prevent or Alleviate Flooding

- Maintain pipelines externally, tighten slack supports and replace broken U bolts on pipe brackets to minimise fretting in way of the supports.
- Operate all of the ship's side valves regularly, so that they can be operated easily when required. Valves such as the fire pump suction valves, which are normally open, should be closed and reopened regularly to prevent a build-up of marine growth.
- Before opening any sea water filters for cleaning, make sure the isolating valves are closed tight by opening the vent in the cover. In any case, break open the cover joint before removing all of the cover bolts. The same applies when opening coolers and pipelines anywhere in the system.
- Care must always be taken when removing covers or opening any part of the sea water pipe system, as valves which are indicated as being closed may not be fully closed. Where gate valves or through cocks are used for draining and venting, prove them clear by rodding before removing covers.
- If the source of ingress is rapid and cannot be identified, shut all remote-operated sea and ship side valves. This action assumes that the levels have reached above the floor plates.
- Personnel should be familiar with the position of the bilge suctions and the pumps that can be utilised for bilge pumping duties. They should also be familiar with the position of the main sea suction and overboard valves and know which main suction is currently in use.
- The emergency bilge suction valve should be operated on a regular basis. Double bottom sounding pipe cocks and caps should be secured after use.

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3.2.3 EMERGENCY FIRE PUMP

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3.2.4 FIRE IN THE ENGINE ROOM

Engine Room Fire Prevention

The best way of dealing with an engine room fire is to prevent one. Oil spills must be cleaned up as soon as they occur and oily waste or rags must not be left lying around. Any leakage from oil pipes must be rectified as soon as possible. Rags, oily waste and similar combustible material must not be stored in the engine room, and plastic containers must not be used for storing such material or used as drip trays. Lagging must be correctly fitted to exhaust manifolds and the dripping of oil onto hot manifolds, even when lagged, must be prevented.

Fire detection equipment must be checked frequently and fire extinguishing appliances must be in an operable state at all times.

Engine Room Fire

Step	Task
1	Immediately sound the fire alarm and muster the crew.
2	If personnel are missing, consider the possibility of searching in the fire area.
3	Determine the location of the fire, what is burning, the direction of spread and the possibility of controlling the fire.
4	If there is the least doubt as to whether the fire can be controlled by ship's crew, warn of the situation on the distress frequencies.
5	If the fire fighting capacity is limited, give priority to fire limitation until the situation is clarified.
6	If substances which are on fire, or close to a fire, may emit poisonous gases or explode, direct the crew to a safe position before actions are organised.
7	Establish the vessel's position and update the communication centre.
8	If any person is seriously injured, request assistance from the nearest rescue centre.

WARNING

Under no circumstances should anybody attempt to tackle an engine room fire alone. It is essential that the fire alarm be raised as soon as an outbreak of fire is detected.

In Port

Step	Task
1	Activate the emergency shutdown system in agreement with the terminal duty personnel.
2	Conduct a crew check.
3	Organise the crew for fire fighting duties.
4	Inform the terminal authorities and the local fire brigade even if the fire appears to be under control.
5	If personnel are missing, consider the possibility of searching in the fire area.
6	Close all accessible openings and hatches to prevent the fire spreading.
7	Prepare to disconnect the cargo and bunkering hoses and release the bunkering barge if applicable.
8	Prepare to vacate the berth, if required, and inform the authorities immediately if there might be problems in vacating the berth.
9	If there is a danger of the release of poisonous gases or of explosion, consider part or total ship abandonment. Ship drawings, cargo plans etc, should be taken ashore. A crew check is to be carried out again.
10	Consider using the fixed fire extinguishing systems, depending on the extent of the fire.

Step	Task
11	On the arrival of the fire brigade, inform the Chief Fire Officer of: - Any personnel missing. - Assumed location of fire. - What is assumed to be burning. - Any conditions that may constitute a hazard.

Battening Down of the Engine Room

Step	Task
1	Stop the main and auxiliary engines and shut down the boiler.
2	Sound the evacuation alarm.
3	Stop all the ventilation fans.
4	Start the emergency generator and put on load.
5	Trip the quick-closing valves and the engine room auxiliary machinery from the fire control station.
6	Count all personnel and ensure that none are in the engine room.
7	Close all fire flaps and funnel doors.
8	Close all doors to the engine room.
9	Start the emergency fire pump and pressurise the fire main.
10	Operate the water mist system or the high expansion foam system depending upon the location and extent of the fire.

WARNING

Fire plans are housed in red cylindrical containers at the port and starboard accommodation entrances on the upper deck. These are positioned to assist outside parties to deal with a fire on board and should under no circumstances be removed.

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3.2.5 ELECTRICAL POWER FAILURE

Authors note: Section in progress.

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3.2.6 ELECTRICAL SAFETY

Electrical Safety

Before carrying out maintenance on any electrical systems, the following conditions are to be complied with:

Step	Task
1	The senior on-watch engineer should be informed of any maintenance on any electrical equipment within the machinery spaces.
2	Approach to the maintenance should be fully discussed at a pre-work planning meeting.
3	Work on electrical systems should be carried out only by suitably trained and equipped personnel.
4	The breaker for the apparatus under maintenance should be opened.
5	Fuses should be removed.
6	The opened breaker should be locked and the key kept in the possession of the person carrying out the maintenance.
7	When working in cubicles such as starter boxes, insulated shields indicate live components and should not be tampered with. Work in this area will require isolating at a more primary level.
8	A caution notice (Caution Men Working) should be attached to the breaker isolating the circuit under maintenance.
9	The circuit should be tested to check that it is dead and that no auxiliary supply is present.
10	All officers and technical staff should be competent in the treatment of those suffering from electric shock.

WARNING
A High Voltage Permit to Work Certificate to be completed as per company QA system.

High Voltage

High voltage circuits are potentially more dangerous than low or medium voltage circuits. This is not only due to the increased voltage, but also the explosion risk and because, under certain circumstances, high voltage circuits can retain a lethal charge even when switched off. In addition, dangerous potentials exist some distance from the actual live high voltage conductors, the distance being determined by the conductor voltage and the dielectric strength of the insulating materials (including air) surrounding the conductor.

Therefore, it is essential that all persons who may be required to work on, or operate, high voltage apparatus are fully aware of the hazards and how to avoid the associated danger.

Personnel carrying out high voltage isolation, earthing, maintenance and inspection should have attended a recognised high voltage safety training course.

High voltage apparatus is classified as any apparatus, equipment and conductors which are normally operated at a voltage exceeding 690V.

Action in the Event of Electric Shock

On discovery of a victim of electric shock, turn off the power source immediately if possible. If this is not possible, the victim must be moved away from contact with the source by use of a high voltage hook (if available), high voltage gloves or by some other means which is suitably insulated.

Carry out resuscitation if this is required using the current/latest approved techniques.

It is advisable that a first aid kit is available when work of this nature is being carried out. This kit should contain equipment to assist in helping a victim breathe, as resuscitation is a common requirement in electrocution from high voltage.

The other common effect of electric shock is burning of the skin, at the site of entry and exit of the electrical current. Suitable treatment for burns should therefore also be available in the first aid kit.

WARNING
On no account must direct contact be made with the person connected to the source.

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3.2.7 HIGH VOLTAGE SYSTEMS ISOLATION AND PERMIT TO WORK SYSTEM

A key interlock system is provided for the 6,600V switchboards, where access to live electrical components can be reached without using tools.

The key interlock system allows safe access to the equipment for maintenance and repair. It ensures that the access to the high voltage areas is prohibited in all cases where the correct switch off/down and earthing procedure of the main breaker is not performed completely or in the wrong order. A specific step by step procedure is required to gain access to the keys for the cubicles.

Where a large circuit-breaker is removed, a shutter system automatically shrouds any live terminals which may otherwise remain exposed.

Prior to any work being carried out, the item of equipment or circuit under maintenance is to be isolated from all sources of possible supply, earthed and tested to ensure that the circuit or equipment is 'dead'.

Earthing

Earthing a circuit, or item of equipment, can be carried out using the earthing device at the circuit-breaker, or locally by the use of portable earthing leads.

After confirmation that the conductor to be earthed is at zero volts by using the appropriate high voltage test equipment, earthing leads should always be connected to the earth before attachment to the conductors.

Upon completion of the work, the leads should be removed from the conductors prior to removing the earth connection. If work is to be carried out on an item of equipment remote from the circuit-breaker, the earth leads should be applied locally at the motor, in addition to using the circuit-breaker earthing device.

Permit to Work (PTW) Procedure

If maintenance or inspection is required to be carried out on any high voltage equipment, a Permit to Work Certificate must be obtained and completed.

All items and parts of the circuit that have been earthed are to be identified on the Permit to Work Certificate. All keys that have been removed and the points of isolation are also to be identified on the PTW.

The electrical officer is responsible for the disconnection of the earthing leads and the return of the circuit or equipment to normal service.

The PTW Certificate is to be signed by the designated staff as indicated on the PTW.

The PTW is in duplicate form. The duplicate copy must be held by the personnel undertaking the work and the top copy is to be kept in the engine control room.

'Caution' and 'Danger Live' notices as appropriate are to be displayed at all points where the work is being carried out and also at any remote isolation points used during the work to prevent any inadvertent switch-on. Additional precautions may be required as indicated on the PTW. These may include reliable communications, eye protection, fire extinguisher and no 'single man' working.

Cancellation of the Permit to Work Certificate must be signed by the person actually carrying out the work and the staff identified on the PTW.

WARNING

A High Voltage Permit to Work Certificate to be completed as per company QA system.

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3.3 Emergency and Safety Procedures - Navigation

3.3.1 Steering Gear Failure

3.3.2 Collision and Grounding

3.3.3 Man Overboard

3.3.4 Emergency Towing and Being Towed

3.3.5 Fire

3.3.6 Flooding

3.3.7 Abandon Ship

3.3.8 Emergency Reporting

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3.3 EMERGENCY AND SAFETY PROCEDURES - NAVIGATION

3.3.1 STEERING GEAR FAILURE

Guidance: Always refer to the relevant Operational Procedure Systems.

The following actions to be carried out:

Step	Task
1	Inform the Master.
2	Inform the Chief Engineer and the duty engineer in the engine control room.
3	Engage emergency steering. This procedure is posted in the steering gear room.
4	'Not Under Command' shapes or lights to be exhibited.
5	Commence sound signalling.
6	Prepare engines for manoeuvring.
7	Take the way off the ship.
8	Prepare for anchoring if in shallow waters.
9	Evaluate the need for tug escort/assistance.
10	Evaluate the need for salvage.
11	Broadcast an URGENCY message to ships in the vicinity.

Emergency Steering Drill

Emergency steering drill should be carried out at least once every three months when traffic and navigational restrictions permit.

The drill is to consist of the direct operation of the main steering gear by using the manual control within the steering flat. This operation is to be directed from the navigation bridge. After each drill, details and the date it is carried out are to be entered in the Deck Log Book.

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3.3.2 COLLISION AND GROUNDING

Minimising Damage

If a collision is inevitable, damage can be minimised by striking a glancing blow.

Collision amidships of either ship must be avoided whenever possible and a bow to bow, quarter to quarter or bow to quarter situation is preferable.

Grounding

Particular attention must be paid to the possible changes in a vessel's stability due to grounding.

A grounded ship is similar to a ship in dry dock. Since the vessel is no longer floating, the pressure on the ship's bottom has exactly the same effect as an equal weight being removed from that level within the ship. Consequently the centre of gravity rises and the GM is reduced.

Grounding at low water on soft mud, in general, presents little danger. This is because the ship will settle into the mud and, as the tide rises, the pressures decrease and the centre of gravity falls once more.

Grounding at high water on a hard sea bed is potentially dangerous. Firstly, the ship cannot settle and secondly, as the tide falls so the centre of gravity will rise and a point may be reached where the GM becomes negative and the ship unstable. The ship may lie over on her side, causing further damage and/or flooding.

The greatest danger arises when the vessel grounds at high water on an uneven hard bottom.

Manoeuvring a Vessel After a Collision

Following a collision or other incident that could result in oil spilling from the vessel whilst under way, the subsequent movement of the vessel could be critical. No two situations will ever be the same and each situation will demand a different manoeuvre to be carried out by the Master. All points listed below are to be considered and are intended to assist in making the right decision. Consider the following:

- Sources of ignition.
- Oil in the water.
- Oil on fire.
- Gas cloud formation and position.
- Location of fire on the vessel.
- Evacuating casualties.
- Current helm/engine situation.
- Momentum of own vessel.
- Availability of helm/engines.
- Proximity of other hazards.
- Effect of the wind on fire/smoke.
- Effect on vessel of currents, tides and wind.

Options to be Considered

- When the vessel is stopped or nearly stopped, putting an astern movement on the vessel to contain the effects of any fire forward of the accommodation block.
- When the vessel is moving ahead, continuing movement ahead to clear any spilled oil and/or altering course to minimise the effects of fire and smoke.

- When aground or locked with another vessel in a collision situation, no movement should be attempted to prevent further damage or spillage, also refer to the vessel's response plan.

Disabled Vessel

In the event of a total power loss, steering failure or both, there are several actions that can be taken to reduce the risk of stranding or collision.

Significant points are:

- The single most effective action which can be taken to influence the direction of drift is to deliberately manoeuvre the ship so that, when stopped, the wind is on a particular side. That is, the direction of drift can vary substantially with the wind on one side of the ship or the other.
- When steering capability is lost, the drift direction and speed can still be influenced by:
 - Giving the ship a list.
 - Changing from stern to head trim.
 - Putting the rudder hard-over to the downwind position (putting the rudder hard-over in the upwind position has virtually no effect at all).

When steering failure occurs, judicious use of the main engine can be helpful in changing the drift direction and speed or even in stopping the ship altogether.

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3.3.3 MAN OVERBOARD

Missing Persons

In the event of a person being suspected missing, the officer of the watch should be informed and steps put in place to determine if they are actually missing or just not readily available.

- Determine where and when the person was last seen.
- Organise a search of the vessel including decks, engine room and all accessible spaces.
- Prepare to turn the vessel round and retrace the track to where and when there was a last sighting of the person.
- Post additional lookouts.
- Prepare the rescue boat for immediate use and have the crew standing by.

Should the on board search not find the person, then a VHF call to other vessels in the area, asking them to keep a good lookout as they transit the area.

Other vessels may join in the search.

On arrival at the last known position, a search of the area will be required, this may involve only your own vessel or possibly others who have come to assist.

Procedure

In the event of a man overboard, the following actions should be taken:

Step	Task
1	Helm hard-over toward the side from which the person fell. Mark the MOB position on the ECDIS and GPS/GPS printer.
2	Release the appropriate bridge wing lifebuoy and light/smoke.
3	Post a lookout to continuously keep sight of the person.
4	Mark the position on GPS and radar equipment.
5	Sound the general alarm and make a PA announcement.
6	Inform the Master of the situation and call additional personnel if required.
7	Continue executing the turn, the type of which will be dictated by the situation and the position of any other vessels in the area.
8	Inform the duty engineer and prepare engine for manoeuvring.
9	As parties close up to emergency stations, the rescue boat can be prepared for lowering.
10	Hoist the appropriate flag signals and use the VHF to warn vessels in the vicinity.
11	Manoeuvre the vessel as close as possible to the target.
12	Launch the rescue boat.
13	Effect a rescue and retrieve the rescue boat.
14	Administer first aid and if necessary obtain medical assistance.

General

The Williamson turn is a proven method of returning the ship, via a reciprocal course, to the position of the person overboard.

Performance of a Williamson turn will considerably reduce the ship's speed. The rapid posting of lookouts is vital in order to locate the person and/or to keep them in sight.

Search Patterns

The recommended search patterns, starting from a common datum point, provide a basis to search for a man overboard. The search patterns for a man overboard, whose position is known approximately, but cannot be seen are as follows:

Note: Refer to the Merchant Ships Search and Rescue Book and the Wheelhouse Poster - Man Overboard Rescue Manoeuvre.

Sector Search Pattern

Step	Task
1	Where the position of a person is known within reasonable limits, the Sector Search Pattern should be used first.
2	All turns are 120° to starboard.
3	The length of the leg is dependent on the state of visibility and the size of the search object, in as much as the length of each leg is the same.
4	The first leg of the search should begin in the direction where the person is most likely to be seen.
5	Should the person still be missing on completion of the initial search pattern and it is known they are within the area, a second pattern should be commenced with the heading 30° to starboard of the initial search pattern heading.

Square Search Pattern

Step	Task
1	If the sector search pattern has failed to find the person, the ship should commence the square search pattern from the same datum point.
2	All turns are 90° to starboard. The length of the leg is dependent on the state of visibility and the size of the search object. After the first square has been completed, the subsequent increase in search leg is increased accordingly.

MERSAR

The IMCO MERSAR manual is a useful source of additional information. Search patterns including those suitable for use with more than one ship and aircraft are described.

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Search and Rescue

Step	Task
1	Check the position of the vessel in distress with respect to own ship's position.
2	Relay the distress message, if no acknowledgement is received from the shore station.
3	Proceed at maximum safe speed to the distress location and inform the RCC.
4	Inform the owners/charterers/operators of the deviation.
5	Maintain a continuous listening watch on all distress frequencies.
6	Consult MERSAR/IMOSAR manuals.
7	Establish communications with all other surface units and SAR aircraft involved in the SAR operation.
8	Follow instructions from the RCC and on-scene commander for executing SAR operations.
9	Plot positions, courses and speeds of other assisting units.
10	Prepare engines for manoeuvring when near to the distress location.
11	Monitor the X-band radar for locating the survival craft transponder (SART) signals using 6 or 12 mile ranges.
12	Post extra lookouts for sighting flares, switch on searchlights at night.
13	Prepare the rescue boat, pilot ladder and nets for assisting the recovery of survivors when sighted.

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3.3.4 EMERGENCY TOWING AND BEING TOWED

The vessel is fitted with a specially designed Emergency Towing Apparatus (ETA). Forward there are two custom-built Panama fairleads, a ten metre section of chafing chain and an OCIMF approved chain stopper. On the poop deck is the automated equipment which allows the towing wire to be released and deployed by one person.

Being Towed

Stern System

To operate this, open the box mounted on the side of the fairlead containing the pick-up gear and messenger lines and deploy over the side. The pick-up gear comprises a marker buoy with self-igniting light attached to a pick-up rope. The pick-up rope is buoyant and brightly coloured to make it easily visible, and is attached to a heavier gauge messenger wire which is used by the other vessel to haul the towing pennant on board. This system can be used when the vessel has lost all power and is dead in the water.

Bow System

Using the bow system will require considerable manpower and time to rig. It will also require the availability of some deck machinery.

It is most likely to be used in conjunction with a salvage tug and for a pre-planned tow with the vessel in no immediate danger. To rig the system it will be necessary first to place the section of towing (chafing) chain in the chain stopper, then using throwing lines and messengers, finally heaving on board the tug's towing wire which is then secured to the vessel's towing chain with the purposely designed shackle. Ensure that the chafing chain, when slackened back, passes through the purpose made Panama fairleads. This will prevent the towing wire from unnecessary chafing. When the ship is totally without power but towage from the bow is still necessary, a messenger can be led from the ocean-going tug through the vessel's towing fairlead and returned to the tug. The tug's winch is then used to heave round the towing wire for connection to the ship's chain.

Towing Another Ship

There are many factors which determine the most suitable method of taking another vessel in tow. The type and size of the ship to be towed, the urgency of the situation, and the duration and route to be taken. Taking into account the size of the vessel, and the equipment fitted, it is extremely unlikely that the towing of another vessel will be undertaken except in the case of extreme emergency. This may occur when trying to prevent a vessel from grounding when either a tug or more suitable vessel is not available. In such situations, the following should be considered:

The initial information required:

- The urgency of the situation and the time available before grounding.
- The size of the other vessel.
- The type of towing equipment available.
- Is power available for deck equipment?
- Available manpower.

Connecting the Tow

- Decision made by Master as to the equipment usage.
- Use the towing vessel's emergency towing apparatus.
- Use towed vessel's emergency towing apparatus.
- Establish continuous radio communication between the two vessels and any other that may be involved in the operation.
- Pass a light line between the vessels.
- Connect to emergency towing apparatus buoy line and deploy when the other vessel is ready.
- Tow wire connected to other vessel.

If picking up the other vessel's tow wire, rig a bridle between two of the poop winches using their wires and connect to the tow wire using a suitable shackle.

Note: The designed brake load on each winch is not more than 80% of the wire breaking strain with the wire at the inner level of the drum. The load at which the brake starts to render will vary depending on the number of layers of wire remaining on the drum and the condition of the brake linings.

Commencing the Tow

- The towing vessel is to make way very gradually, using her engines in short bursts of minimum revolutions.
- Increase speed in stages of five revolutions per minute. Do not alter course until both vessels are moving steadily.
- When altering course, do so in stages of 5°.
- The towing vessel should use its steering gear in conjunction with the towed vessel.
- If the towed vessel's steering is not available, her rudder should be placed amidships and locked.
- The towed vessel should not use her engines unless requested to do so by the towing vessel.

Steering Problems

If towing by the stern and the rudder is not locked, the rudder may assume the hard-over position.

If towing by the bow and the disabled vessel's engines are used, the propeller race can cause the rudder to assume a hard-over position.

The disabled vessel's trim should if possible be as follows:

- Towed by the bow, the trim should be one in one hundred by the stern.
- Towed by the stern, the trim should be one in eighty by the head.
- Steer directly into wind to minimise yaw.
- Some larger vessels yaw the least on a heading 20° to 30° off the wind.

Passing Tow Line Alternatives

Use line-throwing apparatus to pass an initial light line followed by heavier lines.

A helicopter with a lift capacity of two to three tons could be used to facilitate the connection.

It should be remembered that speed and yaw have a considerable effect on the forces acting against a tow. In the case of speed, the forces increase considerably.

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3.3.5 FIRE

Step	Task
1	Sound the fire alarm and initiate the vessel's emergency response procedures.
2	Call the Master, if not already on the bridge, and notify the engine control room.
3	Consider altering course to avoid spreading the fire/danger to personnel.
4	Muster the crew at their emergency stations.
5	Establish communications with the fire control station and the engine control room.
6	Prepare the main engine for manoeuvring.
7	Check for any missing or injured crew members.
8	On locating the fire, inform all on board of the location. Prepare fire fighting/rescue/resuscitation equipment. Operate the appropriate emergency shutdown if applicable.
9	If there is an engine room fire, prepare for engine failure. Start the emergency generator. Shut off all relevant quick-closing valves.
10	Switch off ventilation fans and close all doors, including fire and watertight doors and vents.
11	Switch on deck lighting at night.
12	Start the available fire pumps and commence boundary cooling. Clear adjacent spaces of all combustible materials.
13	Assess the fire situation and fight the fire appropriately.
14	Prepare the lifeboat and liferafts for abandoning the vessel according to the situation.
15	Make the vessel's position known to the communication centre, satellite terminal and other automatic distress transmitters. Update as necessary.
16	Broadcast a Distress Alert and Message if the vessel is in grave and imminent danger and immediate assistance is required, otherwise broadcast an URGENCY message to other vessels in the vicinity.

3.3.6 FLOODING

Step	Task
1	Sound the general emergency alarm and initiate the vessel's emergency response procedures.
2	Tackle the flooding.
3	Sound all bilges and tanks.
4	Identify the location of incoming water and close all intakes to the compartment.
5	If the flood water is in danger of compromising electrical equipment, switch off all electrical power running through the damaged area.
6	If possible, shore up the area to stem the water flow.
7	Consider the transfer of bunkers or ballast to correct an excessive list/trim with due regard to stability of the vessel.
8	Make the vessel's position known to the communication centre, satellite terminal and other automatic distress transmitters. Update as necessary.
9	Broadcast a Distress Alert and Message if the vessel is in grave and imminent danger and immediate assistance is required, otherwise broadcast an URGENCY message to other vessels in the vicinity.

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3.3.7 ABANDON SHIP

Step	Task
1	Broadcast Distress Alert and Message on the authority of the Master.
2	Sound the general emergency alarm on the vessel's whistle/bells.
3	Inform the owners/charterers/operators of the intended abandonment.
4	Instruct crew members to wear adequate and warm clothing.
5	Muster the crew at the lifeboat stations and don their life jackets.
6	Prepare to launch the lifeboat and liferafts.
7	Ensure that liferaft painters are well secured to the vessel.
8	Embark the crew in the lifeboat and launch on the verbal instructions of the Master.
9	Ensure the lifeboat or liferafts remain in safe proximity to the vessel and in contact with each other.
10	Follow the survival instructions, which will be found in the lifeboat and liferafts.
11	Review the situation and consider returning to the vessel if conditions improve.

3.3.8 EMERGENCY REPORTING

The particulars of all accidents or incidents involving the ship and/or her personnel must be sent as soon as possible to Head Office by a secure means. Initial incident reporting may be advised via the telephone, the contents of the call being confirmed by telex. A full report must be made at the earliest opportunity. It would be beneficial to keep a template of the following Initial Incident Report ready for use on each telex communication system.

Initial Incident Report

Step	Task
1	Date and time (UTC) of the incident (a six figure group; first two figures, day of the month, last four figures, the hours and minutes using the twenty-four hour clock).
2	Nature of the incident; collision, grounding, fire etc. Sufficient detail must be given to allow an overall appreciation to be made.
3	Position of ship. Latitude and longitude is preferred along with a general statement of where the vessel is.
4	Name, nationality and type of any other ship involved.
5	Nature and extent of damage: - To own vessel. - To any other vessel or installation involved.
6	Casualties if any, including those missing: - To own vessel. - To any other vessel or installation involved.
7	State of sea and weather.
8	If immobilised and towage or other assistance may be necessary, the following additional information must be included: - Set and drift of current. - An estimate of how long the vessel can safely remain without assistance under the prevailing circumstances. - Nature of any assistance required. - Give the names and positions of any Company ships contacted, and those of associated companies in the vicinity. - Estimated time of readiness to proceed.
9	Details of cargo etc, lost overboard.
10	Any other relevant information.

Note: Nil reports, as applicable, must be made under each heading. This is particularly important in the case of casualty reporting. Plain language should be used in preference to code.

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Checklists for Specific Accidents

In the following it will not always be necessary to report on each item listed and conversely the list will not always cover every item that needs comment. Remember that the reports are used by Head Office to ascertain the particulars of an accident and, therefore, to enable them to act accordingly.

Grounding

- Voyage from - to. Date and time of grounding
- Position of grounding (latitude and longitude for clarity)
- Draught of ship
- Cargo/ballast and bunkers on board, where stored
- Heading of ship in grounded position
- Soundings plan, prior to grounding, at time of grounding and repeated as necessary at various stages between high and low water
- Tidal conditions
- Nature of bottom
- Identify grounded section of ship
- Damage incurred or suspected
- Oil pollution actual or potential
- Weather conditions at time of grounding updated as necessary
- Wind force and direction
- Sea state
- Visibility
- Current, set and drift
- Chart or charts in use
- Positions, courses steered, engine movements prior to grounding
- Heading and estimated speed of ship at time of grounding
- Bridge manning at time of grounding
- Navigation aids in use
- Pilot involvement
- Position and movement of other ships in the vicinity where relevant
- Actions taken after grounding
- Involvement of other parties
- Tugs
- Port Authorities
- Classification Societies
- P & I Club (involved if third party damage incurred)
- View on cause of grounding

Manoeuvring Accident

- Voyage from - to. Date and time
- Position (latitude and longitude)
- Berth
- Describe manoeuvre being carried out

Describe circumstances leading up to the accident:

- Courses steered
- Positions
- Engine movements
- Weather conditions
- Wind force and direction
- Sea state
- Tidal conditions
- Bridge manning, also any witnesses
- Pilot involvement

Describe the damage incurred by own ship, jetty or others and actions taken after the accident:

- Involvement of other parties
- Tugs
- Port Authorities
- Classification Societies
- P & I Club (involved if third party damage incurred)
- Surveyors
- View on cause of accident

Note: All cases of known or suspected hull damage are to be recorded.

Heavy Weather Damage

- Voyage from - to. Date and time of damage
- Position where damage occurred (latitude and longitude for clarity)
- Draught of ship
- Cargo/ballast and bunkers on board

Circumstances leading up to the damage:

- Course steered (include adjustments made because of the weather)
- Engine revolutions (include adjustments made because of the weather)
- Weather
- Wind force and direction
- Sea state and swell
- Visibility

Particulars of damage:

- Any action taken to make good the damage
- Outstanding repairs
- List of damaged equipment requiring replacement

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Loss of Anchors

- Voyage from - to. Date and time
- Position (latitude and longitude), also give additional details such as actual bearings and radar distances
- Depth of water and nature of bottom
- Tidal state and current

Operation in progress:

- Anchoring
- Weighing anchor
- At anchor

Method in use:

- Walking-out
- On the brake
- Which anchor and how much cable lost
- Any other damage

AMVER

The principle of any ship reporting system is to tap the resources of the numerous merchant vessels that are at sea at the time of a marine incident. One or more vessels may offer the earliest possible response if located near the casualty. The purpose of AMVER is to maximise the effectiveness of response to a marine emergency by co-ordinating and controlling the assisting ships.

AMVER (Automated Mutual-Assistance Vessel Rescue) is operated by the United States Coastguard for all merchant vessels of more than 1,000 grt, on voyages in excess of 24 hours, regardless of nationality. AMVER centres located in New York and San Francisco are capable of processing data automatically, and in the event of a marine incident, co-ordinate the vessels most suitable to respond. The data is received through a vessel reporting system, these reports may be made free of charge through participating stations.

The reports are made in the following format:

Sailing Plan

This report may be made well in advance of departure from a port. The report includes the ship's name and call sign, the ports of departure and destination, and the navigational route to be followed between them, along with estimated departure and arrival times. Any special resources such as advanced communication systems should also be included in the report.

Departure Report

This report is transmitted as soon as possible after departure. It should include the ship's name, call sign, and time of departure, and either confirm that the original sailing plan remains valid or update the changes instigated.

Position Report

This report is transmitted within 24 hours of departure and continue to be transmitted within 48-hour intervals during the course of the voyage. It should include the ship's name, time and position, together with the destination and latest ETA.